

NORTHERN KENYA GRASSLAND CARBON PROJECT

Document Prepared By Northern Rangelands Trust, Native, a Public Benefit Corporation, Soils for the Future, LLC, and The Nature Conservancy

Project title	Northern Kenya Grassland Carbon Project
Project ID	1468
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Monitoring period	01 January 2021 – 31 December 2021
(CCB) GHG accounting period	01 January 2013 – 31 December 2042; 30-year total period
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Most recent date of issue	06-April-2021 is the date the monitoring report was most recently submitted
Version	1.0
VCS Standard version	4.1
CCB Standards version	2.0
Project location	Kenya
Project proponent(s)	Northern Rangelands Trust (NRT) Mohamed Shibia mohamed.shibia@nrt-kenya.org
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History of CCB Status	Validated as CCB Gold on 29 April 2020. Verified as CCB Gold for 2013-2016 monitoring period on 20 November 2020. Ongoing CCB Gold verification for 2017-2020 monitoring period.
Gold Level criteria	Climate Change Adaptation Benefits - increased productivity and water storage in soils; Exceptional Biodiversity Benefits - increased populations of trigger and endemic wildlife species and conservation of sky islands; Community Benefits – better livestock health, improving household livelihoods
Prepared by	Native, a Public Benefit Corporation

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1 SUMMARY OF PROJECT BENEFITS

The Northern Kenya Grassland Carbon Project aims to remove greenhouse gases from the atmosphere by implementing sustainable grazing management over approximately 2 million

hectares of northern Kenya savannas and grasslands. Active planned grazing and livestock movements in each of 13 different Conservancies, replaced baseline unrestricted grazing. Rapid movement allows the recovery of perennial grasses, which foster greater carbon inputs, higher retention of carbon as lignin and cellulose, and greater belowground production, all of which lead to the removal of greenhouse gases from the atmosphere to soil carbon.

Climate Benefits

The planned rotational grazing project activity will remove 0.6 to 1.1 tonnes CO₂e per hectare each year for at least thirty years. The project will be implemented across nearly 2 million hectares of rangeland.

Community Benefits

Past overgrazing by pastoralists from more than six indigenous ethnic groups in the region has depleted soils of organic matter, greatly reduced perennial vegetation cover, and the potential production of forage for livestock. The sustainable planned grazing implemented in the project is restoring grass and other livestock forage, which will provide buffers against variable rainfall and reduce the need for long and risky migrations during droughts. Restoration of grass is increasing the climate adaptability of local communities because soils with more carbon store more water and sustain greater forage production through dry periods. Water retained in soils can improve surface water availability. Improved forage may also allow livestock owners to more easily sell livestock into food markets. Revenues from sale of carbon credits shared with communities and salaries/wages will benefit project employees and their families.

Biodiversity Benefits

Restoration of perennial grasses and soil water lead to increased productivity and cover, which will increase populations of four endemic, endangered wildlife species: reticulated giraffe, Grevy's zebra, beisa oryx and black rhino. Associated with the improvements in productivity will be increases in the diversity of grassland and savanna biodiversity indicator species from groups including plants, trees, dung beetles, grasshoppers, and birds.

1.1 Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
1) Local communities, oriented around 13 wildlife conservancies organized under an NGO called the Northern Rangelands Trust, engage in new	The implementation of project activities continue to result in a halt in the decline of forage productivity that occurred prior to the start of the project. After accounting for the effect of annual variation in rainfall,	3.2.2	Same as the current monitoring period

planned rotational grazing practices, as opposed to repeated, permanent grazing simultaneously on all grazing lands.	vegetation data supports that forage production has nearly doubled in the project area during the period when the project activities were fully implemented across all 13 Conservancies (2015-2021).		
2) Multiple communities and ethnic groups participate in an annual regional grazing plan that allows livestock to move over long distances and efficiently use wet and dry season range	Planning through a hierarchy of grazing decision-makers was implemented in October 2015 and led to the explicit inclusion of grazing plans in overall conservancy management plans for 8 conservancies. Since 2015, conservancy managers have utilized these annual conservancy management plans in order to effectively carry out grazing plans for their respective conservancy. It also extended the decision-making process to all ethnic groups regardless of the conservancy in which they live. Each project conservancy continues to utilize Rangeland Coordinators to coordinate and implement regional grazing plans between project conservancies and non-project areas.	3.2.2	Same as the current monitoring period
3) Long-term benefits to both financial, social, and natural capital for both communities and individuals.	Data from social surveys (Social CoMMs and project-wide surveys) continue to demonstrate that the project has improved access to milk and water and greater indicators of security under the project scenario.	4.1.3	Same as the current monitoring period
4) Enhancement of the region's role as a High Conservation Area for key endangered species.	Black rhinos (<i>Dicoris biceris</i>) were successfully introduced into the Sera Conservancy in 2016. The number of illegally killed elephants has sharply declined since the start of the project, and monitoring data from 2021 supports that multiple wildlife species have	5.1.1	Same as the current monitoring period

	continued to expand their range and increase in density since the start of the project.		
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1.2 Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
GHG emission reductions & removals	Net estimated emission removals in the project area, measured against the without-project scenario	1,473,639 tCO ₂ e (to be confirmed by the VVB)	3.2	9,472,988 tCO ₂ e
	Net estimated emission reductions in the project area, measured against the without-project scenario	Not applicable	N/A	Not applicable
Forest¹ cover	For REDD ² projects: Number of hectares of reduced forest loss in the project area measured against the without-project scenario	Not applicable	N/A	Not applicable
	For ARR ³ projects: Number of hectares of forest cover increased in the project area measured against the without-project scenario	Not applicable	N/A	Not applicable
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices have occurred as a result of the project's activities, measured against the without-project scenario	Not applicable	N/A	Not applicable
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities,	1,993,075 (to be confirmed by VVB)	2.1.7	1,933,075

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

² Reduced emissions from deforestation and forest degradation (REDD) – Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

³ Afforestation, reforestation and revegetation (ARR) – Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

⁴ Improved forest management (IFM) – Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (*VCS Program Definitions*)

	measured against the without-project scenario			
Training	Total number of community members who have improved skills and/or knowledge resulting from training provided as part of project activities	No data available from latest project-wide survey (2020).	N/A	N/A
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities	No data available from latest project-wide survey (2020).	N/A	N/A
Employment	Total number of people employed in of project activities, ⁵ expressed as number of full-time employees ⁶	No data available from latest project-wide survey (2020).	N/A	N/A
	Number of women employed in project activities, expressed as number of full-time employees	No data available from latest project-wide survey (2020).	N/A	N/A
Livelihoods	Total number of people with improved livelihoods ⁷ or income generated as a result of project activities	Over 86% of people surveyed (latest in 2020) said they experienced improved livelihoods (assets) from the project activity. Thus 12,706 people benefitted.	4.1.3	For livelihood: Increase in 1,527 people since baseline surveys.

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers, and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region, or economic territory (adapted from UN System of National Accounts (1993) paragraphs 17.14[15.102]; [17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

	Number of women with improved livelihoods or income generated as a result of project activities	Over 86% of women surveyed (latest in 2020) said they experienced improved livelihoods (assets) from the project activity. Thus 3,414 people benefitted.	4.1.3	For livelihood: Increase in 1,086 people since baseline surveys.
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	No data available from latest project-wide survey (2020).	N/A	N/A
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	No data available from latest project-wide survey (2020).	N/A	N/A
Education	Total number of people for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	No data available from latest project-wide survey (2020).	N/A	N/A
	Number of women and girls for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	No data available from latest project-wide survey (2020).	N/A	N/A
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	In latest project-wide 2020 survey, mean time to gather water has declined by 20% under the project scenario.	4.1.3	Same as the current monitoring period.
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project	Over 68% of women surveyed (latest project-wide in	N/A	Increase in 2,149 people since baseline surveys.

	activities, measured against the without-project scenario	2020) said they experienced improved access to drinking water from the project activity. Thus 2,894 people benefitted.		
Well-being	Total number of community members whose well-being ⁸ was improved as a result of project activities	Over 82% of people surveyed (latest project-wide in 2020) said they benefited from the project activity through increased cattle ownership. Thus 12,677 people benefitted.	4.1.3	Increase in 1,513 people since baseline surveys.
	Number of women whose well-being was improved as a result of project activities	In latest project-wide 2020 survey, 90% claimed the grazing plan was “Helpful” or “Very Helpful” (Figure 6).	4.1.3	Same as the current monitoring period.
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation, ⁹ measured against the without-project scenario	In latest project-wide 2020 survey, 80% of the project area experienced successfully implemented grazing plans, therefore 1,594,460 ha were better managed.	3.2.2	Same as the current monitoring period.
	Number of globally Critically Endangered or Endangered	6 under the project scenario.	5.1.3	Same as the current

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Health, Education, Water, etc.), but could also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural significance, etc.

⁹ Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.

2 PROJECT DETAILS

2.1 Summary Description of the Implementation Status of the Project

2.1.1 Summary Description of the Project (VCS, 2.1, 3.6; CCB, G1.2)

The project was implemented in stages from portions of 3 Conservancies that adopted planned rotational grazing in 2012 to all 13 Conservancies in the project area (1.99 million ha project zone) participating in a region-wide drought response grazing plan by February 2014. With expected annual removals of approximately 0.88 metric ton CO₂e/ha on average, the project should remove more than 1.7 million tonnes CO₂e annually and 50 million tonnes CO₂e over its lifetime.

Implementation of the project proceeds by:

1. Determination of grazing plans through initial interactive meetings between NRT personnel, Conservancy grazing coordinators, and Conservancy managers, followed by meetings between community members and Conservancy grazing coordinators and other leaders. Community meetings make full attempts to encourage attendance by women and ethnic minorities within Conservancies.
2. Individual herders, under the guidance and supervision of Conservancy grazing coordinators, move their herds among different grazing blocks within Conservancies, and during dry seasons among different areas of other participating Conservancies.
3. The number of livestock from particular conservancies present in different grazing blocks and in different areas of other participating conservancies is monitored by project-employed grazing coordinators. These coordinators supply monthly reports of locations and numbers to NRT and other professionals involved in the current project.
4. Conservancy grazing coordinators also monitor when herders from a Conservancy move their livestock off project area and record the number of days a number of livestock over which this occurs. These data are used to determine leakage.

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

5. Impacts of livestock are monitored using a project–developed protocol that employs satellite imagery to assess how frequently and with what intensity livestock grazing occurs throughout the project area. The project has an algorithm that converts the satellite imagery data into an assessment of whether project implementation goals were met across the entire project area. Essentially satellite data allows project administration staff to assess whether livestock graze an area once or not at all during a year. If livestock are determined to graze an area more than once, the satellite imagery “pixel” is classified as “not successfully implemented,” and removals are not claimed from the area represented by the pixel.
6. The area of successful implementation is determined for each Conservancy and multiplied by the expected mean carbon removals for that Conservancy to yield the total annual removals for that Conservancy. This is then repeated for each of the 13 participating Conservancies.
7. Leakage is managed by project-employed game scouts, which provide input from grazing coordinators to herders in remote sections of the project area and act as reminders that herders are expected to remain on the project area if forage is available. Herders are made aware of the negative consequences of going off the project area during annual grazing planning meetings. Finally, when herders do go off the project area, game scouts also act as first responders to help avoid conflicts and to signify when NRT personnel are needed to broker reciprocal agreements that allow project herders to legally and respectfully use pastures outside the project area.
8. Further feedback to control leakage is provided by annual distribution of revenues which is in part influenced for individual conservancies by whether their herders followed project grazing plans and/or left the project area. This mechanism issues economic penalties to whole communities as a consequence of individual herders’ decisions and creates social pressure to follow Conservancy grazing plans.
9. Non-permanence threats from encroachment by herders from non-participating communities are also reduced by project border activities of game scouts and the conducting of peace meetings by NRT personnel and Kenya County government agents.
10. The total net greenhouse gas removals claimed by this project in the current monitoring period are 1,473,639 tonnes CO₂e (to be verified by the VVB).

2.1.2 Audit History (VCS, 4.1)

(Section non-applicable to the CCB-only Public Comment Period)

Audit Type	Period	Program	VVB Name	Number of years
Validation/ Verification	(DD-Month-YYYY -- DD-Month-YYYY)		VVB Name	One year
				

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

(Section non-applicable to the CCB-only Public Comment Period)

Sectoral Scope	
AFOLU Project Category	
Project Activity Type	

2.1.4 Project Proponent (VCS, 3.7; CCB, G1.1)

Organization name	Northern Rangelands Trust (NRT)
Contact person	Mohamed Shibia
Title	Rangelands & Carbon Program Director
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Telephone	+254 (0) 64 31405
Email	mohamed.shibia@nrt-kenya.org

2.1.5 Other Entities Involved in the Project

Organization name	Native, a Public Benefit Corporation
Role in the project	Marketing, sales transactions, MRV assistance
Contact person	Kyle O'Keefe

Title	Asset Manager
Address	3 Main Street, STE 212, Burlington, VT 05401 USA
Telephone	+1-802-861-7707
Email	kyle.okeefe@native.eco

Organization name	Soils for the Future, LLC
Role in the project	Technical partner and advisor
Contact person	Mark E. Ritchie, Ph.D.
Title	President
Address	6125 Ketch Drive, Cicero NY 13039 USA
Telephone	+1-315-447-1612
Email	markritchie@soilsfuture.com

2.1.6 Project Start Date (VCS, 3.8)

Project start date	10 December 2012
Justification	This date reflects the timing of the initiation of yearly rotational grazing plans by the first three Conservancies (Il Ngwesi, Kalama, and Westgate), and the signing of a joint MOU between NRT and the Nature Conservancy to administer and finance the implementation of the project.

2.1.7 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Crediting Period	The project crediting period is 30 years from the implementation of yearly rotational grazing plans: from 1 January 2013 to 31 December 2042. The crediting period start date was moved from 10 December 2012 to 01 January 2013 as project monitoring period activities were not fully implemented until 01 January 2013.
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Start Date of First or Fixed Crediting Period	01-January-2013 to 31-December-2042
Total Number of Years of Crediting Period	30 years
CCB Benefits Assessment Period	30 years

2.1.8 Project Location (VCS, 3.11; CCB, G1.3)

☒ KML file has been provided

2.1.9 Title and Reference of Methodology (VCS, 3.1)

(Section non-applicable to the CCB-only Public Comment Period)

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Example: Methodology	Example: VM0007	Example: VM0007 REDD+ Methodology Framework (REDD+MF),	Example: 6.0
...	...	...	...

2.1.10 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB, G5.9)

2.1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes ☒ No

2.1.10.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

2.1.10.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

2.1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the VCS Program Definitions for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

2.1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."?

☐ Yes

☒ No

If yes to all:

2.1.12 Sustainable Development Contributions (VCS, 3.17)

(Section non-applicable to the CCB-only Public Comment Period)

Provide a brief description that includes the following (no more than 100 words):

- A summary description of project activities implemented during the monitoring period that result in SD contributions (i.e., technologies/measures implemented, activity location).
- An explanation of how project activities result in the SD contributions described in Table 1 of this report.
- Identification of which SD contributions described in Table 1 of this report contribute to achieving any nationally stated sustainable development priorities, including any provisions for monitoring and reporting these.
- Evidence of the project's SD contributions shall be provided as appendices to this report.
- Activities implemented during previous monitoring periods shall not be described in this report. Where no activities were implemented during the monitoring period, state as such.

Using Table 1 below, provide the project's quantifiable contributions to specific targets and indicators of the Sustainable Development Goals (SDGs) for the monitoring period. Use the official list of SDG Targets and Indicators (available [here](#)) to identify the SDG Targets to which the project has contributed. Evidence for each contribution shall be identified in accordance with Section 2.1.12.

Contributions should be aligned with the SDGs, as follows:

- Where possible, relate all contributions to official SDG targets and indicators. Refer to the SDG metadata repository (available [here](#)) for guidance on the definitions and concepts included in the SDG indicators (see the examples in rows 1 and 2 in the table below).
- While climate change and mitigation activities relate to SDG 13, they do not align with any SDG 13 target. For climate change mitigation impacts, write "13.0" in the SDG target column and use the indicator "Tonnes of greenhouse gas emissions avoided or removed" (see the example in row 3 in the table below).

Where a project's self-defined measure for tracking a benefit does not align with an official SDG indicator, do not provide an indicator number. Instead, write a project-specific indicator that relates to the most appropriate SDG target (see the example in row 4 in the table below).

Document total project contributions since the project start date, previous SD contribution monitoring period, or VCS monitoring period in the “Current Project Contributions” column and the cumulative contributions over the project lifetime in the “Contributions Over the Project Lifetime” column in Table 1 below. The cumulative impact should be calculated by summing the current project contributions with all impacts included in previously approved VCS monitoring reports or Sustainable Development Contribution Reports.

Remove rows 1-4 of Table 1 below, which serve as instructions and examples. Add or remove other rows from the table as necessary.

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
Sequential row number	SDG Target number	Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator	Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)	Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.	Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.
1)	1.1	1.1.1 Proportion of population below the international poverty line	Implemented activities to decrease	No further changes this monitoring period	The project has increased the 65 participants' total daily income from 1.20 USD/day to 2.57 USD/day, bringing them above the international poverty line
2)	3.2	3.3.3 Malaria incidence per 1,000 population	Implemented activities to decrease	Lowered the malaria incidence per 1,000 to 98 by distributing 200 additional bed nets and conducted malaria prevention workshops.	Lowered the malaria incidence per 1,000 from 157 to 98

3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By conserving 400 ha of tropical rainforest, Project X has prevented the release of 250 thousand tonnes of carbon into the atmosphere during the monitoring period	Prevented the release of 750 thousand tonnes of carbon into the atmosphere
4)	6.1	Proportion of the rural population who have easy access to a safe water supply	Implemented activities to increase	Completed construction of 4 additional improved wells to provide potable water to 230 people	Provided at least 10 liters of potable water per day to 1,200 people, a 40% increase in the catchment area, over the project lifetime by constructing improved wells

2.2 Project Implementation Status

2.2.1 Implementation Schedule (VCS, 3.2; CCB, G1.9)

The implementation of the project features three main components: Planning and monitoring grazing activities and impacts, monitoring biodiversity, and monitoring community benefits and/or costs. Together with periodic re-sampling of soil carbon to verify the SNAP soil carbon model on which removal calculations are based, the following schedule has been implemented:

Table 1. Key dates and milestones in the project's development and implementation.

Date	Milestone(s) in the project's development and implementation
1 Feb 2011	Project baseline soil data collected
10 Dec 2012	Project start date, MOU signed between NRT and TNC
1 Jan 2013	Implementation of planned grazing for first three conservancies (Westgate, Il Ngwesi, Kalama) and crediting period start date
1 Nov 2013	Implementation of planned grazing for six additional conservancies (Sera, Meibae, Namunyak, Oldonyiro, Biliqo Bulesa)
1 March 2014	Formation of Nasuulu, Leparua, Nakuprat-Gotu conservancies
1 June 2014	Implementation of planned grazing for Melako conservancy
1 July 2014	Baseline biodiversity monitoring in Kalama, Westgate, and Sera conservancies
1 Feb 2015	Baseline community monitoring in Il Ngwesi, Kalama, Nakuprat-Gotu, and Melako conservancies
1 March 2015	Validation and verification begin
15 May 2015	Initial conservancy solicitation of participation and public comment period, with announcement and information about the project distributed by NRT grazing coordinators and other staff in the form of pamphlets, and an announcement on the NRT website (http://nrt-kenya.org). All responses to the proposed carbon project during this round were made by cell phone text to a standing repository, which was then downloaded for review and archiving.
1 June 2015	Implementation of planned grazing in Nasuulu, Leparua, Nakpurat-Gotu, and Naibunga conservancies
15 Sep 2015	VVB conducts first site review
1 Oct 2015	Project-wide grazing planning process and grazing implementation report
1 May 2016	SocialCoMMs monitoring in Westgate, Sera, Meibae, Namunyak, Oldonyiro, Biliqo Bulesa, Melako conservancies

15 Aug 2016	Second round of conservancy solicitation of participation and public comment period, with announcement and information about the project distributed by NRT grazing coordinators and other staff. All responses to the proposed carbon project during this round were made by cell phone text to a standing repository, which was then downloaded for review and archiving.
Sept-Oct 2016	2015 and 2016 announcements rounds followed by community meetings, open to all members in each Conservancy, in each of the 13 participating Conservancies. Meeting times and dates were posted at all Conservancy headquarters and were communicated verbally in villages in each Conservancy by NRT SocialCoMMs supervisors.
1 Oct 2016	Expanded baseline soil carbon sampling in Melako, Biliqo Bulesa, Sera, Namunyak, and Meibae conservancies
1 Oct 2016	Project-wide grazing planning process and grazing implementation report
7 Nov 2016	VVB conducts second site review
1 Oct 2017	Project-wide grazing planning process and grazing implementation report
15 April 2018	VVB conducts third site review
1 Oct 2019	Project-wide grazing planning process and grazing implementation report
29 April 2020	Validation finalized by VVB and Verra
1 May 2020	Biodiversity Monitoring in Kalama, Westgate, Sera, Melako, Meibae
1 May 2020	SocialCOMMs Monitoring - 6 Conservancies, randomly selected
1 Oct 2020	Project-wide Grazing Planning Process and Grazing Implementation Report - All 13 Conservancies.
2017-2020	At varying times, Annual Governance Meetings (AGM) were held with representatives from each project conservancy. Carbon project updates were discussed as part of each AGM's agenda.
2021	Resampling of all project soil carbon sampling sites carried out by Soils for the Future.
2021	Biodiversity Monitoring in Kalama, Westgate, Sera, Melako, Meibae.
2021	At varying times, Annual Governance Meetings (AGM) were held with representatives from each project conservancy. Carbon project updates were discussed as part of each AGM's agenda.
2021	All conservancy and regional grazing planning and committee meetings respectively occurred monthly and quarterly for each conservancy throughout the year.
2021	Ongoing stakeholder engagement and consultation took place at various intervals with all conservancies throughout the year.

28-30 March 2022	Soil carbon sampling at a subset of sampling sites to test 2021 sampling results for 2021 Verification, 2017-2020 Verification VCS and CCB site visit by VVB
December 2022	2017-2020 VCS Verification Complete, finalized by VVB and Verra
May 2024	Anticipated 2017-2020 CCB Verification Completion, finalized by VVB and Verra
May 2024	Anticipated initiation of 2021 Verification by VVB
June-July 2024	Anticipated 2021 Verification CCB site visit by VVB

2.2.2 Baseline Reassessment (VCS, 3.2.6, 3.2.7)

(Section non-applicable to the CCB-only Public Comment Period)

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☐ No

If yes, provide a summary of the baseline reassessment applied during the monitoring period. Include a summary on the following:

- *Details of the use and applicability of the latest approved version of the methodology or its replacement, where relevant.*
- *Sections in the project description that have been updated to reflect changes in the new baseline.*
- *Indicate whether the baseline scenario is still valid. If the previous baseline scenario is no longer valid, summarize the new baseline scenario as described in the updated project description.*
- *Describe the impact of new relevant national and/or sectoral policies and circumstances on the validity of the baseline scenario, where relevant.*
- *Include the percentage change between the revised baseline emissions provided in the updated project description and the previous baseline emissions.*

2.2.3 Methodology Deviations (VCS, 3.20)

(Section non-applicable to the CCB-only Public Comment Period)

Describe and justify any methodology deviations applied, including any previous deviations. Include evidence to demonstrate the following:

- *The deviation does not negatively impact the conservativeness of the quantification of GHG emission reductions or carbon dioxide removals.*
- *The deviations relate only to the criteria and procedures for monitoring or measurement, and do not relate to any other part of the methodology.*

2.2.4 Minor Changes to Project Description (CCB Program Rules, 3.5.6)

There is one minor change to the Project Description. In the Project Description, it is stated that for biodiversity monitoring, “Surveys of key indicator species, sampling in savanna/grassland habitats will take place every 3 years in five permanent biodiversity monitoring platforms, or ‘camps.’” Baseline surveys were conducted in 2014 and the first project surveys were conducted in 2021. A survey did not take place in 2017 as initially stated in the Project Description. The justification for this change in surveying cadence is that the project did not generate VERs (from 2013-2016) to be sold until 2022, and therefore the project did not have revenue to pay for the biodiversity sampling.

2.2.5 Project Description Deviations (VCS, 3.21; CCB Program Rules, 3.5.7 – 3.5.10)

There are six ongoing project description deviations during this monitoring period:

1. Section 2.1.15 of the PD lists the project crediting period as beginning on 10 December 2012. For the first verification, the monitoring period was moved from to 01 January 2013. Project monitoring period activities were not fully implemented until 01 January 2013.
2. While Section 2.1.9 of the PD identifies project stakeholders based on occupation, socioeconomic status, and role in the community, Section 4.2.1 of the PD proposes to monitor impacts on the basis of ethnic group. During the first monitoring period, we identified stakeholders based on occupation, socioeconomic status, and role in the community and monitored impacts only on those stakeholder groups (Section 4.1.1 of this report).
3. The parameter $ED_{c,t}$ (number of days spent by encroaching [off-project] livestock of category c on the project area during year t) was not mentioned in the PD but has been added to Section 3.1.2 of this Monitoring Report. This parameter is used along with an approach modified from the “Tool for testing the significance of GHG emissions in A/R CDM project activities” to show that project emissions from encroaching livestock for the current monitoring period are de-minimis (<5% of project methane emissions) in each year of the monitoring period and can therefore be excluded from the final calculation of net emissions and removals.
4. The project area has decreased by 3 ha due to technological improvements in analyzing conservancy boundaries. The change from 1,993,078 ha to 1,993,075 ha is reflected in all calculation files and was reviewed by the VVB during the 2017-2020 verification event.

For deviations 5.) & 6.), please reference Appendix 2 for a summary and full methodological details for the assessment of successful pastoralist grazing plans.

During the previous 2017–2020 monitoring period, there were two deviations (5 & 6) to the remote sensing method as compared to the 2013–2016 Monitoring Report. First (deviation 5), an adjusted set of ungrazed reference polygons is used for calculating annual relative change in biomass. This is justified due to conservancies with lower mean annual rainfall (relative to wetter reference polygons) exhibiting high relative changes in NDVI and thus being judged in error as unsuccessful. Secondly (deviation 6), the threshold value of annual relative change in NDVI is adjusted from 0.05 to 0.1. This is justified because a threshold of 0.1 more accurately reflects the occurrence of single grazing events in both wet and dry season grazing areas.

5. Justification for Adjusted sets of reference polygons for calculating successful implementation of grazing plans during 2017-2020 and future monitoring periods.

The method of assessing annual grazing plan implementation depends on comparing annual relative changes in NDVI, $SUM\Delta NDVI_{rel}$, determined by summing monthly $\Delta NDVI_m$ across the twenty-four 16-day intervals for each year, $SUM\Delta NDVI_m$ (see Eq. 2 in Appendix D) at each MODIS pixel, as compared with mean annual sums of $\Delta NDVI_m$ across the same time intervals ($meanSUM\Delta NDVI_{REF}$) from reference polygons that are not grazed or only slightly grazed by livestock (see Eq. 2 in Appendix D). These reference locations are embodied in multiple polygons that span the project area, including “Core” areas of Il Ngwesi and Kalama Conservancies, the tops of plateaus in Kalama Conservancy that are relatively inaccessible to livestock, and a large ~1,000 ha fenced rhino sanctuary in Sera Conservancy. In most years, the mean annual relative change in NDVI ($meanSUM\Delta NDVI_{REF}$) in these reference areas is relatively similar despite differences in mean annual rainfall. Consequently, the Google Earth Engine (GEE) calculations of annual relative change in NDVI ($SUM\Delta NDVI_{rel}$) for individual pixels represent a good estimate of livestock impacts in drought years (2017, 2018). However, in wet years following droughts, such as occurred in 2019 relative to 2018, the $meanSUM\Delta NDVI_{REF}$ in Il Ngwesi Core area and Kalama Plateaus are much bigger than the values for Kalama Core and Sera Rhino Sanctuary reference polygons. Averaging in these higher $meanSUM\Delta NDVI_{REF}$ from those reference polygons virtually guarantees that conservancies with lower mean annual rainfall will exhibit frequent pixels with high $SUM\Delta NDVI_{rel}$ and thus be judged in error as unsuccessful. Therefore, reference polygons appropriate to the mean annual rainfall of the different Conservancies should be used for the relevant year (2021). Reference polygon sets are broken into higher rainfall (wetter) and lower rainfall (drier) conservancies:

Reference set 1 is to be used with the higher rainfall conservancies – Il Ngwesi, Naibunga, Meibae, Nanapishu (Oldonyiro), Leparua, Nasuulu, Namunyak

Reference set 2 is to be used with the lower rainfall conservancies – Kalama, Westgate, Sera, Biliqo Bulesa, Melako, Nakuprat-Gotu

Thus, we calculated relative change in NDVI, $\text{SUM}\Delta\text{NDVI}_{\text{rel}}$ (Eq. 1- see Appendix 2) for the drier Conservancies Melako, Sera, Namunyak, Biliqo Bulesa, Nakuprat-Gotu, Nasuulu, Kalama, Westgate, and Leparua Conservancies using only the reference polygons from Kalama Core and the Sera Rhino Conservancy. Calculations for wetter Conservancies Il Ngwesi, Naibunga, Meibae, and Oldonyiro include all the reference polygons, as is coded in the current GEE model.

6. Justification for a Shift in the Threshold Value of Annual Relative ΔNDVI

The threshold for unsuccessful implementation was obtained from two logistic regressions, which indicate the estimated ΔNDVI values that correspond to whether substantial livestock were present in a grazing block (Fig. 3- see Appendix 2), and whether substantial livestock were present for more than one two-week period during the year (unsuccessful implementation).

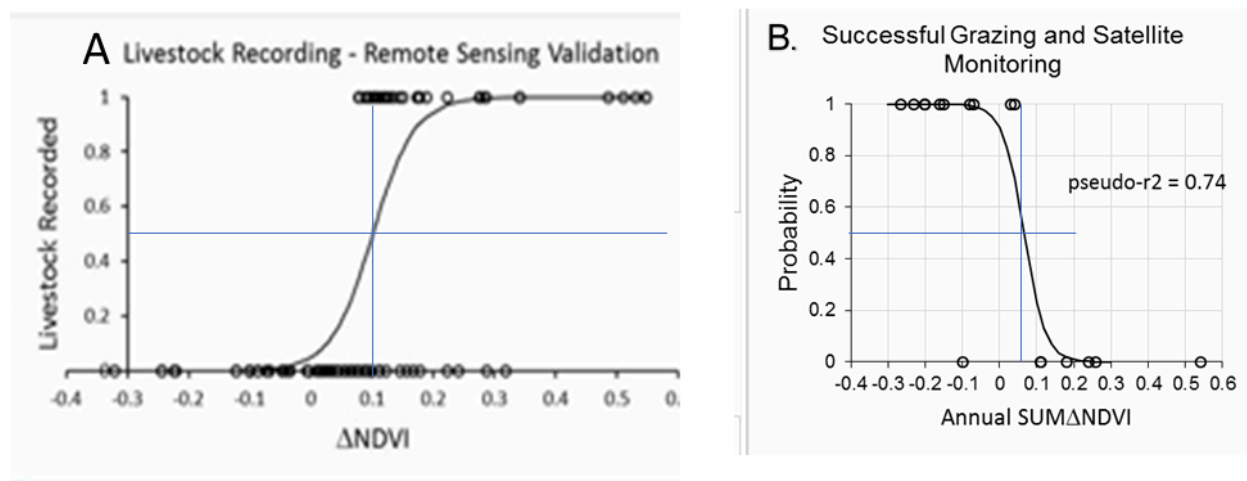


Fig. 1. Logistic regressions to assess successful implementation of grazing plans in the Northern Kenya Grasslands Carbon Project. Fig. 1.A. The logistic regression of livestock presence (livestock recorded = 1) and absence (livestock recorded = 0) versus calculated **monthly relative ΔNDVI** ($\Delta\text{NDVI}_{\text{m,rel}}$, Eq. 1- see Appendix D), the value of monthly relative ΔNDVI at which there was greater than a 50% probability (livestock recorded > 0.5) of recording substantial livestock present is 0.1 (intersection of two perpendicular lines). This logistic regression was previously shown in Fig. 3C (see Appendix D). Also, livestock were only rarely encountered in map locations where $\Delta\text{NDVI}_{\text{m,rel}} < 0.1$. Fig. 1.B. At 17 grazing blocks where presence/absence of livestock was monitored over an entire year, a logistic regression of the occurrence of only a single grazing event (probability = 1) or more than one grazing event (probability = 0) versus the relative annual sum of ΔNDVI for the 2014 calendar year (**annual relative ΔNDVI** , $\text{SUM}\Delta\text{NDVI}_{\text{rel}}$ from Eq. 2 - see Appendix D). This logistic regression was previously shown in Fig. 4B- see Appendix D.

The threshold of 50% probability of unsuccessful implementation occurs at approximately annual relative $\Delta\text{NDVI} = 0.08$, but, due to the small sample size of grazing blocks, there is a large confidence interval around that estimate. Since monthly relative ΔNDVI ($\Delta\text{NDVI}_{\text{m,rel}}$) must be > 0.1 for livestock to be detected at a location then it is reasonable that the annual sum, or annual relative ΔNDVI ($\text{SUM}\Delta\text{NDVI}_{\text{rel}}$) should be > 0.1. This is especially true for dry season grazing areas because they do not regrow immediately after grazing. Since it requires a relative change in NDVI of 0.1 to detect a grazing event (Fig.

1.A.), it is difficult for dry season grazing areas to show a relative delta NDVI < 0.1 even if they are grazed only once. Thus using a threshold of 0.1 more accurately reflects the occurrence of single grazing events in both wet and dry season grazing areas.

This justifies a threshold of annual relative Δ NDVI ($SUM\Delta NDVI_{rel} = 0.1$) for assessing whether grazing plans were successfully implemented at a location. Thus, where pixels have $SUM\Delta NDVI_{rel} \leq 0.1$, these areas are deemed and verified to be successfully grazed in each year 2017-2020 and will be assessed for the monitoring period of 2021. Pixels with $SUM\Delta NDVI_{rel} > 0.1$, are deemed to be unsuccessfully grazed in each year 2017-2020 and for future monitoring periods, including this monitoring period of 2021.

2.2.6 Grouped Projects (VCS, 3.6; CCB, G1.13-G1.15, G4.1)

This section is not applicable, as the project is not a grouped project.

2.2.7 Risks to the Project (CCB, G1.10)

Long-term benefits beyond the project timeline are the ultimate goal of the project. Restoration of perennial grasses and re-building of soil carbon should allow community members the opportunity for sustainable livelihoods from pastoralism and to dramatically improve livelihoods and well-being from livestock husbandry. These benefits would not depend on carbon revenues, as the capacity to sustain livestock should at least double from its current value, and wildlife and tourism opportunities should be higher than current. As highlighted in NRT's Strategic Plan 2018-2022, Conservancy governance structures put in place with carbon revenues to support and manage grazing within Conservancies will be funded beyond the project lifetime by interest from conservation trust funds built from carbon revenues. Newly established regional grazing plans and drought mitigation strategies will be coordinated by county governments and be funded by devolved national tax revenues rather than by carbon. Risks of encroachment from herders from outside the project area may continue. However, regional governance, which includes local elders and women trained in peace-making, sharing of forage agreements among Conservancies and mobile security teams drawn from across ethnic groups and managed at the County level will be well established and capable of resolving conflicts quickly (see Mid-Year Technical Report on NRT Operational Plan 2018, as supporting document).

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Short-term: Failure to implement	Failure is most likely if drought is severe, which would cause large-scale movement of livestock	Rapidly build grass banks, particularly in extensive unsettled areas typically

planned rotational grazing	outside the project area and other grazing management decisions that might lead to over-use of rangelands and an incremental reversal of accumulated soil carbon stocks.	used only during the dry season, that can be used as forage during droughts. • The accumulation of soil carbon will lead to greater infiltration of rainfall, which will make communities progressively less vulnerable to climate variability and drought as the project proceeds. • Further build the livestock-to-market program instituted by NRT to provide cash compensation for de-stocking in advance of intense droughts, where droughts can be anticipated by using early warning systems such as the USDA FEWS network http://www.fews.net/. • Through education associated with monitoring social benefits of the project, continue to convey the community and personal revenue advantages linked to continued participation in the project. • Convey the risks of encroachment from herders from off-project communities, and train and implement strategies to handle encroachment and thus reduce risk. For example, game scouts are trained to deal with and communicate about poachers and
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		encroaching livestock herders.¹² • Progressively entrain sustainable livestock management processes as a new “tradition” of planned rotational grazing. • Continue to demonstrate the indirect positive influence of rangeland vegetation and soil recovery on biodiversity and the concomitant benefits of biodiversity to people.
Long-term: Persistence of community willingness to participate in the project	Potential inability for project participants to adapt to climate change and climate variability, etc.	Community willingness to participate will depend on several factors. Lack of land tenure can discourage participation, but the Kenya Community Lands Act of 2016 creates drivers to encourage people to remain and care for land and to invest in management and governance strategies that yield sustainable productivity from it. Rotational grazing management can be challenging to govern, but with carbon revenues to support governance, it should be able to continue long enough for community members to see the returns of implementation in the form of improved forage, greater surface water distribution, development of livestock value chains, etc. To that end, the objective of the project to meet CCB climate,

¹²

https://static1.squarespace.com/static/5af1629f12b13f5ce97ca0b5/t/5b6330d9575d1f66316bf5f0/1533227321252/Wildlife-CoMMS+guide_complete_low_res.pdf

		biodiversity and community standards, and the need for auditors to gather social data, was communicated to community members prior to both the 2015 and 2017 site visits during public comment periods, as well as a February-March 2019 public comment period. In addition, the specific objectives of the standards and the need for the social community managed monitoring program (Social CoMMs) was communicated to Conservancy chairpersons and grazing coordinators from all 13 Conservancies during a September 2017 workshop.
Climate Change	External risk that has the potential to cause irreversible degradation of the soil and grass resources if baseline management strategies of unrestricted grazing are pursued.	The longer the project is operational, the participating communities will continue to create buffers against drought and will be equipped with additional management options to deal with increased frequencies of droughts and more frequent incursions of people seeking livestock forage.

2.2.8 Benefit Permanence (CCB, G1.11)

The project is part of a larger effort by NRT, TNC, and numerous donors, such as USAID and DANIDA (US and Danish international aid agencies) to restore rangelands and improve human well-being in the northern Kenya landscape. The project activity, improved grazing management, is at the core of educational efforts in the 13 Conservancies aimed at making long-term livelihood and land management changes that improve well-being. These changes have occurred at the Conservancy administrative and public decision-making level and are designed to help communities conduct sustainable grazing practices over a 30-50 year time horizon beyond the project lifetime. Incentives to continue these grazing practices include healthier and more valuable livestock, improved forage production, surplus production that can be used as grass banks against drought, improved small stock and milk production during dry seasons, and lower physical risks when migrating to dry season ranges with livestock. These incentives act as barriers against the return to continuous grazing near settlements that generated past degradation.

Describe the measures needed and implemented since the last CCB validation or verification to maintain and enhance the climate, community, and biodiversity benefits beyond the project lifetime.

2.3 Stakeholder Engagement & Safeguards

2.3.1 Stakeholder Identification (VCS, 3.18, 3.19; CCB, G1.5)

The stakeholder makeup has not changed since validation and all stakeholders have been identified during each verification review.

2.3.2 Stakeholder Access to Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

Various meeting types, including AGMs, were held to review the project's performance and status with stakeholders in 2021. During these meetings, the stakeholders were notified where further documentation could be found and provided the documentation during the meeting. During the monitoring period, the documentation could be found at NRT headquarters, each participating Conservancy headquarters, and posted at public kiosks in the major towns of Isiolo, Archer's Post, Sereolipi Wamba, and Merille. This process was also carried out for the dissemination of the translated monitoring report summary for the 2017-2020 monitoring period. This will be the case for the 2021 monitoring period once the monitoring report is approved and finalized.

2.3.3 Dissemination of Summary Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

Given the expanse of the project, an initial summary of project documents was provided at multiple locations throughout the project area, including NRT and participating conservancy headquarters during three different public comment periods in September 2015, April – May 2017, and April – May 2018. Stakeholders were informed of where the documents could be found as hard copies and online through stakeholder meetings and postings on the NRT website, conservancy headquarters, and at kiosks in the major towns of Isiolo, Archer's Post, Sereolipi Wamba, and Merille. As was the case for 2017-2020, project documentation for 2021 will be posted at conservancy HQ. This will include a 2021 monitoring report summary translated into Kiswahili. Additionally, links will be provided to 2021 project documentation posted on the NRT website in both English and Kiswahili.

2.3.4 Informational Meetings with Stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

In a given year, numerous opportunities are provided for stakeholders to engage in informational meetings and events. Each conservancy hosts an Annual Governance Meetings (AGMs) with the purpose of conservancy leaders providing conservancy and community updates, receiving questions and feedback, and aligning on goals for the upcoming year. Special AGMs, held at village or zonal level across the conservancies, were utilized in 2021 to brief the wider community and ensure that there was consensus within the community to continue participating in the project. During 2021, each of the following meetings was held, at least once, some virtually due to the COVID-19 pandemic, within or for each project conservancy: Annual General Meetings (AGMs), Grazing Awareness Meetings (educational visits to other project conservancies), Conservancy Board Meetings, Council of Elders Meetings, Grazing Committee Trainings, Grazing Committee Meetings (monthly), Regional Grazing Meetings (multiple times per year), Grazing Planning (Wet Season) Meetings (quarterly or bi-annually).

2.3.5 Risks from the Project and No Net Harm (VCS, 3.18, 3.19)

(Section non-applicable to the CCB-only Public Comment Period)

Describe actions needed and implemented to mitigate likely natural and human-induced risks resulting from project activities during this monitoring period, including risks to stakeholder participation, working conditions, safety of women and girls, and safety of minority and marginalized groups including children.

The Project Risks Table (see Appendix 2) may be used if appropriate. Delete the table in the Appendix if not used.

2.3.6 Community Costs, Risks, and Benefits (CCB, G3.2)

The project involves community members in planning and management operations in multiple ways, and includes efforts to involve **vulnerable and marginalized groups**, notably women and the poorest 25% of households, most of which lack livestock. Several different indigenous groups occupy the project zone (Figure 1), and the risks and values to “the community” are fully applicable to each group. The success of the project depends on the joint cooperation of the participating communities and the financial and livestock management benefits of the project apply equally to all the Conservancies.

Communities within the conservancies that are a part of the Northern Kenya Rangelands Carbon Project have been engaged since the inception of the project. Methods of engagement have ranged from education and support for changing grazing behavior to preserve grasslands, to participation in informative carbon discussions, and the signing of the agreements which have provided the permission for the sale of the credits.

At the outset, the engagement was primarily based around the improvement of soils and grasslands through sharing rotational grazing techniques. As new grazing techniques were successfully implemented across the conservancies, NRT, TNC, Soils for the Future and Grevy’s Zebra Trust introduced to communities the potential for enhanced carbon sequestration through improved grazing management. The engagement with the communities on the benefits of these new grazing methods shifted to a conversation about the potential for the conservancies to generate revenue from soil carbon. In 2015, NRT held a series of public meetings and announcements in the 14 conservancies (in local languages) describing the carbon project, its potential to create revenue, how it linked to the grazing programs, and what the costs and benefits of embarking on this project would be. After these meetings each of the participating conservancies, through their democratically elected Boards, signed Letters of Intent to participate in the carbon project. In September of that year, the first VVB site visit commenced where just under 200 community members were interviewed either in focus groups or in individual households, and public meetings were held in six different towns across the project area to assess and promote project awareness.

In 2021, NRT hosted a workshop whereby all the details were laid out of how the project would run, its key activities, and how its success and benefits depended on good governance and adhering to the grazing plans. Issues of leakage, encroachment, and revenue sharing were discussed with the members of conservancy leadership and rangeland coordinators. This was well received and NRT, TNC and Soils for the Future, acting on behalf of the conservancies, continued to engage with VCS as they pursued further auditing work to support the validation of the proposed project methodology. A second VVB site visit at this time and a further 50-100 community members were interviewed by the visiting team and public meetings were held at a further (and different to the previous) three locations in the project area. Between 2017 and 2018 participating conservancies formally agreed for NRT to represent their carbon rights for the project.

In 2021, public notices were made at every conservancy HQ and in every major settlement area in the project area to publicly announce the project and provide a summary of the project monitoring report (2013-2016) to that point.

As leadership in the conservancy changes every three years, through the conservancy board elections, NRT felt it was critical to host another briefing to ensure that current leadership was in a position to fully understand the project and for communities to decide whether they were still interested in pursuing the sale of carbon credits through their conservancies. Thus, in May and June 2021, NRT undertook a widespread briefing and socialization of the project for all participating conservancies and their member communities. NRT presented a revised and renewed 'Project Implementation Agreement'* to set out a clear understanding between NRT and participating conservancies as the project launches the first issue of carbon credits. As part of the community consultation process, NRT held briefings on the project for each conservancy's board chair, manager, grazing committee chair, and rangelands coordinator. These members then briefed their Boards and Grazing committees, and following that, 'Special Annual Governance Meetings' were held at the village or zonal level across the conservancies, to brief the wider community and ensure that there was consensus within the community to continue participating in the project. These meetings were extensively documented both through written minutes and visually recorded through photographs. Based on a strong consensus, all communities approved a resolution for the conservancy chairperson to sign the Project Implementation Agreement on behalf of their community. The 14 chairpersons met on 24 June 2021 and formally signed the Project Implementation Agreement. Since this time, the Carbon Project Oversight Committee (CPOC) formed by representatives of each of the participating conservancies, has met at least every three months to oversee the management of the project, and full minutes of these meetings are shared with each participating conservancy.

It is important to note that NRT as the project proponent fully recognizes that free, prior, and informed consent for communities is not something that is delivered on a one-off basis and is something that must be continually invested in by the project on behalf of the participating communities throughout the project lifecycle. NRT seeks to continually make improvements to this process and seeks to encourage conservancies to support best practices for free, prior, and informed consent in all communications with community members, including giving appropriate notice for public meetings and endeavoring to seek representation/endorsement from those members of their pastoralist communities who may be far from home at the time when such public meetings are held.

**The Project Implementation Agreement outlines both the project activities involved and provides details outlining the compliance and reporting requirements of the project. It provides a schedule of the reporting requirements a conservancy agrees to partake in and it includes details on access for project monitoring, the validation process, verification, registration, records, and other reporting relating to CCB Gold Standard, and sets out the benefit sharing mechanism for revenue.*

2.3.7 Information to Stakeholder on Verification Process (VCS, 3.18.6, 3.19; CCB, G3.3)

Prior to the 2021 verification auditor site visit, documentation will be sent and translated to inform stakeholders of the upcoming site visit and verification process in 2024. Stakeholders have been informed of each auditor site visit and subsequent verification review since the project's initial validation/verification review kicked off in 2018.

Within-project stakeholders were identified by monthly community meetings held by the Conservancy managers, bi-annual meetings of grazing chairmen and the NRT grazing coordinators with herders, and workshops held in villages and at Conservancy headquarters.

Off-project stakeholders were identified from annual County public meetings in the towns of Baragoi (Samburu County), Isiolo (Isiolo County), Laisamis (Marsabit County), and Nanyuki (Laikipia County), from individual interviews with government representatives, and from oral reports of Conservancy chairman who negotiate grazing rights for herders from off-project communities

Meetings were advertised by email, calls and texts on networks of personal cell phones, and posting on the NRT website. Some meetings were announced with public postings of flyers and in public meetings within communities prior to project stakeholder meetings.

2.3.8 Site Visit Information and Opportunities to Communicate with Auditor (VCS, 3.18.6, 3.19; CCB, G3.3)

Stakeholders have been informed of each auditor site visit and subsequent validation/verification review since the project's initial validation/verification review in 2015.

The objective of the project to meet CCB climate, biodiversity, and community standards, and the need for auditors to gather social data, was communicated to community members prior to both the 2015 and 2017 site visits during public comment periods, as well as a February-March 2019 public comment period. In addition, the specific objectives of the standards and the need for the social community-managed monitoring program (Social CoMMs) was communicated to Conservancy chairpersons and grazing coordinators from all 13 Conservancies during the September 2017 workshop. Up to this time, as mentioned in the section above, in February 2022, prior to the 2017-2020 verification auditor site visit, documentation was sent and translated to inform stakeholders of the March 2022 site visit.

Prior to the 2021 verification auditor site visit, documentation will be sent and translated to inform stakeholders of the upcoming site visit and verification process in 2024. The auditor will utilize an independent translator for all interactions with community members and other stakeholders. Neither NRT nor other project partners will be present during the auditor's interviews with community members and other non-NRT stakeholders. NRT and project partners will be available to assist the auditor in the facilitation of logistics, including ground transportation, lodging, and scheduling of NRT staff interviews. Additional information will be provided in this section following the site visit.

2.3.9 Stakeholder Consultation (VCS, 3.18; CCB, G3.4)

Stakeholder consultation within the project area is multi-layered; it happens at multiple levels to ensure points raised from the management team, or points raised from a herder and/or community member, are known and shared with all entities involved.

Grazing planning continues to be undertaken quarterly - it is based around seasonal plans (wet vs dry). The planning starts at the village / zonal level where herders sit together to develop their local plans - these are then communicated at a conservancy level through the conservancy grazing committee (representatives from each zone) to ensure the zonal plans are coordinated within the greater conservancy plan. The coordinated conservancy plan is then relayed back to the zonal level (through the grazing committee) to confirm the plans with the herders who will be the ultimate enforcers- as in, herders are the ones who either comply or go against their own plans and it is the herders who decide when penalties / by-laws should be imposed.

The conservancy grazing plan is then relayed at a regional level between neighboring conservancies to coordinate the larger landscape - and also outside of the project area where applicable. Herders and communities continue to influence the project implementation as they are the “users” of the grazing plans and all plans evolve from provided input at the village / zonal level. As mentioned above, these grazing plans are self generated and supported plans which originate from the herders and communities and continue to be critical for project implementation.

Every meeting held will continue to be minuted and filed for record keeping and future VVB review.

Ongoing consultation	Stakeholder consultation within the project area is multi-layered; it happens at multiple levels to ensure points raised from the management team, or points raised from a herder and/or community member, are known and shared with all entities involved.
Date(s) of stakeholder consultation	01-January-2021 - 31-December-2021

Communication of monitored results	Various meeting types, including AGMs, were held to review the project's performance and status with stakeholders in 2021. During these meetings, the stakeholders were notified where further documentation could be found and provided the documentation during the meeting. During the monitoring period, the documentation could be found at NRT headquarters, each participating Conservancy headquarters, and posted at public kiosks in the major towns of Isiolo, Archer's Post, Sereolipi Wamba, and Merille. This process was also carried out for the dissemination of the translated monitoring report summary for the 2017-2020 monitoring period. This will be the case for the 2021 monitoring period once the monitoring report is approved and finalized.
Consultation records	All consultation meetings held were minuted and filed for record keeping and are available for stakeholder access.
Stakeholder input	Input received from stakeholders during the consultation period focused on the benefit-sharing mechanism and was addressed in part of the May and June 2021 design of the project implementation agreement (PIA). Reference the consultation table below for PIA-specific consultation and outcomes.
Ongoing consultation	In relation to the provision of a long-term sustainable revenue stream, an extensive benefit-sharing and distribution mechanism was designed with members of the participating conservancies, which began in May 2021. This was specific to the sharing of anticipated revenue from the project, which after many years of development had now been both validated and the first issue of credits verified and available for sale. In May and June 2021, NRT undertook a widespread briefing and socialization of the project for all participating conservancies and their member communities. NRT presented a revised and renewed

	Project Implementation Agreement¹³ to set out a clear understanding between NRT and participating conservancies as the project launched the first issue of carbon credits. Crucially, the PIA also set out proposed draft benefit-sharing mechanisms for the first three years of the project (three years was determined as the first issue of credits from the project represented a crediting period from 2013-2016. Fewer credits were generated in 2013 as it was the first year of the project, and therefore the crediting period better represented a crediting period of three years. At the time, it was envisaged that distributing revenue on an annual basis reflective of the crediting period would help create a mechanism by which a sustainable revenue stream could be built over time, and subsequently participating conservancies and their communities would have visibility of the funds available to them in coming years).
Date(s) of stakeholder consultation	Various dates throughout May - June 2021
Communication of monitored results	As part of the community consultation process, NRT held central briefings on the project for each conservancy's elected board chair, appointed manager, elected grazing committee chair, and rangelands coordinator. During these briefings, the proposed agreement was presented in English and Swahili and translated to local languages; and all who were present had access to legal advice and a legal advocate who presented the draft agreement to them, line by line (again, in English and Swahili and local languages.) The legal advocate present at these briefings was present at the initial briefing and was available to present advice to conservancies, and to answer any and all queries raised about the agreement.

¹³ The **Project Implementation Agreement** outlines the project activities involved and provides detail outlining the compliance and reporting requirements of the project. It provides a schedule of the reporting requirements a conservancy agrees to partake in and it includes detail on access for project monitoring, the validation process, verification, registration, records and other reporting relating to CCB Triple Gold, and sets out the agreed benefit sharing mechanism for revenue for the first three years of the project.

	These meetings were extensively documented both through written minutes and visually recorded through photographs and in some places, film. Special 'AGM's held at village or zonal level across each conservancy were considered the most appropriate way to socialize this specific information to as many community members as possible and for those community members to discuss the agreement and provide direct feedback to their Board and Grazing Committee representatives. In some conservancies, there may have been as many as 24 village/zonal meetings. Further, by hosting Special meetings, the agenda was very specific to ensure clear communication and messaging and additionally, hosting these meetings locally to each zone was aimed at ensuring the highest possible level of participation from community members at that time. Based on a strong consensus from those at the time, all communities approved for the conservancy chairperson to sign the Project Implementation Agreement on behalf of their community.
Consultation records	All consultation meetings held were minuted and filed for record keeping and are available for stakeholder access.
Stakeholder input	Input received from stakeholders during the consultation period led to the 14 chairpersons meeting on 24 June 2021 and formally signing the Project Implementation Agreement (PIA). The same legal advocate was also present and made the requested changes based on community and conservancy feedback to the draft agreement prior to the PIA signing.

2.3.10 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

Changes during the 2021 monitoring period revolve around the fact that the landscape has become increasingly more coordinated - regional meetings and dialogue between conservancies and communities on the move has improved significantly in the past year. The project has also focused heavily on building capacity at the village level to ensure

understanding and ownership at that level. There is also the development of county government rangelands policies which ensure local ownership but with support from the authorities when it comes to imposing by-laws.

During the first and second monitoring periods, dedicated personnel positions were created within NRT and the Conservancies to continuously monitor herder activity, detect encroachment from off the project area, and increase communication between Conservancy leaders, grazing coordinators, and herders. Efforts have continued to build capacity to ensure greater effectiveness in these areas through ongoing training and structured mentoring from the rangelands department within NRT as well as better coordinate regional level planning ensuring project activity success.

In addition, the project plans to devote some project revenues to establishing livestock finishing infrastructure (protected pastures and fencing, transportation hubs, etc.) to help shift community values from the number of livestock to the quality of livestock. Over time, this adaptation should reduce grazing pressure and increase the ability to reserve forage as grass banks and increase revenues and improve livelihoods of community members, including women and the poorest 25%.

Information about project activities continues to be disseminated from conservancy leaders and grazing coordinators throughout the local communities through established communication channels. Conservancies are owned and run by the communities that they represent; as such the governance structure within a conservancy focuses on the representation of the communities that live within the conservancy boundaries. Conservancies are led by an elected Board typically 12-14 members strong. Each of these Board members represent a village or zone (cluster of small villages) within a conservancy. A Chairperson is elected from within the Board and he / she represents the conservancy within the broader NRT Council of Elders. Once appointed, Board members undergo Leadership and Management Planning (LAMP) training and are tasked to not just represent the voices of the members of community they were elected by, but also to convey information concerning the management of the conservancy back to their communities. This is a critical step in the "communication loop".

A similar structure is applied to each conservancy's Grazing Committee, where each village or zone elects a representative, but instead of LAMP training, the committee undertakes more specific training on grazing management and the implementation of rotational grazing (project activity focussed) for the project which leads to improved rangeland management and subsequent increased sequestration of carbon.

Summary of comments received	Actions taken
In June 2021, based on a strong consensus from those at the time, all communities approved for their conservancy chairperson to sign the Project Implementation Agreement on behalf of their community.	Input received from stakeholders during the consultation period led to the 14 chairpersons meeting on 24 June 2021 and formally signing the Project Implementation Agreement (PIA). The same legal advocate was also present and made the requested changes based on community and conservancy feedback to the draft agreement prior to the PIA signing.

2.3.11 Stakeholder Consultation Channels (CCB, G3.5)

Outside of both the Board and Grazing Committees, the Conservancy Manager and the Rangeland Coordinator (who are appointed by the Board and are formal conservancy employees) also provide channels for information flow to communities, but also back to conservancy management and leadership. Examples of this would be around carbon project updates, grazing coordination with neighboring conservancies, in advance of livestock movement, results from routine surveys (e.g: Veg-CoMMS).

As part of baseline social data gathering, community elders, women, and young herders (morani) were asked a series of questions in focus groups in three Conservancies, and over 1000 households were interviewed in a household survey of four Conservancies in October 2013. Baseline data were gathered from the most populous ethnic groups.

These social data illustrated community concerns about encroachment by herders from outside the project area and the problem of security in effectively implementing planned grazing. These comments increased project costs devoted to paying grazing assistants, who patrol in designated grazing blocks to inform herders from outside the project area about expectations and rules for grazing within the Conservancy and to otherwise diffuse any conflicts.

Both baseline and project scenario data were collected through NRT Social CoMMs surveys and focus groups. It is important to note Social CoMMs surveys were designed for use throughout all NRT conservancies and are not exclusive to the conservancies participating in the project. Baseline data were collected in 2014 from conservancies that had just joined the project and had not yet implemented grazing plans. Project scenario data were collected in 2015 and 2016 from conservancies that had implemented planned grazing for at least two years. SocialCoMMs is an outgrowth of training of community members to conduct social surveys by professional social scientist Dr. Jessica Musengezi of The Nature Conservancy, supported by permanent NRT staff. Through the guidance of TNC and NRT personnel, focus groups are chosen by community chairmen to represent the public, rather than by the grazing committee. Household surveys are conducted at randomly selected households, and

every effort is made to interview at least the matriarch of each household. The results of groups and surveys were (are and will continue to be) incorporated in the management plan for the project as follows:

- a. Responses to household surveys collected, with emphasis to include vulnerable and marginalized groups, notably women and the poorest 25%.
- b. Data are analyzed with regards to perceived risks and detriments of project activities.
- c. Actions to mitigate risks or detriments, as discovered from Social CoMMs data are originated and discussed in quarterly carbon project management meetings.
- d. Operating procedures are defined and updated on the basis of suggested actions from management meetings.

2.3.12 Stakeholder Participation in Decision-Making and Implementation (VCS, 3.18, 3.19; CCB, G3.6)

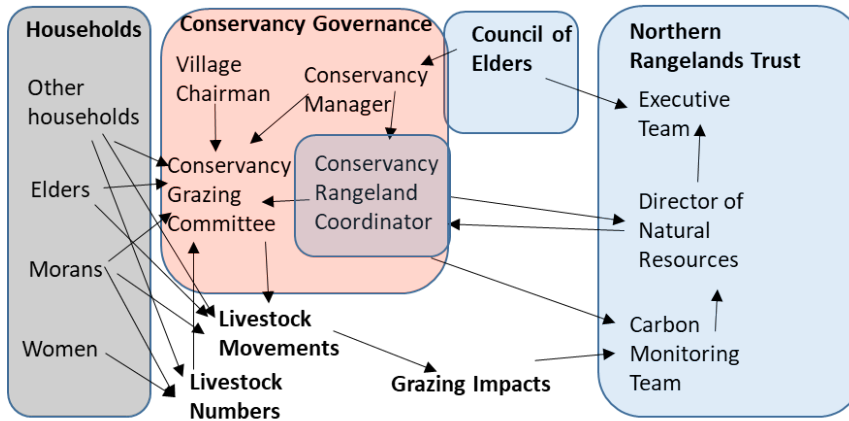
Extra efforts are made by community leaders to invite women, the poorest 25%, ethnic minorities, and other vulnerable groups to within-Conservancy public and grazing planning meetings. These invitations are both general, in the form of public notices or verbal public announcements as well as personal. Community leaders identify vulnerable households and communicate directly with the heads of these households to emphasize the importance of their participation.

This planning process is now established to operate throughout the project's lifetime.

Planned livestock movements include some movements outside the project area. Mainly these movements are transitions through HCV areas such as movements through the Mathews and Mukugodo Forests during wet to dry season transitions. These are kept as brief as possible but are nevertheless counted as leakage.

Cultural division of labor places grazing issues in the purview of men, so including women and households without livestock in the decision process can be challenging. However, focus groups that included women and morani revealed that women influence decisions about livestock sales and the distance and timing of livestock movements within their household, and thus receive information in an additional within-household communication channel beyond step 4. Women heads of household, many of which have significant livestock herds, are also contacted by village elders, Conservancy grazing committee members, NRT grazing coordinators, and NRT Social CoMMs supervisors.

Planning through a hierarchy of grazing decision-makers was implemented in October 2015 and led to the explicit inclusion of grazing plans in overall conservancy management plans for 8 conservancies. Since 2015, conservancy managers have utilized these annual conservancy management plans in order to effectively carry out grazing plans for their respective conservancy. It also extended the decision-making process to all ethnic groups regardless of the conservancy in which they live.



Schematic of how different stakeholders influence grazing management decisions. These stakeholders operate in a hierarchy of organizations from NRT down to the household level and are centered around grazing committees and the Conservancy grazing coordinator. The carbon monitoring team and NRT operate as a key information feedback loop.

2.3.13 Anti-Discrimination Assurance (VCS, 3.19; CCB, G3.7)

All NRT and Conservancy endeavors are required by their charter to follow the 2007 Kenya Occupational Safety Act. Compliance is monitored by NRT human resources staff. Field employees are provided with functioning 4WD vehicles with licensed, trained drivers and/or motorcycles for navigating on and off the road with helmets and other protective gear. Staff work in pairs for safety and the ability of a partner to transport another to health care providers distributed among different Conservancies. Game scouts are highly trained and provided uniforms, rifles, operating radios, and other appropriate field gear, and also work in teams transported, if necessary by trained and licensed drivers.

Particular attention is placed on avoiding discrimination and sexual harassment. Procedures and policies are already established to avoid sexual harassment are in place for conservation game scouts, which regularly mix genders in isolated field settings, and the same policies apply to other workplaces, as specified in sections 11.1-11.5 of the NRT Human Resources policy. This policy is accompanied by a grievance procedure in sections 11.6-11.9 that include provisions for reporting and resolving incidents, representation by other employees, and sanctions against offenders.

2.3.14 Grievances (VCS, 3.18.4; CCB, G3.8)

We did not receive any grievances from community members or off-project stakeholders related to project design and activities during the 2021 monitoring period. Pre-existing conflicts have existed in the project area for many years prior to the project design and start date. However, these conflicts are not exacerbated by the project design or activities. The project design and activities aim to reduce conflicts through the reduction of regional ethnic tensions.

The grievance mechanism in place, as stated in the PD, allows for any issue arising to be communicated - this can come from anyone who is a member of the conservancies, and therefore the project. An updated grievance mechanism was approved by conservancy representatives in 2023 and has been adopted for the project.

There is also an "informal / undocumented" grievance mechanism where grazing committees, who represent every herder within the project through their zonal distributions, can be approached by their respective communities with any issue they have. Grazing committee members can then raise the issue to the conservancy level, who can then raise it to the carbon project leadership team for further review at that level.

In regard to reporting disputes and/or grievances related to cattle-raiding and unplanned grazing within the project zone, it is important to note that cattle-raiding is theft of someone else's property - and it is therefore dealt with by the national government of Kenya. At a project level, when there is a case of cattle-raiding the incident is reported through the formal conservancy chain of command; from the reporter (most likely a community member, or conservancy ranger) to the conservancy warden, to the conservancy manager and then to the JOCC (Joint Operations Command Center) based at Lewa. The JOCC has Kenya national police embedded within, and it is their responsibility to address every incident of cattle-raiding reported, using lawful means.

When it comes to unplanned grazing, this is dealt with at a project level; conservancy grazing committees and relevant elders representing the respective communities involved in the incident are mobilised to engage one another in advance of any conflict arising. Dialogue is the mitigating factor - where the herders are then addressed to ensure they comply with the grazing plan in question. If the incident escalates and conflict arises, the incident is reported as per the example of cattle-raiding above, and the relevant national authorities intervene where they feel necessary.

Grievances received	Resolution and outcome
N/A	The grievance procedure has been communicated with stakeholders and is easily accessible for grievance submissions.

2.3.15 Worker Training (VCS, 3.19; CCB, G3.9)

Worker training occurs for monitoring employment in Social, Veg, and Wildlife CoMMs. Training occurs as guided by the most recent version of the NRT Human Resource Policy Manual. Training is implemented by Conservancy members to others, as these are Conservancy-managed programs. All procedures have evolved from experienced practice by local community members and are specifically designed to build local capacity that transcends personnel turnover. More details are available in the guide "Wildlife CoMMs: Ranger-Based Monitoring of Wildlife and Illegal Activities" Grazing coordinator training has involved university courses and off-site training, such as the US Forest Service rangeland

management training or programs at the Savory Institute. The grazing coordinator team has survived almost complete staff twice since 2013.

Formal training is encouraged by NRT, leave is allowed, and explicit policy within its HR manual codifies expectations.

Training in grazing management continues at the level of grazing committee members occurs on an annual basis. In addition, training in grazing management is now occurring at the village level. Three-hour workshops given to primarily women and elders in 10 villages in each of Kalama and Namunyak Conservancies occurred in June of 2016. This village-level training was also expanded to other Conservancies in 2017 and continued in 2022.

Project employment opportunities feature the following jobs, with associated risks and mitigation:

Job	Risks	Mitigation
Rangeland coordinator	Attack by wildlife, herders from rival ethnic group or outside the project area, Traffic accident	Accompanied by armed askari, Enclosed 4-wheel drive vehicle maintained by NRT
Monitoring field assistant	Attack by wildlife, herders from rival ethnic group or outside The project area, Traffic accident	Accompanied by armed askari, Enclosed 4-wheel drive vehicle maintained by NRT
Social survey technician	Attack by belligerent subject Traffic accident	Surveys made by appointment, Approval of village chairmen Travel in enclosed 4-wheel drive

2.3.16 Community Employment Opportunities (VCS, 3.19.13; CCB, G3.10)

These laws and policies have been and will continue to be implemented in the project through implementation of NRT's hiring policies, which include:

1. Open announcements of jobs associated with NRT and carbon project positions.
2. Coverage of all employees under Workmen's Compensation

3. The National Hospital Insurance Act of 1988, which ensures medical care for NRT and Conservancy employees in the event of accidents or harm related to exposure, wildlife hazards, and violence associated with enforcing livestock protection laws and grazing rights.
4. Occupational Safety and Health Act of 2007 further expanded the range of types of accidents covered and is used as the basis for education and training and employees.

The project will provide numerous employment opportunities for Conservancy members. 45% of NRT staff come from the local Conservancies, and searches for grazing coordinator, monitoring specialist, field assistant, driver, interviewer, data entry, accountant and other positions will actively seek Conservancy members. Some project revenues will be used by village or Conservancy administration to hire other Conservancy members as game scouts or other non-project jobs. These jobs will be announced and posted at Conservancy Headquarters, on the NRT website, and on standard job posting sites, and special attention will be given to applicants from vulnerable or marginalized groups, such as women and members of households without cattle. NRT has a strong track record of hiring women, as 33% of their workforce is composed of women comparable to the national average¹⁴ but well above employment levels in the Conservancies and in most pastoralist communities in Kenya.

2.3.17 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

The summary of worker risks and mitigation is covered in Section 2.3.13. These are indigenous communities living and working where they always have done, so the project does not impose any "new" threats as it supports and promotes their traditional practices. In terms of risks the communities might face; these include:

- Each-other through resource-based conflict (this is not a new thing and the project activities are aimed at increasing peace between the communities through the sharing of a common incentive)
- Wildlife - as wildlife numbers increase due to project activity and improved grasslands, conflict is also likely to arise
- Politics - Kenya continues to face risks around land tenure and use, infrastructure development and taxation
- Pressure from communities outside the project area - this could be called the "green magnet effect" due to improved grasslands This is to be mitigated by the development of new projects in the region.

2.4 Management Capacity

2.4.1 Required Technical Skills (VCS, 3.19; CCB, G4.2)

The project team has experienced personnel covering all the major areas of expertise, including monitoring, modeling, and carbon accounting (SftF, Native, TNC), marketing (Native), financing (Native, TNC, NRT), social surveys (NRT, TNC), community relations and communications (NRT, SftF) and education and training of community members (NRT). Additional expertise is available from other stakeholders for monitoring (ICRAF) education and training (Grevy's Zebra Trust).

2.4.2 Management Team Experience (VCS, 3.19; CCB, G4.2)

The project proponent is **Northern Rangelands Trust (NRT)**, **NRT** facilitates and coordinates a governance relationship with multiple indigenous communities across the 2 million ha project zone. NRT helps different communities establish wildlife and biodiversity conservation activities in the form of Conservancies that cooperate with private vendors to benefit from tourism in return for setting aside land from year-round cattle grazing. The communities engaged in these Conservancies are prepared to collectively participate in planned rotational grazing of livestock in both protected areas of their lands as well as open lands used communally by their members. Consequently, the communities of NRT are full participants in the carbon project. NRT also brokers peace agreements between different ethnic groups within and outside the project area and is instrumental in conflict resolution, all of which clearly contribute to community well-being. **Skills Provided:** community engagement, livestock management advice, peace-making and conflict resolution, administrative support, and social, vegetation, and wildlife monitoring.

Soils for the Future, LLC (SftF) is a carbon project developer who was responsible for the design and monitoring of project activities in 2021. SftF is working with its associated commercial enterprise **NRT Trading, Ltd, The Nature Conservancy (TNC)** and the 13 Conservancies. **Skills Provided:** biodiversity and soil and vegetation monitoring, carbon accounting, carbon modelling, advice on grazing management.

Native is a project marketer and developer that builds customized projects and purchase **arrangements** with private companies seeking to offset emissions or invest in community development projects. **Skills Provided:** marketing, sales transactions, MRV assistance.

TNC facilitates community-based conservation projects throughout eastern and southern Africa, including two REDD+ projects involving local community protection of forests in Tanzania. They implement a strategy of building local capacity to manage lands and resolve conflicts between competing land uses and conservation. **Skills Provided:** Fund-raising, administrative support, and social monitoring.

NRT Trading is a for-profit commercial enterprise associated with NRT and is responsible for enhancing, financing, and operational oversight of commercial business enterprises facilitated by NRT. **Skills Provided:** sales transactions, financial risk abatement.

The Conservancies are each a set of communities, typically dominated by a particular ethnic group that have joined together to set aside lands for wildlife and other types of conservation. Their goal is to provide security for livestock and wildlife and make better use

of lands to maintain cultural traditions, diversify livelihoods, and reduce poverty. **Skills provided:** livestock management, wildlife, and social monitoring.

2.4.3 Project Management Partnerships/Team Development (VCS, 3.19; CCB, G4.2)

Grevy's Zebra Trust (GZT) has been and will continue to be instrumental in education of pastoralist communities regarding the rotational grazing of bunched herds as a desired management objective. Their training led to adoption of rotational grazing in the first two Conservancies in NRT, and subsequent workshops helped expand rotational grazing concepts to additional Conservancies.

Contact: Belinda Low Mackey, Executive Director, Grevy's Zebra Trust, Nairobi, belinda@grevyszebratrust.org

International Centre for Tropical Agriculture (CIAT). This organization has and will provide the analyses of soil carbon and plant tissue lignin and cellulose for the project. CIAT also conducted some of the soil sampling used to determine the baseline carbon stocks in the project area.

Contact: Dr. Leigh Winowiecki, Soil Scientist I, International Centre for Tropical Agriculture, Nairobi, leigh.winowiecki@gmail.com

International Centre for Agroforestry (ICRAF) conducted some of the soil sampling used to determine the baseline carbon stocks in the project area and has developed a remote-sensing based model for estimating soil carbon stocks.

Contact: Tor-Gunnar Vagen, Senior Scientist, International Centre for Agroforestry, Nairobi, tvagen@cgiar.org

2.4.4 Financial Health of Implementing Organization(s) (CCB, G4.3)

The principal implementing partner in this project is NRT, whose annual budget is approximately \$10 million from both donor and commercial sources (for more details see NRT 2017 Annual Report¹⁵). NRT has grown substantially since its inception in 2004 and has a sustainable network of donors for livestock management and rangeland restoration featuring USAID, DANIDA, and The Nature Conservancy (TNC). TNC provides more than US\$1 million annually in support of NRT management and arranged \$7 million in low-interest loans to support the livestock-to-market program operated by NRT Trading, a commercial adjunct of NRT <http://www.nrt-kenya.org/nrt-trading/>. Project implementation costs not covered by these donor sources will be re-paid by revenues from the first issuance after the first monitoring period (2013-2016).

15

https://static1.squarespace.com/static/5af1629f12b13f5ce97ca0b5/t/5b727401575d1f3a6fb55fec/1534227491888/LOW_Res_2017_State_Of_Conservancies_Report.+FINAL.pdf

2.4.5 Avoidance of Corruption and Other Unethical Behavior (VCS, 3.19; CCB, G4.3)

The Project Proponent and any of the other entities involved in project design and implementation are not involved in or are not complicit in any form of corruption such as bribery, embezzlement, fraud, favoritism, cronyism, nepotism, extortion, and collusion. Proceeds of sales will be distributed according to pre-established contracts signed by each project proponent and the managers of each Conservancy.

2.4.6 Commercially Sensitive Information (VCS, 3.5.2-3.5.4; CCB Program Rules, 3.5.13 – 3.5.14)

No commercially sensitive information has been excluded from the public version of this Monitoring Report.

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

In 2021, no additional laws were introduced with impacts to the project participants or project activity. Grazing and grazing resource (water, soil) rights are conferred by land ownership associated with Group Ranch status via Kenya Land Act of 1968 (revised in 1970) or by Africa customary law for communities on Trust lands via the Kenya Trust Land Act of 2010. These rights are further conferred and superseded by the Kenya Community Lands Act of 2016. Rights to benefit from soil conservation are conferred by the right to register lands as Conservancies by the Kenya Wildlife Conservation and Management Act of 2013 and the Kenya Community Lands Act of 2016 (see also section 2.5.1).

The relevant labor laws include the Employment Act (Cap.226); the Regulation of Wages and Conditions of Employment Act (Cap. 229); the Trade Unions Act (Cap. 233), the Trade Disputes Act (Cap. 234), the Workmen's Compensation Act (Cap. 236). These have been and will continue to be implemented in the project through implementation of NRT's hiring policies, which include:

1. Environmental Management and Co-ordination Act, 1999. The project has been registered with and reviewed by the National Environmental Management Authority. Additional laws protect the integrity of Conservancies and their ability to engage in contracts
2. Companies Act, (Law of Kenya Cap. 486). NRT is a registered non-profit organization and the Conservancies are registered as limited liability corporations and are in good standing to operate in Kenya.
3. The Trespass Act (Law of Kenya Cap. 294). This is a law that is seldom enforced but could be used by Conservancies to help enforce the project area borders and reduce project livestock encroachment into the project area.

4. The National Hospital Insurance Act of 1988, which ensures medical care for NRT and Conservancy employees in the event of accidents or harm related to exposure, wildlife hazards, and violence associated with enforcing livestock protection laws and grazing rights.

The Northern Kenya Rangelands Carbon Project started in 2013, before the enactment of the Kenya Community Land Act 2016.

There are no requirements in either the act, or the regulations, applicable to NRT's development and conduct of a voluntary carbon sequestration project with the participating conservancies. The County Government holds unregistered community lands in trust on behalf of communities until they are formally registered. This does not preclude such communities from creating and operating conservancies under the Wildlife Conservation and Management Act of 2013, nor does it preclude such conservancies from participating in a voluntary carbon sequestration project.

Eight out of the 14 community conservancies participating in the Carbon Project were already registered group ranches which have since transitioned to community lands. All the participating conservancies in Samburu County and Laikipia County have transitioned to the Community Land Act 2016 through the support of NRT, County Governments, and other partners. They all have community title deeds. In these areas, The Community Land Management Committees doubles as Conservancy Board as per Community Land Act of 2016.

The remaining Isiolo County and Marsabit County participating conservancies are in the process of transitioning to the new Community Land Act 2016. The conservancies in these two counties are governed by democratically elected local boards from the community.

All community conservancies, whether in community land or trust land, conduct democratic elections for their board of directors. All but the Isiolo project conservancies have received Letters of No Objection from their respective County Governments, ensuring the support the government has for this project (letters provided to VVB). The Isiolo County Government has acknowledged receipt of notice of the project (attached), has raised no objection, has made public statements in support of the project, and has accepted project revenues provided by the conservancies. With the change in administration from the last election, NRT has restarted the formal approval process.

The Kenya 2010 Constitution and the Land Act of 2016 legally recognize communities' customary land rights regardless of whether they are registered or not. The Community Land Act of 2016 enables the local communities to legally claim, own and manage the land that they depend on for their livelihood, culture, and identity. The Community Conservancies falling under Trust Lands have the constitutional right to be accessing any resources within their traditional land, and neither the Acts nor regulations negate any investments. The Isiolo County communities from which the cited petition arose are not participants in the project.

2.5.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18, 3.19; CCB, G3.11)

The National Hospital Insurance Act of 1988 and Occupational Safety Act of 2007 ensures medical care for NRT and Conservancy employees in the event of accidents or harm related to exposure, wildlife hazards, and violence associated with enforcing livestock protection laws and grazing rights. How the project addresses compliance of this Act is covered in Section 2.3.14.

Per section 12 of the NRT HR Policy Manual (p. 49), employees are reminded of their rights to medical care in the event of job-related accidents at hiring. Employees have access to basic understanding of policy; responsibilities for Health and Safety (H & S) at work are all documented. There are up-to-date Health and Safety risk assessments and fire risk assessments conducted annually. The provision of necessary health and safety information, instruction and training is systematically undertaken along with these assessments and there is provision and correct use of necessary personal protective equipment (PPE) at all times.

Per the National Hospital Insurance Act, NRT has a Health and Safety Committee (NRT HR Policy Manual p. 50). NRT arranges for emergency response/evacuation, first aid and occupational health and employees are trained in documented procedures/safe systems of work, where necessary to establish and maintain a safe and healthy working environment.

Health and Safety protocols are extended to clients and visitors. These protocols are documented and known and communication and consultation with employees on H & S issues is consistent throughout the organization. There is consistent monitoring, investigation and reporting of any incidents, accidents or occupational ill-health; and corrective and preventative actions where any incidents, accidents or occupational ill-health occur are a priority. Accidents are required to be reported on a standard form.

Third-party contractors perform regular audits of all safety measures, equipment, fire exits, among others, to ascertain the extent to which NRCL's Health and safety measures meet desired standards and legislation. Staff may be required to participate in safety-related activities, whenever called upon to do so.

2.5.3 Human Rights (VCS, 3.19)

(Section non-applicable to the CCB-only Public Comment Period)

Demonstrate how the project continues to recognize, respect, and promote the protection of the rights of IPs, LCs, and customary rights holders in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples.

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

(Section non-applicable to the CCB-only Public Comment Period)

Demonstrate that the project continues to preserve and protect cultural heritage as part of project activities.

2.5.5 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

Disputes over rights to territories and resources	N/A
Respect for property rights	Grazing and grazing resource (water, soil) rights are conferred by land ownership associated with Group Ranch status via Kenya Land Act of 1968 (revised in 1970) or by Africa customary law for communities on Trust lands via the Kenya Trust Land Act of 2010. These rights are further conferred and superseded by the Kenya Community Lands Act of 2016. Rights to benefit from soil conservation are conferred by the right to register lands as Conservancies by the Kenya Wildlife Conservation and Management Act of 2013 and the Kenya Community Lands Act of 2016 (see also section 1.3.2). Rights to benefit from soil conservation as a component of wildlife conservation are conferred by the right to register lands as Conservancies by the Kenya Wildlife Conservation and Management Act of 2013¹⁶ and by the Community Lands Bill of 2016¹⁷. The former law confers the rights of communities to form Conservancies and benefit from conservation activities such as wildlife management and tourism and land restoration, and the latter confers this right on any public land occupied by communities, including trust lands and designated community lands. Conservancies are defined as a “community-owned and community-run institution which aims to improve biodiversity conservation, land management and the livelihoods of its constituents over a defined area of land traditionally owned, or used, by that constituent community.” As Conservancies and members of the Kenya Wildlife Conservancy Association (KWCA), communities have the unrestricted, and unencumbered right to benefit financially from conservation activities, which would include restricted,

¹⁶

http://www.kws.org/export/sites/kws/info/publications/acts_policies/The_wildlife_conservation_and_management_bill_2013.pdf

¹⁷ http://info.mzalendo.com/media_root/file_archive/Community_LandBill_2015.pdf

	planned and managed grazing and building soils through carbon storage ¹⁸ . The Kenya Wildlife Service administers the formation of Conservancies and all 13 Conservancies have submitted applications for the formal formation of Conservancies.
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2.5.6 Benefit Sharing Mechanism (VCS, 3.18, 3.19)

(Section non-applicable to the CCB-only Public Comment Period)

Where the project impacts property rights as described in Section 2.5.5 above, use the table below to describe the benefit sharing that occurred during the monitoring period, in accordance with the validated agreement.

Summary of the benefit sharing plan	
Benefit sharing during the monitoring period	

2.5.7 Free, Prior, and Informed Consent (VCS, 3.18, 3.19; CCB, G5.2)

Participation in this project is completely voluntary. Participation by community members is presumably motivated by the prospect of benefits to forage availability, livestock health and marketability in addition to benefits to community infrastructure and retention of land rights. The decision about where and when to move livestock will be made by a community decision-making process through existing mechanisms of livestock management, and not by NRT, TNC, or other stakeholders. The voluntary nature of the decision has been made clear by the announcement of the project during the public comment period during consultations with key informants, opinion leaders, and community administrative leaders.

Encroachment onto private lands is discouraged by expected payment of lease fees by project-participating community members to the landowners. Excursions on lands outside the project area traditionally is not compensated as per cultural norms, but expectations of paying leases in those cases are increasing^[6]

The need for geographical integrity of the project for carbon accounting and coordinated, planned grazing at a regional scale creates a project of core participating Conservancies surrounded by Non-participating communities (NPC's). As the project develops and rangelands and soils are restored, the project area may become a magnet for encroachment from NPCs during droughts. Likewise, members of participating Conservancies members may also choose to move livestock onto range used by NPCs (this is treated as leakage in project carbon accounting); project protocols request such excursions to be legal and

¹⁸ <https://kwcakenya.com/conservancies/conservancies-policies/>

justified through compensation (leases or rents) to NPCs. Such negotiations thus far have been mostly informal (no legal documents or paper trail).

Individual conservancies may on their own volition choose to relocate settlements within their Conservancy, but this will follow established mechanisms of decision-making within the community. At no time will NRT, SFF, or TNC request community members to relocate. This will be made clear by the announcement of the project during the public comment period through consultations with key informants, opinion leaders, and community administrative leaders.

The Project's principal values are to improve pasture and forage for livestock and wildlife, improve resilience to drought conditions, and provide new revenue streams for conservancies and the communities that live within them. These values are fully aligned with the principle of safeguarding local communities and indigenous and marginalized people's interests' and further, all of the carbon project activities seek to support and follow FPIC legislation, and promote the indigenous and human rights of the communities involved in the project.

The project proactively seeks to ensure that FPIC is a key part of ongoing project activities and is embedded in the operational framework for the project over the remaining project life span (project activity end date is 2042.) NRT acknowledges that the requirements for FPIC will change over time, and as it has endeavored to do over the project lifespan to date, it will continue to evolve the ways in which project engagements, activities, benefits sharing, and reporting are communicated and carried out to fulfill the highest possible standards in regards to FPIC.

This speaks particularly to wider efforts being made by NRT within the project area to promote maximum participation in conservancy decision making processes and governance to help ensure representation from all indigenous and marginalized community members, including the inclusion of women and youth and all other key stakeholders.

Consent	In 2021, as leadership in the conservancy changes every three years, through the conservancy board elections, NRT felt it was critical to host another briefing to ensure that current leadership was in a position to fully understand the project and for communities to decide whether they were still interested in pursuing the sale of carbon credits through their conservancies. Thus, in May and June 2021, NRT undertook a widespread briefing and socialization of the project for all participating conservancies and their member communities. NRT presented a revised and renewed Project Implementation Agreement¹ to set out a clear understanding between NRT and participating conservancies as the project launches the first issue of carbon credits. (The project briefing and PIA briefing were translated
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	during the presentation to local languages). As part of the community consultation process NRT held briefings on the project for each conservancy's board chair, manager, grazing committee chair, and rangelands coordinator. These members then briefed their Boards and Grazing committees, and following that, 'Special AGMs' were held at the village or zonal level across the conservancies, to brief the wider community and ensure that there was general consensus within the community to continue participating in the project. These meetings were extensively documented both through written minutes and visually recorded through photographs. There was strong consensus by the communities with all communities approving the conservancy chairperson to sign the Project Implementation Agreement on behalf of their community. The 14 chairpersons met on 24 June 2021 and formally signed the Project Implementation Agreement. (Minutes of these meetings and a copy of the fully signed PIA are available on request. NRT also has copies of the minutes of Special AGMs providing a record of community endorsement of the Agreement and the discussions that were held around this.)
Outcome of FPIC	During the signing of the Agreement, it was agreed by the participating conservancies that the Carbon Project Oversight Committee (CPOC) be formed. The Committee includes seven representatives from participating conservancies; two from each of Laikipia, Isiolo, and Samburu Counties and one from Marsabit (given that only Melako Conservancy in Marsabit County is part of the project.) These representatives form the CPOC along with a number of NRT administrative functions (CFO, Director of Governance, Carbon Project Lead, Carbon Project Officer, and Director of Natural Resource Management) and provide oversight of NRT's role as project proponent and administrator of the project. This committee meets five times per year and minutes of each meeting are shared with all participating conservancies to provide regular ongoing information about the project and management of particular issues that arise. The first meeting was held in October 2021 (Minutes of these meetings are available on request.).

2.5.8 Property Right Protection (VCS, 3.18, 3.19; CCB, G5.3)

Each of the participating conservancies signed a Letter of Intent to participate in the Project in 2015. In 2017, each signed a binding Consent and Waiver, consenting to NRT entering into the marketing and sale agreement with Native with respect to the Project Verified Emissions Reductions, and waiving their rights, in favor of NRT and Native, to all VERs that had been or would be produced by the Project, other than their rights vis-à-vis NRT to

proceeds from the sales thereof. Sales occurred only after all such Consents and Waivers were signed. Copies are to be provided to the VVB.

Land tenure in northern Kenya is community-based, in which government-owned land allows occupying communities ownership rights to resources. Some communities (Il Ngwesi, Kalama, Leparua, Meibae, Naibunga, Namunyak, Sera, Westgate) exist as or have applied for registration as Group Ranches under the Kenya Land (Group Representatives) Act of 1968¹⁹. This law conferred rights through group representatives who are “authorized to hold land and other **assets** on behalf of the group, and to act on behalf of and for the collective benefit of all group members.” (Section II.8.). Soil carbon would be considered an asset for the group members since it has value in the voluntary carbon market.

Other communities, such as those near Biliqo Bulesa, Melako, Oldonyiro, Nasuulu, and Nakuprat Gotu Conservancies are located on Trust lands, which confer rights to grazing and by inference resources that promote grazing (such as water, soil carbon) under customary law conferred by Section VIII item 59 of the Kenya Trust Land Act of 2010²⁰. These rights are now further conferred by the passage of the Community Lands Act of 2016, which confers rights to resident communities to graze and conduct conservation activities even on Trust lands²¹.

Unfortunately, all communities have suffered historically from an inability to defend ownership or customary rights from either government initiatives or land use by non-member individuals but NRT efforts and support have helped reduce conflicts and organize governance and management activities in these communities. As part of this process, all the participating NRT communities are now registering their lands, including settlements, as Conservancies. These are defined by the Kenya Wildlife Conservation and Management Act of 2013 as a “community-owned and community-run institution which aims to improve biodiversity conservation, land management and the livelihoods of its constituents over a defined area of land traditionally owned, or used, by that constituent community.” As Conservancies, communities have the right to benefit financially from conservation activities, which would include restricted, planned, and managed grazing and building soils through carbon storage.

Most Conservancies are dominated by a single ethnic group, but the dominant group varies among Conservancies. Generally, interactions among ethnic groups in the project area are congenial, as evidenced by perennial dry season movements of livestock from Conservancies dominated by one ethnic group (e.g. Samburu from Kalama, Westgate and Namunyak) to lands on Conservancies dominated by others (Rendille in Melako and Borana in Biliqo Bulesa). Within-community distribution of resources generally proceeds without bias towards ethnic groups, as evidenced by a lack of indicated bias about revenue distribution among different ethnic groups within a Conservancy in baseline household surveys.

As part of community property protection rights, community members are never required to relocate. Movement of habitation is voluntary, such as when families voluntarily moved to

¹⁹ <http://faolex.fao.org/docs/pdf/ken62430.pdf>

²⁰ <http://faolex.fao.org/docs/pdf/ken62437.pdf>

²¹ http://kenyalaw.org/kl/fileadmin/pdfdownloads/Acts/CommunityLandAct_27of2016.pdf

villages and along the borders of Westgate Conservancy to facilitate implementation of Conservancy grazing plans in 2014. Likewise, ethnic minorities within Conservancies are allowed to select their home location.

2.5.9 Identification of Illegal Activity (VCS, 3.19, CCB, G5.4)

The project depends on voluntary actions by community members to herd livestock according to planned rotational grazing principles. By communal grazing rights afforded to each Conservancy, altered grazing management is a legal activity. Monitoring of biodiversity involves no capturing or handling of vertebrates, and sampling of plants and insects is not covered under Institutional Animal Care regulations. Human surveys by NRT staff are covered under research methods approved by an external Institution Review Board covering TNC and NRT-sponsored social science research.

Traditional pastoralist practices of stealing livestock, armed takeover of grazing areas or water points, or killing pastoralists of other ethnic groups will not be tolerated. Conflicts, when they do exist have been and will be addressed by combined NRT, Conservancy, and County government response teams. When response team intervention fails to resolve the conflict, NRT calls a peace meeting, such as occurred in July of 2015. Peace meetings have been highly successful in reducing conflict in the area since the advent of NRT in 2008.

No significant illegal activities were identified in 2021 and all activities were tracked through “conflict incident reports.” These reports are generated monthly.

2.5.10 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

The decline in traditional governance and wider insecurity in the Horn of Africa have combined to make illicit firearms readily available. There is a general willingness throughout the region for people to use them to enforce resource access rights. The proliferation of firearms has combined with tribal traditions in which raiding livestock from a neighboring tribe is a male rite of passage and means by which a *moran* (warrior) may obtain social status and marry. However, in recent years, the availability of firearms has led to more frequent raids and inter-tribal killings. In Samburu District, 88% of respondents reported that they had used firearms in their possession for cattle raiding with comparable figures likely elsewhere in the region. Cattle-raiding is estimated to result in the loss of US \$1 million annually in Samburu District, and is a significant factor constraining economic development in the region^[7]. Insecurity from the threat of cattle-raiding has fed the cycle of increased environmental degradation, which in turn fuels the conflict ^[7]. Because much of the rangeland has become unsafe for grazing livestock and nomadic herders, pastoralists have become further sedentary in secure areas^[7]. Furthermore, stealing cattle may be the only viable way for many impoverished people to rebuild herds lost from droughts. Drought further exacerbates conflict because herders may move to areas distant from their home Conservancies to find water or grass, thus bringing historically antagonistic ethnic groups into proximity with little governance as to how limited resources are allocated.

Ethnic groups with the most frequent history of conflict are Samburu with Turkana and Maasai and with Borana. Two major sources of conflict highlight certain localities in the

project area that have received more peace meetings and pre-emptive consultation during droughts when conflicts are more likely to occur. First Conflicts between Turkana, Maasai and Samburu date back 400 years^[8, 9], and both maintain cultural traditions of livestock theft from each other. These three ethnic groups are juxtaposed along the Ewaso Nyiro River where Turkana have settled among otherwise Maasai-dominated communities but are in close proximity to Samburu north of the river. This confluence of three antagonistic ethnic groups (Turkana, Maasai, and Samburu) around a critical water resource (the river) fosters persistent livestock theft, especially during drought conditions.

The other ethnic conflict “hotspot” is along the Sera- Biliqo Bulesa border, where Samburu herders bring livestock from the west simultaneously with Borana bringing livestock from the east during the dry season. Borana, together with Gabra are a subgroup of the larger group of Oromo people centered in north central Kenya and in Ethiopia and differ sharply in religion, livestock preferences (camels vs cattle) and language from the Samburu. Conflicts over forage and water intensified between these two groups in the late 19th Century, and animosity persists.

These geographic foci allows NRT staff, Conservancy grazing coordinators and game scouts, and County government staff to more efficiently deal with conflicts as they arise, with effort put into specific localities at different times. For example periods of persistent conflict occur in July, when Samburu herders typically cross Sera Conservancy and approach two locations with reasonably predictable availability of water, Kisima Hamsini (50 wells in Kiswahili) and Qom, placing them in proximity to Borana and Gabra herders approaching from the east^[1]. The other major conflict period is September- October near the end of the dry season, when livestock of Samburu in Kalama, Turkana in Nakuprat-Gotu and Nasuulu often have significant herds in juxtaposition^[1]. In 2017 and 2018, NRT, working with Samburu County government attempted to anticipate conflict in these localities with peremptory peace meetings, but severe drought conditions and erratic movements of herders from different Conservancies required repeated post-conflict resolution from NRT and Samburu County.

Project activities are highly unlikely to either cause or influence the outcome of such disputes. First, most conflicts occur because of livestock theft by members of different ethnic groups, and resolution of these disputes involves returning stolen livestock, peace meetings, intervention by County officials, etc. Project activities involve movement of livestock according to local Conservancy grazing plans or regional grazing plans in the event of extended dry seasons or drought. These plans allow anticipation of livestock movements 2-4 weeks ahead of schedule so that conflict areas can be cleared of unscheduled livestock or range-sharing verbal agreements can be arranged. Conflicts are more likely when herders are not following planned project activities, e.g., arrive at grazing areas or watering holes too early or too late, or if herders steal livestock, which is never tolerated and is prosecuted by Conservancy game scouts as part of the project activities.

Such meetings are largely at the site of conflict, impromptu, and conducted in local unwritten languages, so meeting records are not available. However, since May 2015 grazing coordinators meetings are attended by project proponents, Conservancy managers have increasingly mentioned the outcome of meetings in their reports, and these are available as

supporting documents. Conservancy manager reports from early 2018 describe multiple peace meetings with successful resolution in which stolen cattle were returned, agreements were reached with encroaching herders from communities outside the project zone, and disputes were settled between elders and morani (young adult males) from within the same Conservancies. In addition, programs in which morani are interviewed and involved in grazing committee decisions began to be implemented in early 2018, which further reduced conflicts during the failed short rains of November 2018.

There is a history of conflict between ethnic groups in the project area based on access and proximity to water leading to encroachment of herders from other conservancies and outside the project area and persistent livestock theft between groups. Conflicts are more likely to occur during droughts. The Project is unlikely to either cause or influence the outcome of these conflicts. Project activities involve movement of livestock according to conservancy or regional grazing plans and conflicts are more likely when herders are not following planned grazing activities. As a result of the Project, there are more game scouts within the conservancies who patrol conservancy borders and deal with conflicts as they arise.

3 CLIMATE

Note that Section 3 relates exclusively to data and parameters for monitoring GHG emission reductions and removals. No community or biodiversity parameter should be entered here with the exception of any relevant information for fulfilling the optional Gold Level criteria.

3.1 Monitoring GHG Emission Reductions and Removals

(Sections under 3.1 are non-applicable to the CCB-only Public Comment Period)

3.1.1 Data and Parameters Available at Validation (VCS, 3.16)

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data
Value applied	Provide the value applied

Justification of choice of data or description of measurement methods and procedures applied	<i>Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.</i>
Purpose of data	<i>Indicate one of the following:</i> • <i>Determination of baseline scenario (AFOLU projects only)</i> • <i>Calculation of baseline emissions</i> • <i>Calculation of project emissions</i> • <i>Calculation of leakage</i>
Comments	<i>Provide any additional comments</i>

3.1.2 Data and Parameters Monitored (VCS, 3.16)

Complete the table below for all data and parameters monitored during the project crediting period (copy the table as necessary for each data unit/parameter). The values provided are used to quantify the reductions and removals for the project crediting period in Section 3.2 below.

Data / Parameter	
Data unit	<i>Indicate the unit of measure</i>
Description	<i>Provide a brief description of the data/parameter</i>
Source of data	<i>Indicate the source(s) of data</i>
Description of measurement methods and procedures to be applied	<i>Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).</i>
Frequency of monitoring/recording	<i>Specify measurement and recording frequency</i>
Value monitored	<i>Provide an estimated value for the data/parameter</i>

Monitoring equipment	Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of the data	Indicate one of the following: • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Calculation method	Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.
Comments	Provide any additional comments

3.1.3 Monitoring Plan (VCS, 3.16, 3.20)

Describe the process and schedule followed during this monitoring period for obtaining, compiling, and analyzing the monitored data and parameters set out in Section 3.1.2 (Data and Parameters Monitored) above during this monitoring period, including details on the following:

Include details on the following:

- The methods used for measuring, recording, storing, aggregating, collating, and reporting on monitored data and parameters. Where relevant, include the processes used for calibrating monitoring equipment.
- The organizational structure, responsibilities and competencies of the personnel that carried out the monitoring activities.
- The processes used for handling any internal auditing performed and any non-conformities identified.
- The implementation of sampling approaches, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures. Where applicable, demonstrate whether the required confidence level or precision has been met.

Where appropriate, include line diagrams to display the GHG data collection and management system.

3.1.4 Dissemination of Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

Given the expanse of the project area and the paucity of reliable telecommunications, the monitoring plan is still disseminated in a series of separate actions. 80% of monitoring is performed by communities and is subdivided into wildlife, vegetation, and social, with each having a separate community-based management system (CoMMs); the results of these surveys are shared annually with the conservancies who then communicate the results to communities. Livestock monitoring is performed by grazing coordinators who are members of the communities they monitor, and their activities are communicated to the public in grazing committee meetings within Conservancies; grazing committee meetings occur quarterly.

Larger and broader monitoring plans, such as those involving satellite imagery and cross-Conservancy issues such as seasonal migratory movements, have not been presented since May 2016 and September 2017 as their relevance to the communities is limited without the context of the verification and issuance of VERs which can then be subsequently sold. During a briefing meeting with Conservancy Leadership and Management teams in late January 2022, monitoring plans including the use of satellite imagery, were appropriately conveyed to communities through each respective conservancy AGM held during the course of the 2022 (most occurred between February and April). It was important that these results were communicated to communities in the context of the project to enhance the understanding and application of the project activities, and now visible results.

There were additional meetings during the 2017-2018 drought period focused at the regional level to bring all conservancies together across the region. Individual and regional plans were shared during these meetings and plans outlined for the movement of livestock during the drought. Such meetings were mostly held quarterly during this time and areas of high livestock concentrations (such as in Sera and Biliqo) were identified as high priority and afforded additional planning so that the surrounding conservancies could support them achieve their own grazing plans, as well as foster peaceful coexistence among different ethnic communities grazing in those areas.

Most of the conservancies in the project area were impacted by the locust invasion 2019-2020 in Northern Kenya. However, communities themselves cannot do much to control the movement of locust except to report the location of the swarms to authorities to facilitate a coordinated control effort between the government and the Food and Agriculture Organization of the United Nations. This included spraying swarms of locust in affected areas. However, communities did not carry out any additional meetings on the locust invasion itself.

3.2 Quantification of GHG Emission Reductions and Removals

(Sections under 3.2 are non-applicable to the CCB-only Public Comment Period)

3.2.1 Baseline Emissions (VCS, 3.15)

Quantify the baseline emissions and/or carbon stock changes for the monitoring period in accordance with the applied methodology. Baseline emissions may be negative where carbon stock increases (sinks) exceed baseline emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

3.2.2 Project Emissions (VCS, 3.15)

Quantify project emissions and/or carbon stock changes for the monitoring period in accordance with the applied methodology. Project emissions may be negative where carbon stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

3.2.3 Leakage Emissions (VCS, 2.5, 3.2, 3.6, 3.15, 4.3)

Quantify leakage emissions for the monitoring period in accordance with the applied methodology. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

3.2.4 GHG Emission Reductions and Carbon Dioxide Removals (VCS, 3.15, 4.1)

Quantify the GHG emission reductions (reductions) and carbon dioxide removals (removals) for the monitoring period. Include all relevant equations.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCUs for reductions and removals separately where the applied methodology provides procedures and equations to do so.

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	
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Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO₂e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No <i>If no, provide justification.</i>
State, in tCO₂e, the expected total GHG benefit to date.	
If a loss occurred (including a loss event or reversal), state the amount of tCO₂e lost:	

ii) Complete the table below for the project crediting period. Note that the buffer pool allocation is split proportionally between the reductions and removals. (For example, if a project achieves 20,000 tCO₂e removals and 80,000 tCO₂e reductions and has a buffer contribution of 20%, or 20,000, the removal VCU's would be 16,000 and reduction VCU's would be 64,000).

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Buffer pool allocation (tCO ₂ e)	Reductions VCU's (tCO ₂ e)	Removals VCU's (tCO ₂ e)
DD-MMM-Y YYY to 31-Dec-YYY Y	Example: 50,000	Example: 20,000	Example: 10,000	Example: 4,000	Example: 8,000	Example: 8,000
01-Jan-YYYY to 31-Dec-YYY Y						
01-Jan-YYYY to DD-MMM-Y YYY						
Total						

For all projects, state the estimated ex-ante GHG emission reductions and carbon dioxide removals and the achieved reductions and removals for the monitoring period. Report the percentage difference and explain any difference. The quantities of reductions and removals are the total quantities before any deductions for buffer credits.

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
DD-MMM-YYYY to 31-Dec-YYYY				
01-Jan-YYYY to 31-Dec-YYYY				
...				
01-Jan-YYYY to DD-MMM-YYYY				
Total				

3.3 Optional Criterion: Climate Change Adaptation Benefits

Project activities provide climate change adaptation benefits to high conservation value ethnic groups as well as women in the poorest 25% of community members.

3.3.1 Activities and/or processes implemented for Adaptation (CCB, GL1.3)

The project zone includes several community High Conservation Values that are threatened by climate change.

1.) Sustaining livelihoods and cultural identity of indigenous groups. (General Benefits)

The project zone includes more than half the remaining people of several indigenous ethnic groups that have practiced pastoralism in northern Kenya since at least the mid-19th Century. The project helps sustain the livelihoods of endemic groups that include the Samburu (speakers of a Nilo-Saharan language) in Kalama, Oldonyiro, Meibae, Westgate, Sera, and Namunyak Conservancies, the Rendille (speakers of a Cushitic language) in Melako Conservancy, and Borana (a subgroup of Oromo (also Cushitic) speakers in Biliqo Bulesa Conservancy. These people account for more than 80% of the people in the project zone.

2.) Support economic viability on traditional livestock grazing lands.

The ability of people in the project zone to continue pastoralism is threatened by development and land privatization, mineral exploration, and the potential for the

government to designate lands as protected game reserves or other types of conservation areas without human occupancy.

3.) Provision of ecosystem services.

Livestock production, if sustainably managed, could meet an ever-growing demand for meat in Nairobi and elsewhere in Kenya, thus providing an important ecosystem service from land that has too poor of soils and too little rainfall to support intensive agriculture. The project itself will lead to removal of more than fifty million tonnes (CO₂ equivalent) of greenhouse gases, thus providing a valuable global ecosystem service.

4.) Regional peace and security.

Finally, successful, sustainable occupancy of the land by pastoralists provides the basis for peace among traditionally antagonistic groups such as Turkana or Borana and Samburu and serves as a bulwark against the import of terrorism from Somalia on Kenya's western border.

The project activity, planned rotational grazing both within Conservancies and across Conservancies as part of a regional grazing plan, will have positive effects on all four of these community HCVs (Figure 2).

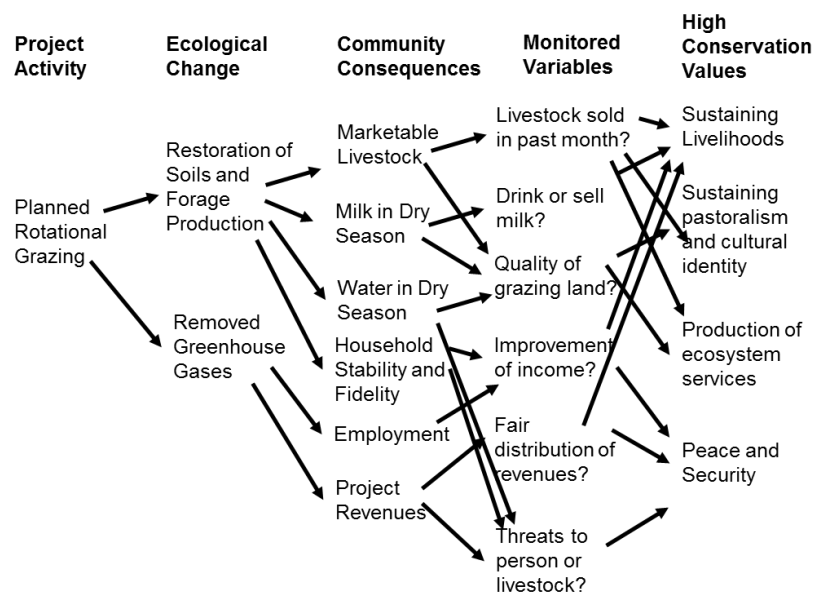


Figure 2. Causal model linking project activity to community High Conservation Values (HCVs) under the project scenario, which are threatened by climate change. The project activity should produce key ecological changes that lead to multiple community consequences. These consequences can be monitored and demonstrated with SocialComms (See Section 4) to improve all four HCVs for three principal endemic indigenous groups.

HCV 1: Sustaining livelihoods and cultural identity of indigenous groups

Samburu

High Conservation Value	Sustaining livelihoods and cultural identity of indigenous Samburu (similar to Maa Nilotic language, < 120,000 speakers)
Qualifying Attribute	The project will provide, as it develops, forage reserves and greater productivity within areas managed for grazing. These changes derive from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. produce more robust, healthier animals, more animals that can be sold on meat markets, milk for family use and sale. These benefits can be experienced during extended dry periods and droughts and dramatically improve food security and cash revenues to households. Improvements to forage production should reduce conflicts between Samburu and traditional enemy groups such as Borana and Pokot
Focal Area	Oldonyiro, Westgate, Kalama, Sera, and Namunyak Conservancies

Rendille

High Conservation Value	Sustaining livelihoods and cultural identity of indigenous Rendille (unique Cushitic languages < 100,000 speakers)
Qualifying Attribute	The project will provide, as it develops, forage reserves and greater productivity within areas managed for grazing. These changes derive from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. These changes produce more robust, healthier animals, more animals that can be sold on meat markets, milk for family use and sale. These benefits can be experienced during extended dry periods and droughts and dramatically improve food security and cash revenues to households. Improvements to forage production should reduce conflicts between Rendille and traditional antagonist groups such as Borana and Pokot
Focal Area	Melako Conservancy

Borana

High Conservation Value	Sustaining livelihoods and cultural identity of indigenous Borana (Oromo language, Cushitic indigenous to Ethiopia, Muslim)
Qualifying Attribute	The project will provide, as it develops, forage reserves and greater productivity within areas managed for grazing. These changes derive from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. These changes produce more robust, healthier animals, more animals that can be sold on meat markets, milk for family use and sale. These benefits can be experienced during extended dry periods and droughts and dramatically improve food security and cash revenues to households. Improvements to forage production should reduce conflicts between Borana and traditional enemy groups such as Samburu
Focal Area	Biliqo Bulesa and Nakuprat-Gotu Conservancies

HCV 2: Support economic viability on traditional livestock grazing lands. This support enables the unique cultures in the northern Kenya landscape to persist in the face of climate change rather than dissolve under pressure for land development and loss of livestock due to drought.

High Conservation Value	Support economic viability on traditional livestock grazing lands
Qualifying Attribute	The project, as executed from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies, will promote a shift in the economic valuation of forage and livestock production over previous wealth valuation of livestock numbers. This will allow communities to operate increasingly on a cash basis, maintained by remote cell phone banking mechanisms like M-Pesa ²² and allow greater flexibility in managing herd sizes during drought. These adaptations will reduce the risk of catastrophic livestock loss during severe droughts and

²² <https://www.safaricom.co.ke/personal/m-pesa/do-more-with-m-pesa/m-pesa-global>

	perhaps reverse the shift in livestock holdings to sheep and goats that pressure even very low forage production.
Focal Area	Entire project area, all indigenous groups

HCV 3: Provision of ecosystem services

The principal ecosystem services provided by the project are increased surface water availability, wildlife to support ecotourism. All these aid communities in adapting to climate change.

High Conservation Value	Ecosystem services provided by the project are increased surface water availability, wildlife to support ecotourism
Qualifying Attribute	The project will, as it develops, will build up soil organic matter, thereby increasing forage productivity and water-holding capacity of the soil. These outcomes result from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply across all Conservancies. These changes should alter hydrological cycles, holding soil moisture longer into dry seasons and rejuvenating springs and groundwater sources, which will allow broader distribution of livestock on the landscape and provision of milk and meat through extended dry periods and drought. Wildlife populations are expected to increase, providing greater opportunities for and revenues from ecotourism. These changes will increase food security and reduce risk of ethnic conflicts over solitary sources of water and forage during dry seasons. The need for risky, long-distance migrations may decrease as well.
Focal Area	Entire project area, all indigenous groups

HCV 4: Peace and Security

High Conservation Value	Peace and security
Qualifying Attribute	Greater forage and water availability from improved rangeland conditions will result from project activities derived from community-level decisions at public meetings to decide grazing plans within Conservancies and plans that apply

	across all Conservancies. Outcomes should allow herders from ethnic groups with histories of conflict to avoid convergent space use and temptation for historical cultural activities like stealing livestock or dominating resource locations at gunpoint. Encroachment of livestock outside regional grazing plans should also decrease. Security has been a major driver of sedentary settlement, and greater peace should allow for more peaceful dry season migrations. Greater game scout presence supported by project revenues will also help decrease deep encroachments of herders from outside the project area and quickly enforce justice on livestock thieves. Greater security and peace should also facilitate programs to develop infrastructure for finishing livestock for cash markets.
Focal Area	All Conservancies, but especially dry season ranges in Sera, Melako, and Biliqo Bulesa, as well as dry season refuges near the Ewaso Nyiro River (Kalama, Westgate, Oldonyiro, Nasuulu, Nakuprat-Gotu Conservancies)

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CCB, CM2.1)

The project provided substantial economic benefits to community households, as measured by the Social CoMMs surveys of over 2000 households in 2014 and 2015 and again in 2020 (project). Between baseline and project surveys, a higher proportion of households reported increased income over the previous year and ownership of cattle but a lower incidence of illness. Climate change adaptation also increased, as the proportion of households with access to milk increased by 20% and time spent gathering water declined.

Ongoing SocialCOMMs surveys show the project provided benefits to women, as women-led households surveyed also showed the increases in income, ownership of cattle, and lower incidence of illness.

Community Group	Women and Poorest 25%
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Impact	% of households that claimed to benefit from implementation of grazing plans on home Conservancies and on other participating Conservancies
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	90% claimed grazing plan was “Helpful” or “Very Helpful” (Table 2).

Community Group	Women
Impact	% households whose income decreased in past year:
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Decreased from 55% in the baseline scenario to 26% in the project scenario (Table 2)

Community Group	Poorest 25%
Impact	Employment
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Employment among poorest 25% increased from 5% in the baseline scenario to 12% in the project scenario (Table 2)

Community Group	Women
Impact	% of households headed by women requiring less than 2 hours/day to acquire water
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Increased from 23.0% in the baseline to 68.9% in the project scenario

Community Group	Women
Impact	% of households headed by women that consumed milk in the previous week
Type of Benefit/Cost/Risk	Actual Benefit
Change in Well-being	Increased from 45.5% to 63.8%

Community Group	Women
Impact	% Households with livestock
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Increased from 68.4% in the baseline scenario to 80.8% in the project scenario (Table 2).

Community Group	Women
Impact	% Households headed by women that found grazing lands to be good or very good
Type of Benefit/Cost/Risk	Actual benefit
Change in Well-being	Increased from 38.3% in the baseline scenario to 68.5% in the project scenario (Table 2)

4.1.2 Negative Community Impact Mitigation (VCS, 3.19; CCB, CM2.2)

1. More migration with livestock in order to increase rest may expose herders to encounters with antagonistic ethnic groups or livestock theft.
Mitigation: rapid response teams, anticipatory shared grazing agreements and peace meetings, regional grazing plans, and better communication between grazing coordinators and game scouts to facilitate conflict resolution.
2. Family members away from home for longer time periods.
Mitigation: Many herders are already away from home for up to 9 months of the year in response to drought and low forage availability. Proper management, with accompanying recovery of vegetation on home Conservancies, may actually increase time at home.
3. Carbon revenues may be converted into more livestock, creating little net positive benefit.
Mitigation Conservancy revenue distributions go mostly for public goods and are decided at open community meetings. There will likely be little personal return of cash, and thus capital to buy more livestock, to individual households.

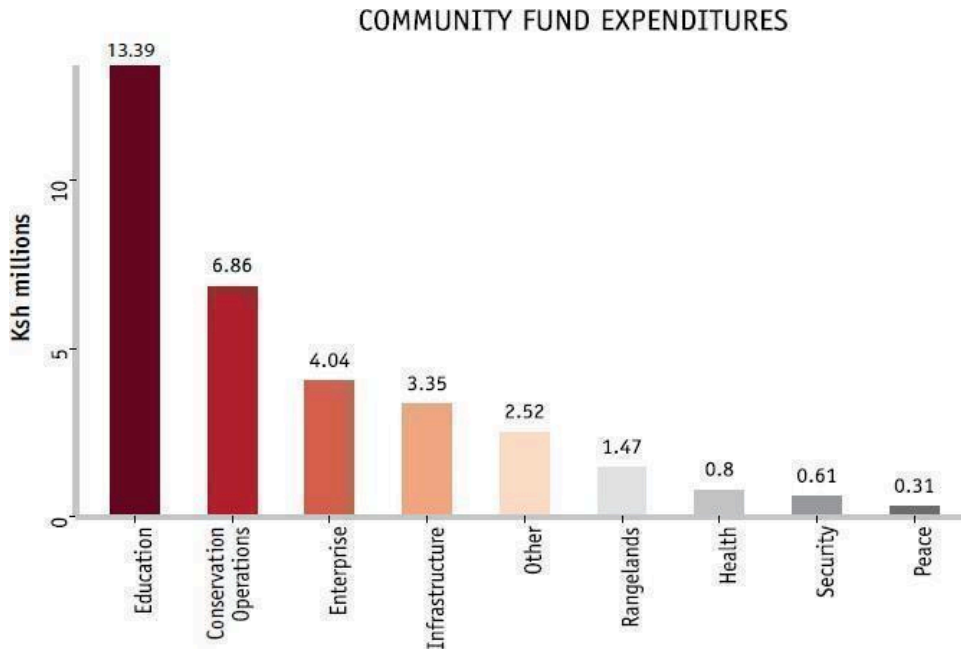


Figure 3. Expenditures for the NRT Pooled Conservancy Fund, showing the majority of allocation for education, including payments from bursaries for school fees for the poorest community members. The majority of the rest is allocated to conservation operations, new enterprises, and infrastructure.

These benefits extend in many different positive ways to the poorest 25% (largely without livestock), women heads of household, young herders (morani) or other young men and women from poor households that may lack capital for livestock but whose education (funded by bursaries) may make them employable. The benefits extend to communities in Conservancies throughout the project area and thus help people from multiple indigenous ethnic groups.

4.1.3 Net Positive Community Well-Being (VCS 3.19; CCB, CM2.3, GL1.4)

The project introduces multiple positive community benefits. However, insufficient time has passed since the project start to the current crediting period for most of these benefits to be realized, but here several positive indicators for key population segments indicating exceptional community benefits are shown.

First, Livestock ownership provided a clear metric of the poorest 25% of people (Figure 3), and our household survey identified opinions of women, including households with female heads.

Benefits were classed as to whether they affected livelihood, health, or ability to adapt to climate change. Household survey questions generally are framed in terms of a choice of 2-4 alternatives from which respondents must choose. Consequently, data were analyzed

according to the proportion of households choosing one or more responses to specific questions. Livelihood indicators were the proportion of households that (1) increased in income over the past year and (2) owned cattle. Health indicators were the proportion of households in which (3) an adult or (4) a child had been sick in the past month. Adaptation indicators were the proportion of households that (5) found the grazing program very helpful, (6) judged the quality of grazing lands to be “good” or “Very Good,” (7) drank milk the previous day, (8) found security to be “Improved” versus “Stayed the Same” or “Worsened,” and (9) spent less than two hours/day gathering water.

These assessments were based on household interviews at 2663 households in 8 different Conservancies during 2014 and 2015 (Baseline), and 2,637 households across five Conservancies in 2020 (Project). The results (Table 2) showed that overwhelmingly, across all households, the project activity was associated with dramatic improvements in all three classes of benefits.

Benefits to Women

Table 2 also clearly supports the idea that women benefit in multiple ways under project activities. Most strikingly, female-led households report increases in family income much more frequently, perceive grazing lands as “Good” or “Very Good” more often, and perceive improved security in 80% of cases relative to about a third of households under Baseline conditions. Interestingly they demonstrate larger increases in perceived security than men, but men-led households experience sickness less frequently and show larger increases in perceived quality of grazing lands and access to milk under project activities. Overall, the results are highly consistent with strong exceptional benefits to women.

Table 2. Proportions of households receiving benefits in the Northern Kenya Grasslands Carbon Project in 2014 and 2015 (Baseline conditions) and in 2020 (Project conditions) for all, female-led, and male-led households. Bold numbers indicate significant changes between baseline and project survey respondents.

Head of Household		All		Female		Male	
Number of Households		2663	2637	609	756	2054	1881
Time		Baseline	Project	Baseline	Project	Baseline	Project
Livelihood	Greater Income	30.3%	52.4%	29.2%	51.9%	30.6%	52.6%
	Own cattle	75.1%	86.2%	68.4%	80.8%	77.1%	88.4%
Health	Adult Illness	49.3%	28.9%	28.9%	29.0%	55.3%	28.9%
	Child Illness	25.3%	19.3%	24.3%	19.3%	25.6%	19.3%
Adaptation	Grazing Program Helpful	62.0%	65.6%	64.0%	63.6%	61.4%	66.3%
	Grazing Lands Good or Very Good	33.0%	62.4%	38.3%	61.5%	24.3%	62.7%
	Access to Milk	36.8%	69.3%	45.5%	64.8%	34.2%	71.1%
	Security Improved	37.1%	73.2%	33.0%	80.7%	38.3%	70.1%
	Hours to Water < 2	23.0%	68.9%	21.9%	68.5%	23.3%	69.0%

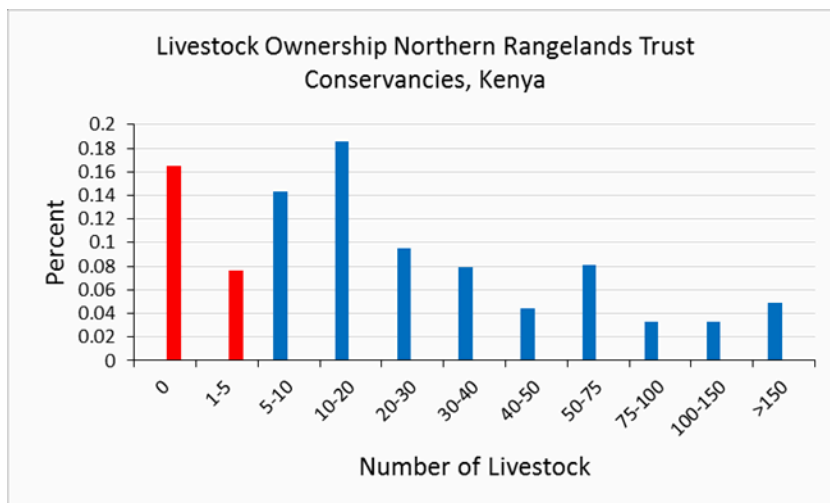


Figure 4. Livestock ownership distribution within four Conservancies in the project area. Red bars indicate the cumulative 25% of households with fewer than 5 livestock and no cattle that comprise the 25% poorest people in the project communities.

Second, the project scenario conveyed numerous financial benefits. Project Conservancies (> 2 years history with a grazing plan) employed 30% more people in total, employed twice the % of employees from vulnerable groups (Figure 6) than Baseline Conservancies (grazing plan in effect < 1 year). There was essentially no difference in employment of women between Baseline and Project Conservancies. Project Conservancies had fewer people than Baseline Conservancies and experienced lower income than the previous year (Figure 6) and a greater proportion of households that experienced increased income (Table 2). Project Conservancies also featured a significantly lower proportion of households that thought security had improved or stayed the same than Baseline Conservancies (Figure 5). Households in communities under the project scenario drank milk at a higher frequency than in the baseline scenario (Table 2, Figure 6) and were less likely to be sick.

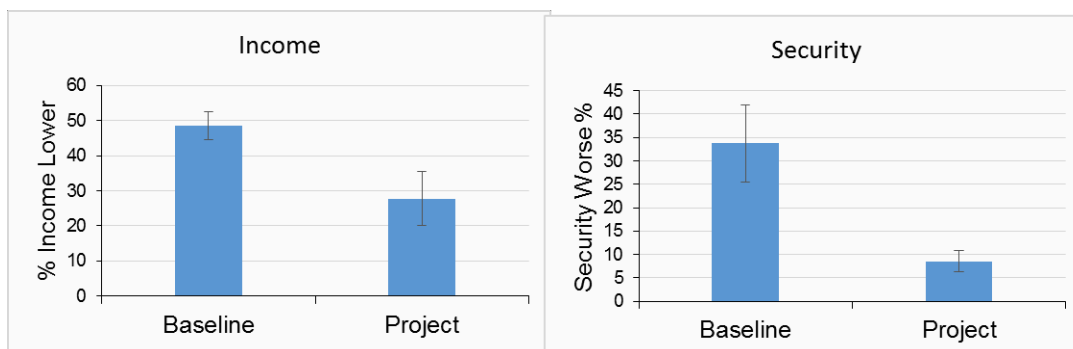
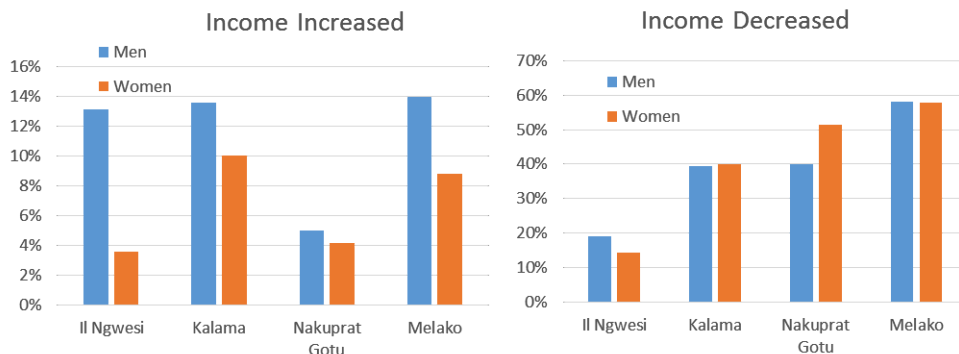


Figure 5. Comparison of benefits associated with Baseline (grazing plan in first year) and Project (Grazing plan > 2 years): A. percent that experienced lower income in the past 12 months, B. Percent that experienced reduced security in the past 12 months.

Financial Capital



Human Capital and Health

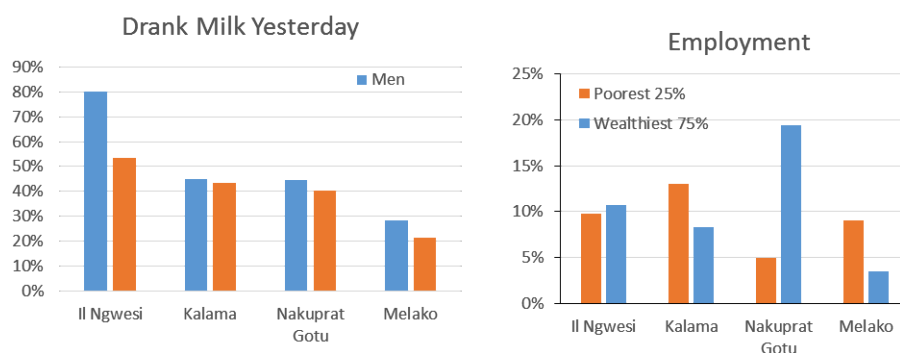


Figure 6. Economic and health indicators for **Project** Conservancies (Il Ngwesi, Kalama) with a longer history of planned grazing) versus **Baseline** (< 1 year with a grazing plan) (Nakuprat-Gotu, Melako) in the Northern Kenya Grasslands project area.

Finally, response of communities to planned grazing is overwhelmingly positive, as the vast majority (> 65%) of households find their Conservancy's grazing plan "very helpful" as compared to "not helpful," "somewhat helpful," or "don't know" (Figure 7). This positivity was expressed across households with men or women as heads, among the poorest 25% (largely not cattle owners) and owners of more than 5 livestock, and among people from Conservancies with a longer history of planned grazing versus not.

All these data indicate some positive trends that have accompanied the onset of project activities. Future data collected across additional Conservancies will allow a better assignment of cause and effect than is possible now, but the results suggest that the project activities are providing some social benefits, even as the resource benefits in livestock and grass await better weather conditions with higher rainfall.

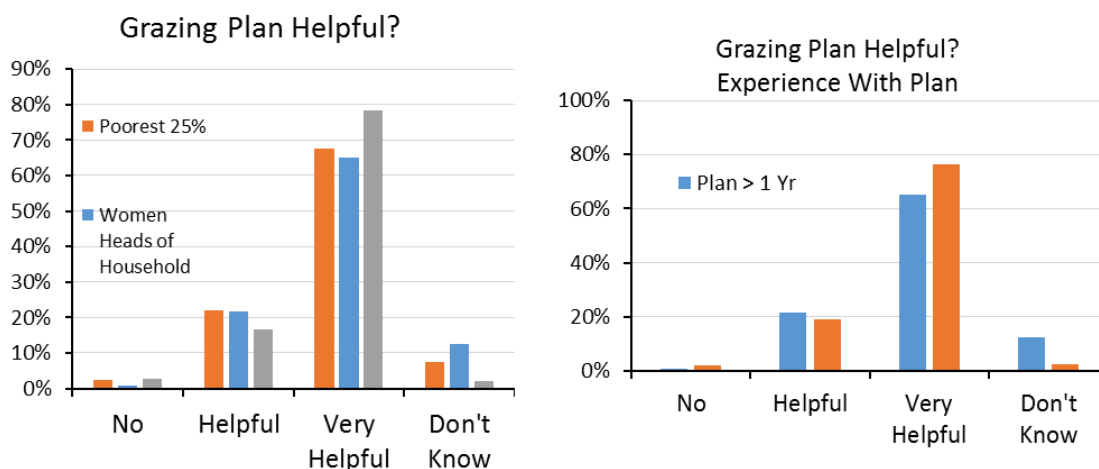


Figure 7. Responses to the question “Is the grazing plan helpful?” across different key groups of households in the project area. Data indicate overwhelming positive support for the project activities across the population at large and for vulnerable groups (women and poorest 25%). Enthusiasm for the grazing plan was not significantly different between **Baseline** (Plan < 1 Yr, blue bars) and **Project** (orange bars).

Anticipated climate change in the project zone is somewhat uncertain, as climate models differ in their predictions of whether mean annual rainfall will increase^[26] or decrease^[27]. However, climate variability is expected to increase regardless^[27, 28]. Increased variability and warmer conditions are expected to increase the frequency and intensity of droughts, shrink the length of the season soil moisture is available, reduce overall mean forage production, and potentially reduce livestock capacity by 10%^[26, 29, 30]. Assuming that temporal variability extends to spatial variability, climate change may increase the frequency of periods where pastoralists are in conflict over water or forage^[27, 31]

The rebuilding of soil organic carbon from the project activities of widespread PR grazing practices will likely induce dramatic improvements in the ability of people in the project area to adapt to climate change. Recent climate trends in the region^[32, 33] suggest a reduction in mean precipitation, with longer periods of consecutive years with either above- or below-average precipitation. The principal defense for pastoralist communities against such projected changes is to build heterogeneity in grazing use of the landscape so as to create opportunities for “chasing” spatially and temporally variable precipitation within a season and the establishment of grass banks as a hedge against drought. The PR grazing practices are fully expected to restore perennial grass cover and increase soil organic carbon. The occurrence of greater soil carbon across the project area would result in the following aids in climate adaptation

1. Increased infiltration and retention of limited precipitation in shallow soil layers^[34, 35].
2. Green, higher quality forage available for more months during dry seasons/years.

3. Reproduction by livestock will extend further into the dry season or occur during dryer years, which will supply milk to families year-round^[4, 36]. Baseline social surveys indicate that milk available for household use or to sell is available in communities.

Figure 8. Climate adaptation benefits A. Mean ($\pm$ SE) liters of milk sold per household under **Project** Conditions (grazing plans implemented for ≥ 2 years, $N = 5$ Conservancies) compared to **Baseline** conditions ($N = 4$ Conservancies, > 250 households interviewed/ Conservancy) (t-test, $df = 7$, $P < 0.003$). B. **Project** conditions produce significantly shorter mean ($\pm$ SE) travel distances to water (t-test, $df = 7$, $P=0.03$).

Grazing and Dry Season Soil Water Content, Idaho USA (Weber & Gokhale 2011)

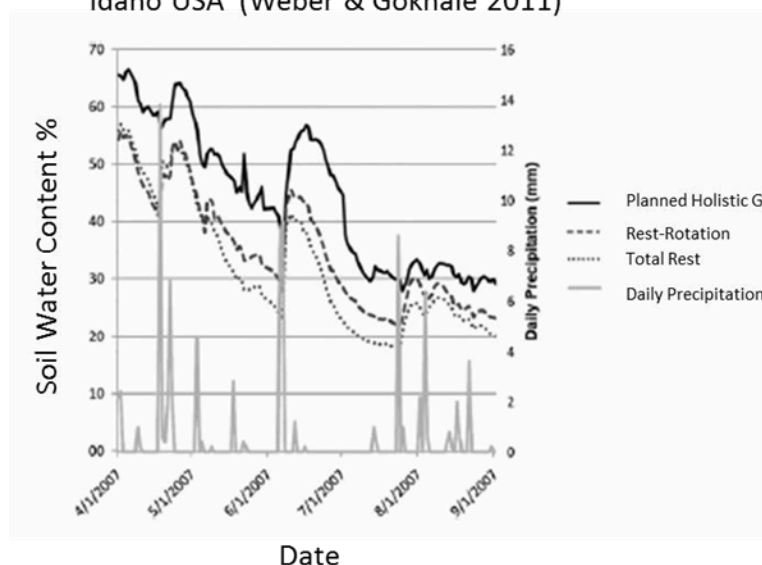


Figure 9. Soil water content to a depth of 1 m in areas receiving different livestock grazing management during the growing season in a temperate shrub steppe^[37]. Note the persistently higher water content in planned “holistic” rotational grazing treatments as compared to total rest or limited rest-rotation management.

4. Greater retention of water in shallow layers due to higher soil carbon^[37-39] (Figure 9) supports the advent of standing water in a greater number of places across the landscape to support cattle, which have a greater dependence on water than smaller livestock, particularly goats. These effects are evident even in tropical savannas that form hardpan surfaces following degradation^[40, 41]
5. Greater standing biomass and water in the dry season allows greater allocation of livestock to cattle and away from sheep and goats.

Overall, the project activities should greatly enhance communities' ability to cope with climate change relative to their baseline condition.

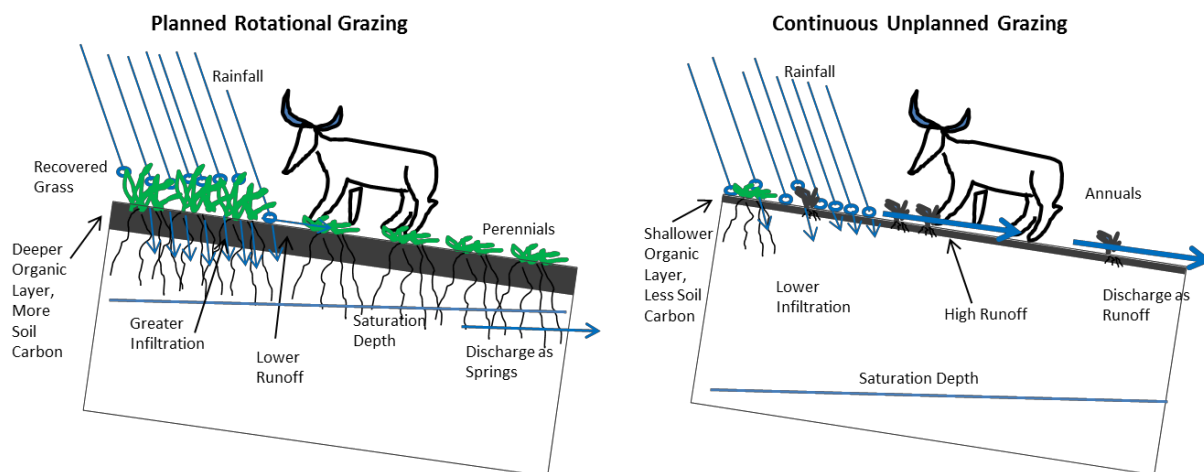


Figure 10. Diagrams to contrast the surface hydrology of rainfall in landscapes with planned rotational grazing (project activity) and continuous unplanned grazing (baseline activity).

4.1.4 Protection of High Conservation Values (CCB, CM2.4)

Implementation of planned grazing has not increased and in some cases has even reduced incursions of project livestock into HCV areas. For example, satellite-based assessments of livestock impacts using MODIS imagery and calculations of NDVI show that project activities reversed a trend of increasing use of the Mathews by livestock during the baseline period (Figure 11).

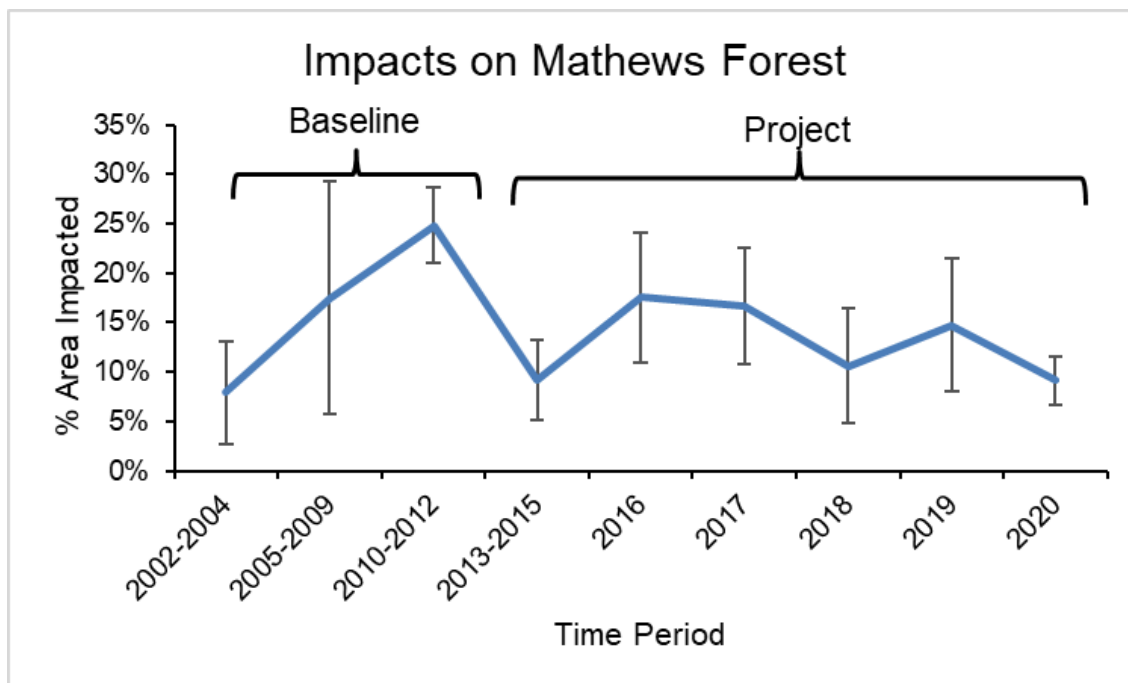


Figure 11. Mean ($\pm$ SE) extent of area experiencing net negative impacts ($SUM\Delta NDVI > 0.5$) of livestock in the Mathews Forest, averaged over different time periods. Note the trend of increasing impact on the Forest over time during the Baseline period followed by a dramatic reduction to near 2002-2004 levels under the project activities. These data likely reflect increased awareness of the need for herders to avoid the Mathews among grazing committees and managers in Namunyak Conservancy (which surrounds the Mathews) and more rapid movements through the Forest in grazing plans.

Livestock monitoring data from the Wildlife CoMMs data covering 2011-2012 (Baseline) and the initial monitoring period of 2013 – 2016 (Project) already indicate a clear reversal of trend in reducing encroachment into the Mathews Forest (Figure 12). Even in a drought year, when encroachment is usually at its worst, livestock produced negative impacts on only 20% of the Mathews Forest in 2014 (Figure 12). Impacts during the project (e.g., 2014) have mostly occurred on the edges where degradation has already occurred, compared to over 30% of the forest negatively impacted, and mostly in the high biodiversity rainforests at high elevation, that occurred in 2008, a comparatively wet year (Figure 11). As rotational practices continue and grass banks are established across a wide range of locations in the project area, encroachment should decrease further, thus addressing the number one threat to the Mathews Forest.

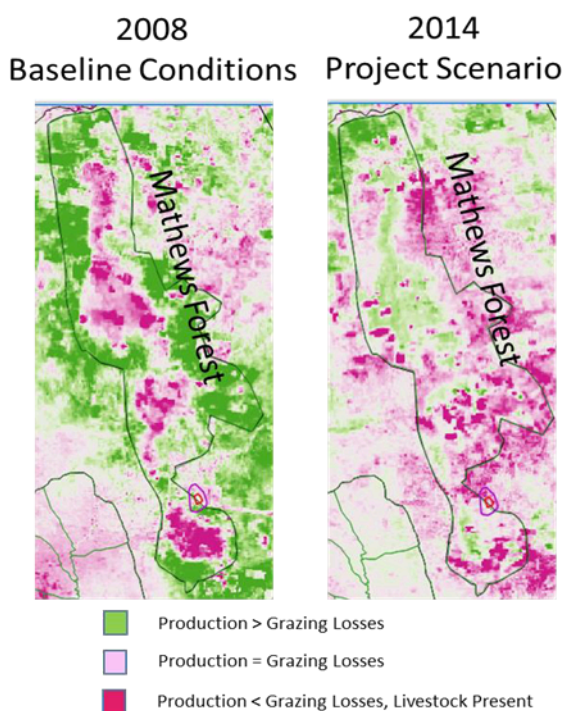


Figure 12. Satellite based monitoring of net annual grazing impacts, using the sum of monthly $\Delta NDVI$ (see section 5.4.3) on the Mathews Forest, comparing those in a wet year (2008), conservatively representing baseline conditions, and those in drought year under a preliminary regional grazing plan to mitigate effects of drought in western side of the project area. Note that impacts in 2014 were mostly along the edge of the forest, while impacts in 2008 were concentrated in the interior high elevation rainforest areas of the Mathews.

4.2 Other Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

The principal offsite stakeholders are herders from other Group Ranches or Trust lands, and these stakeholders may be negatively impacted by the enforcement of project boundaries and the perception that traditional use of other communities' grazing lands during droughts or dry seasons is no longer permitted. A key mission in designing planned rotational grazing will be to continue to have negotiated agreements among grazing committees of different Conservancies or communities both on and off the project area. These agreements will allow continuance of cultural traditions while maintaining the integrity of the project. They also may enhance social cohesion because acceptance of off-project livestock will be easier once sufficient grass banks are established.

To this end, NRT has established a **regional grazing plan** to manage movements and

numbers of livestock across Conservancies during droughts or extended dry seasons. This plan uses near real-time information on rainfall in the previous 1-3 months from USDA-USAID Famine Early Warning System (FEWS) <http://earlywarning.usgs.gov/fews/> and from the University of Reading's (UK) Tropical Applications of Meteorology using SATellite data, or TAMSAT <http://www.met.reading.ac.uk/~tamsat/data/>, to identify areas experiencing drier and/or wetter than normal conditions. From this information system, a large-scale inter-conservancy rotation of livestock can be designed that eases grazing pressure in drier areas while sustainably using forage available within the project area. This regional grazing plan can also accommodate movements of herders from outside the project area as a way to limit encroachment and better manage leakage.

4.2.2 Net Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

Community feedback

The carbon project, whereby yet another benefit of planned rotational grazing is presented, has been communicated much more recently due to high uncertainty in the carbon market and a reluctance to raise hopes about potential revenues. However, the Conservancies began on 15 May 2015 and then again on 15 August 2016, a solicitation of participation and public comment period, with announcement and information about the project distributed by NRT grazing coordinators and other staff in the form of pamphlets, and a posted announcement on the NRT website (<http://nrt-kenya.org>). Responses to the proposed carbon project were made by cell phone text to a standing repository, which was then downloaded.

These announcements were followed by community meetings in September and October 2016, open to all members in each Conservancy, in each of the 13 participating Conservancies. Meeting times and dates were posted at all Conservancy headquarters and were communicated verbally in villages in each Conservancy by NRT Social CoMMs supervisors.

In 2021, Annual Governance Meetings were held with representatives from each project conservancy. In 2021, a Carbon Project Oversight Committee was formed from conservancy representatives and meetings were held to focus on developing additional internal governance procedures based on community feedback.

Information about the project's participation in the voluntary carbon market (VCM) inclusive of all aspects of the project (costs, risks, benefits and requirements) was communicated with stakeholders through signed agreements with conservancies including, consent and waiver forms and letters of intent. Both of these agreements include language on carbon credits and anticipated revenue including, "we intend to participate in these actions, with the expectation of revenues from the sale of carbon credits, for at least the next 30 years." Examples of these signed agreements will be provided to the VVB for review. As the project was not validated until 2020 and verified/issuing credits until late 2021/early 2022, there was uncertainty if the project would take off prior to 2020 and thus the delayed development of the benefit sharing mechanism until 2022.

Adaptation to community impacts

In meetings, community grazing committee chairmen communicated how grazing plans had been implemented or not in the past and reviewed past compliance with grazing plans, identified barriers to compliance, and communicated expected benefits of project activities as rotational or migratory grazing. The response of communities, especially after September 2016 meetings and workshops, were improved efforts to make, write, and implement grazing plans within Conservancies. This provides evidence that community members understood the benefits of project activities and approved intensified efforts to conduct grazing plans.

Evidence from social surveys indicate that a majority of members thought that they or the community were benefited by the implementation of grazing plans. These baseline data were, and future data will be, collected through NRT Social CoMMs surveys and focus groups. SocialCoMMs is an outgrowth of training of community members to conduct social surveys by professional social scientist Dr. Jessica Musengezi of The Nature Conservancy, supported by permanent NRT staff. Through the guidance of TNC and NRT personnel, focus groups are chosen by community chairmen to represent the public, rather than by the grazing committee. Household surveys are conducted at randomly selected households, and every effort is made to interview at least the matriarch of each household.

As an example of how this worked, our baseline SocialCoMMs survey in 2014 identified the risk of encroachment onto Conservancies as the number one concern. Implementation of grazing management training and education, which at that point was the main emphasized activity, was reduced in effort, and effort was increased to work with grazing coordinators, grazing chairmen, Conservancy managers and community chairmen to resolve disputes due to encroachment from off-project herders. More effort was also applied to developing a regional grazing plan to allow herders from within-project Conservancies to use grazing lands from other within-project Conservancies. This shift allowed NRT to have a rapid response to conflicts between herders from different ethnic groups and avoid fatalities that occurred under similar circumstances in the same period in 2014. The formation of a regional grazing plan also facilitated discussions of County-wide grazing plans in October of 2015, which delivered information to and coordinated activities. Since 2017, quarterly grazing committee meetings have been held to discuss, review and plan for rangeland matters. In addition, grazing committees have continuously been trained on rangeland management and their roles as rangeland committee members.

In 2021, all net impacts on other stakeholders continued to be tracked through annual Social CoMMs reports. Project Social CoMMs results are shared with the respective conservancy manager. Conservancy managers are tasked with discussing the Social CoMMs results with their conservancy board. The Social CoMMs results are also fed into 5-year Conservancy Management and Development Plans aimed at linking results to socio-economics and social development projects such as access to water, education centres and healthcare.

4.3 Community Impact Monitoring

4.3.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The key variables determined from the three types of surveys reflect specific aspects of

financial, human, natural, physical, and social capital that may be influenced by the carbon project. Some very important variables to consider for NRT and the NKCP are given below. Social data collection involves gathering information from approximately 5 groups of key informants from Conservancies with representation from women, elders, and morani and at least 10-15 focus groups [2 or 3 in each of five Conservancies] from within the entire project area.

Key informants were identified from existing NRT administrative conduits, including village chairmen, grazing chairmen, and community public health professionals (if available). In addition key informants included community elders from various community committees, such as grazing and settlement allocation committees. Typically at least five key informants were interviewed from each Conservancy. Interview questions were similar in theme to those in household surveys, except that informants were asked to reflect their opinion of community-wide issues and needs.

Focus groups were conducted by extending a general invitation within each of 3-4 villages (100-500 households in size) within each Conservancy to attend a meeting 5-10 days after the initial announcement in each village to engage in open discussions of issues related to well-being and grazing, the core project activity. In addition, village chairmen were asked to identify and extend invitations to individuals from vulnerable and marginalized groups, in particular women and the poorest 25% (nearly always single head-of-household households with few or now livestock)

Household Surveys

The goal was a sample size sufficient to demonstrate significant change over some period of time during the implementation of the project. Given typical standard errors in human data of 10-20% and frequent non-normal distributions, at least 25 households is necessary to ensure a sample large enough to successfully compare different subgroups^[42]. This sample size was selected based on reducing the coefficient of variation to less than 30%, based on standard deviations in responses from Louise Glew's thesis^[1] comparing socio-economic results from communities with and without Conservancies. This sample size should allow 15% improvement in conditions during the project crediting period to be found statistically significant ($P < 0.05$).

Sufficient effort will be made to include some of the same individuals upon re-sampling (samples taken after the start of the project) in order to reduce sampling error.

Surveys (N=1030) were conducted in November and December in 2015 in each of four Conservancies. Two Conservancies had been implementing grazing plans since 2013 (Kalama and Il Ngwesi) and represented the project scenario. The other two Conservancies (Melako, Nakuprat-Gotu) had just been organized and were still developing grazing plans and represented the baseline scenario. These Conservancies were selected because they encompassed most of the ethnic groups represented in the project area (Il Ngwesi, Maasai; Kalama, Samburu; Nakuprat-Gotu, Turkana, Pokot, and Borana; Melako, Rendille). Village chairmen and Conservancy managers were contacted and interviewed to identify key stakeholders, women, and underrepresented groups for focus groups and to plan the random selection of households for household surveys. Details of survey questions are

provided below in subsection 4. Outcomes of surveys are summarized in section 4.1.3.

Focus Groups: Wealth or well-being ranking of local or community stakeholders

The tendency in most project types is for 'elite capture' and for the poorest to lose out, as has happened with the majority of community forestry experiences (McDermott & Schreckenbergs 2009). A wealth ranking exercise should also lead to a better understanding of local perceptions of well-being and poverty, and can also generate a useful sampling frame for household surveys. As described by PROFOR (2008) a wealth or well-being ranking exercise involves the following stages:

- Undertake a community mapping exercise to identify and list all households
- Write down the names of the household heads on cards, one per card
- Consult with key informants and focus groups on poverty or well-being categories. In the case of the Nepal Swiss Community Forestry Project (PROFOR, 2008), this resulted in six wealth categories: capable, improving poor, coping poor, declining poor, extreme poor and incapable poor. In other studies, the number of months of household food reserve is a common wealth 'numeraire' (note: this stage could be undertaken first)
- Select 'representative', respected and knowledgeable key informants, including some women
- Get the key informants to classify the cards (households) into the wealth or well-being categories .

Use of Venn Diagrams for Stakeholder Analysis

Participants firstly cut three sizes of circles – at least two sets of circles using different colored cards. One color is for 'insider stakeholders' and another is for 'outsider stakeholders.' For each 'outsider stakeholder', the participants need to decide the importance of their involvement in the project, and select the corresponding size of circle:

- * Little importance = small circle
- * Some or significant importance = middle sized circle
- * Very important = large circle

The name of the 'insider' stakeholder or stakeholder groups is then written on the appropriate sized circles/cards. This is repeated for all 'insider' stakeholders using the other colored card. When all stakeholders have been represented, the circles are organized and stuck to a flipchart, grouping and placing the circles according to the relationships between the stakeholders: the closer the relationship between two stakeholders, the closer the circles should be on the flipchart.

The next stage is to cut three sizes of triangles from different colored cards. For each stakeholder (group), a small, medium or large triangle is chosen to represent the degree of influence that the stakeholder has on the project. The triangle is then stuck on the edge of the stakeholder circle. A stakeholder with a small 'importance circle' could have a large

'influence triangle' and vice versa. The overlap of the circles represents the extent of the relationship between stakeholders (see Figure 5). Once the diagram is complete, it is reviewed by the wider group, which should continue to discuss the relative importance and influence of each stakeholder or stakeholder group until a consensus is reached.

Conduct of Household Surveys

Surveyors use the following introductory script:

Hello, my name is and I am involved in a project sponsored by the Northern Rangelands Trust to investigate the livelihoods and well-being of people who live in this conservancy. The goal is to better understand the impact of the conservancy on your daily life.

<ASK TO SPEAK TO THE HOUSEHOLD HEAD, OR IF THE HOUSEHOLD HEAD IS NOT THERE OR BUSY, ASK TO SPEAK TO ANOTHER ADULT MEMBER OF THE HOUSEHOLD. THE PERSON INTERVIEWED MUST BE AT LEAST 18 YEARS OLD.>

I would like to ask a number of questions related to you and your household. The interview will last about 30 minutes. Please note that all the information you provide will be kept confidential and will not be shared with anyone else. The information will only be used to characterize the area in which you live. All individual information will be added together to determine the average for the whole community, so nobody will be able to identify individual participants. There are no risks from participating and there are no right or wrong answers. It is your opinion that is important to us. Do you have any questions?

Thank-you.

Can we find a quiet and comfortable place where I can ask you the questions?

Questions will be asked to both heads of household (mostly men) as well as head women of households so as to monitor project impacts on women as well as different groups of men.

Twelve indicators evaluated project impacts on women:

1. Have you been employed by the Conservancy in the past 12 months
2. Do you feel safe enough?
3. Does the Conservancy have a grazing plan?
4. Is the grazing plan helpful?
5. How has your household income changed in the last 12 months (1=decrease, 2 = stay the same, 3= increased)
6. How far to travel to water?
7. What were sales of livestock products in the past month?
8. Did anyone in the household sell milk yesterday?
9. Did anyone drink milk yesterday?
10. Did anyone in the household eat meat yesterday?
11. Were any adults sick?

12. Were any children sick?

These indicators were chosen to compare project scenario with no project scenario (prior to implementing project activities) on economics (questions 1, 5, 7), security (2), access to resources (6, 8-10), household health (11, 12), and recognition of benefits of the project activity (3, 4).

For assessing past livestock numbers, the methodology VM0032 requires surveys of at least 30 households, or 10% of households, whichever is greater, in a Conservancy. Surveys, conducted in October-November 2014, solicited the number and kind of livestock households owned, how these are grouped with others or otherwise managed to achieve rotational grazing, and their locations throughout the year. In addition, heads of household were asked for the number and kind of livestock owned in 2004, 2005, 2007, 2008, 2009, and 2011. Interviews were conducted by rangeland coordinators of livestock owners, typically while out herding livestock, and so did not include households without livestock. This bias was corrected by determining the fraction of households without livestock from the general household surveys, and appropriately discounting estimates of livestock numbers.

Livestock numbers for each Conservancy were obtained from monthly grazing reports by rangeland coordinators during 2014, 2015, and 2016 to provide project livestock numbers for the purpose of estimating leakage and methane emissions. For Conservancies where household surveys were conducted during any of these years, livestock numbers were estimated by multiplying livestock per household by number of households and corroborated against rangeland coordinator reports for that Conservancy.

In total, survey results indicated that exceptional community benefits have already occurred under the project scenario as income, security, and access to milk and water for multiple stakeholder groups have increased relative to the baseline scenario (Section 4.1.3). Further benefits are expected to be realized relative to the baseline scenario as the project continues and more household surveys (post-2021) are carried out. In addition to Social CoMMs surveys, an annual impact survey will be conducted for the project beginning in 2024.

4.3.2 Monitoring Plan Dissemination (CCB, CM4.3)

Each year in September or early October, a Conservancy-wide Project Meeting is held and attended by Conservancy chairs, community grazing chairpersons, grazing coordinators, and the interested public. This meeting is announced on the NRT website and notices are posted in major community kiosks and markets. At the meeting, a report of the previous 12 months' monitoring results will be presented. Data included are livestock counts, leakage and distribution within the project area, any soil or vegetation or biodiversity surveys, and results of the previous Social CoMMs surveys. Following this presentation plans for the ensuing 12 months are discussed, with feedback from managers and the community welcomed. An Annual Report of the project, with a monitoring plan modified in response to community feedback from the Project Meeting and a summary written in Kiswahili, is posted on the NRT website. Links to the posted report are sent by text through the grazing management and grazing coordinator network and be communicated via email to other NGO and government stakeholders.

These meetings were held in September 2016 and each year subsequently, including in 2021.

The translation of the monitoring report to Kiswahili will be done after the monitoring report is approved and sent to Verra. This was done most recently for the 2013-2016 monitoring report that was translated to Kiswahili and thereafter shared as a monitoring report summary in 2021. Once the 2017-2020 monitoring report is approved and finalized (CCB verification pending), a monitoring report summary will be translated and sent to each conservancy.

4.4 Optional Criterion: Exceptional Community Benefits

The project activities will bring considerable community benefits, particularly to women and the poorest 25% of community members.

4.4.1 Short-term and Long-term Community Benefits (CCB, GL2.2)

As demonstrated in section 4.3, project activities have already helped increase employment and mitigate decreases in household revenue. Households have better food security in the form of higher and more consistent milk availability. These benefits have been demonstrated for marginalized groups (poorest 25% and women).

Longer-term benefits of improved forage, higher water tables and greater surface water availability have yet to be realized due to severe droughts in 2011, 2015 and 2017 that prevented substantial increases in grass cover during wetter years (2012, 2014) from persisting. However, the proportion of project area in which grazing of sites only once per year has remained above 80% (Figure 9), which suggests that recovery of forage and rangeland productivity is still expected.

4.4.2 Marginalized and/or Vulnerable Community Groups (CCB, GL2.4)

Community impacts of the project activities were monitored from surveys in 2014 and 2015 of more than 1000 households and were conducted by community members within their Conservancies, as part of NRT-assisted Social CoMMs activities. Data were compared between Conservancies with longer histories of implemented grazing plans (Project Scenario) and Conservancies that had just joined the project (Baseline Scenario). These household surveys contained multiple questions that pertained to the benefits discussed in section 4.1.1.

Community Group	Women and Poorest 25%
Net positive impacts	The project provided substantial economic benefits to community households, specifically women and the poorest 25% as measured by the Social CoMMs surveys of over

	2000 households in 2014 and 2015 and again in 2020 (project). Between baseline and project surveys, a higher proportion of households reported increased income over the previous year and ownership of cattle but lower incidence of illness. Climate change adaptation also increased, as the proportion of households with access to milk increased by 20% and time spent gathering water declined. Reference Table 2 for additional insights. In addition to Social CoMMs surveys, an annual impact survey will be conducted for the project beginning in 2024.
Benefit access	SocialCOMMs surveys showed the project provided benefits to women, including women-led households indicating increases in income, ownership of cattle, and lower incidence of illness. Finally, response of communities to planned grazing is overwhelmingly positive, as the vast majority (> 65%) of households find their Conservancy's grazing plan "very helpful" as compared to "not helpful," "somewhat helpful," or "don't know" (Figure 7). This positivity was expressed among the poorest 25% (largely not cattle owners).
Negative impacts	More migration with livestock in order to increase rest may expose herders to encounters with antagonistic ethnic groups or livestock theft. Mitigation: rapid response teams, anticipatory shared grazing agreements and peace meetings, regional grazing plans, and better communication between grazing coordinators and game scouts to facilitate conflict resolution. Family members away from home for longer time periods. Mitigation: Many herders are already away from home for up to 9 months of the year in response to drought and low forage availability. Proper management, with the accompanying recovery of vegetation on home Conservancies, may actually increase time at home. Carbon revenues may be converted into more livestock, creating little net positive benefit.



Mitigation Conservancy revenue distributions go mostly for public goods and are decided at open community meetings. There will likely be little personal return of cash, and thus capital to buy more livestock, to individual households.

4.4.3 Net Impacts on Women (CCB, GL2.5)

Sections 4.3 and 4.1.3 demonstrate that women benefit from the project in multiple ways, including increased household food security (milk available to drink and sell).

4.4.4 Benefit Sharing Mechanisms (CCB, GL2.6)

During this 2021 monitoring period, no revenues were generated from the project so there is no example of benefit sharing yet. However, in anticipation of revenue generation in 2022, processes were developed in 2021. Revenue will be disbursed to conservancies in each of the years 2022, 2023, and 2024. These processes and subsequent agreements are detailed below.

As described widely in the project description document and subsequent monitoring reports, project benefits are:

1. Improved pasture for pastoralists and wildlife
2. Improved resilience to drought
3. Providing long-term, sustainable revenue from the ongoing generation of VERs from the project and the subsequent sale of the products on the international voluntary market
4. Improved governance and financial management capabilities

In relation to the provision of a long-term sustainable revenue stream, an extensive benefit sharing and distribution mechanism was designed with members of the participating conservancies, which began in May 2021²³. This was specific to the sharing of anticipated revenue from the project, which after many years of development, had now been both validated and the first issue of credits verified and available for sale.

In May and June 2021, NRT undertook a widespread briefing and socialization of the project for all participating conservancies and their member communities. NRT presented a revised and renewed Project Implementation Agreement²⁴ to set out a clear understanding between NRT and participating conservancies as the project launched the first issue of carbon credits.

²³ The project was validated in April 2020 and the first credits available for issuance in December 2020. The revenue share arrangements were only then designed and agreed with communities as revenue from the project became more likely - and over time, certain.

²⁴ The **Project Implementation Agreement** outlines the project activities involved and provides detail outlining the compliance and reporting requirements of the project. It provides a schedule of the reporting requirements a conservancy agrees to partake in and it includes detail on access for project monitoring, the validation process, verification, registration, records and other reporting relating to CCB Triple Gold, and sets out the agreed benefit sharing mechanism for revenue for the first three years of the project.

Crucially, the PIA also set out proposed draft benefit-sharing mechanisms for the first three years of the project (three years was determined as the first issue of credits from the project represented a crediting period from 2013–2016. Fewer credits were generated in 2013 as it was the first year of the project, and therefore the crediting period better represented a crediting period of three years. At the time, it was envisaged that distributing revenue on an annual basis reflective of the crediting period would help create a mechanism by which a sustainable revenue stream could be built over time, and subsequently participating conservancies and their communities would have visibility of the funds available to them in coming years.)

The PIA was socialized in a three-stage process:

Stage 1: As part of the community consultation process NRT held central briefings on the project for each conservancy's elected board chair, appointed manager, elected grazing committee chair, and rangelands coordinator. During these briefings, the proposed agreement was presented in English and Swahili and translated to local languages too; and all who were present had access to legal advice and the legal advocate who presented the draft agreement to them, line by line (again, in English and Swahili and local languages.) The legal advocate present at these briefings was present at the initial briefing, and at Stage 3 (see below) and was available to present advice to conservancies, and to answer any and all queries raised about the agreement.

Stage 2: These members then briefed both their respective Boards and Grazing committees, and following that, 'Special AGMs' were held at the village or zonal level across each of the conservancies, to inform the wider community and ensure that there was consensus within the community to continue participating in the project.

These meetings were extensively documented both through written minutes and visually recorded through photographs and in some places, film. Special 'AGM's held at the village or zonal level across each conservancy were considered the most appropriate way to socialize this specific information to as many community members as possible and for those community members to discuss the agreement and provide direct feedback to their Board and Grazing Committee representatives (minutes to be provided to VVB). In some conservancies, there may have been as many as 24 village/zonal meetings. Further, by hosting Special meetings, the agenda was very specific to ensure clear communication and messaging and additionally, hosting these meetings locally in each zone, coordinated by each conservancy's manager and board, was aimed at ensuring the highest possible level of participation from community members at that time.

Based on a strong consensus from those at the time, all communities approved for the conservancy chairperson to sign the Project Implementation Agreement on behalf of their community.

Stage 3: The 14 chairpersons (in fact now 21 chairpersons following the subdivided units in Naibunga, Namunyak, and Ol Donyiro) met on 24 June 2021 and formally signed the Project Implementation Agreement. The same legal advocate was also present and made the

requested changes based on community and conservancy feedback to the draft agreement prior to signing.

Please note that all conservancies had a hard copy of the PIA given to them after signing and we understand they have also been provided a soft copy via email from the NRT team. Supporting documentation for all engagements and agreements is available and will be provided to the VVB for review.

4.4.5 Governance and Implementation Structures (CCB, GL2.8)

The governance structure of the project that links NRT to community leaders to grazing committees and public hearings (Figure 2, section 2.3.10) allows households with few livestock, women, and underrepresented groups to participate in the formation of grazing plans and therefore in project activities. This participation is facilitated by community leaders who seek out and invite smallholders, women, and minorities to committee meetings.

4.4.6 Smallholders/Community Members Capacity Development (CCB, GL2.9)

The project helped build the capacity of smallholders, women, and minorities to participate in the project. Each year of the monitoring period, families from these groups provided personnel for project-related employment, including game scouts, soil and vegetation surveyors, social surveyors, and livestock herding. Community members continue to be employed each year of the monitoring period as processors of vegetation, wildlife, and social data. Education efforts by grazing coordinators and NRT and StfF personnel were provided to women and children in Kalama, Westgate, Meibae, Naibunga, and Melako Conservancies in 2014 and 2016. Additional education efforts were provided to women and children in multiple conservancies by grazing coordinators and NRT personnel in 2024.

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (VCS, 3.19; CCB, B2.1)

Change in Biodiversity	Increased populations of trigger species Grevy's zebra, elephant, beisa oryx, black rhino, gerenuk, and reticulated giraffe
Monitored Change	Heat maps of sightings of five of these species show expansion of distributions and 3-5 x higher sightings per year of five species (see section 5.1.3 below). NRT, Sera

	Conservancy, and the Kenya Wildlife Service have established a black rhino sanctuary in central Sera Conservancy.
Justification of Change	The change has occurred through greater protection by project-employed game scouts, who also help manage leakage and livestock incursions from outside the project area. Geographic centers of population counts in the project area are associated with areas of more successful implementation of grazing plans, such as central Sera, Kalama, and southern Melako Conservancies
Change in Biodiversity	Increased species richness and species diversity indices of savanna biodiversity indicator taxa, including dung beetles, grasshoppers, herbaceous plants, and birds
Monitored Change	Diversity indices and species counts were sampled at biodiversity camps
Justification of Change	The change has occurred through greater protection by project-employed game scouts, who also help manage leakage and livestock incursions from outside the project area. Geographic centers of population counts in the project area are associated with areas of more successful implementation of grazing plans, such as central Sera, Kalama, and southern Melako Conservancies

5.1.2 Mitigation Actions (VCS, 3.19; CCB, B2.3)

In the initial phases of the project, there is a strong need to create long periods of rest from grazing across large areas in order to allow some vegetation recovery. Given the widespread occurrence of degraded soil and vegetation conditions across the project area, herders, particularly from Namunyak Conservancy, may be tempted to move livestock into the Mathews Forest Reserve to cut limbs and small trees as forage for livestock. Chronic use of the forest for this purpose in the past has led to a reduction in forest cover and degradation, such as dominance of the forest by fast-growing early successional species²¹.

This potential negative impact should disappear as the project proceeds because successful planned rotational and migratory grazing should produce substantial grass banks that can be used to provide short-term rest and drought forage bases. These grazing options should greatly lessen the motivation for herders to enter forests and thus reduce potential impacts. Consequently, any potential negative effects should be temporary and produce little lasting effects on biodiversity. The impact of grazing management and movements of herders and

avoidance of protected areas when possible is evident in the net impacts of grazing on the Mathews Forest, a major contributor to the High Conservation Value of the project area (see section 7.3).

The lack of protection of wildlife and continued rangeland degradation outside the project zone may further deplete biodiversity found there relative to that found inside the project zone, where rangeland restoration, funding for community conservation and monitoring activities, and community-based protection and law enforcement have already increased wildlife numbers and biodiversity. Such a result represents an *unmitigated* negative effect of the project activity. Despite ready access to firearms, improved socioeconomic conditions associated with the project do not lead to increased wildlife poaching among the principal pastoralist groups, as their cultural traditions largely taboo the consumption of wild meat.

Unmitigated negative impacts on biodiversity outside Project Zone	Net Positive Biodiversity benefits with project
Degraded soil and vegetation from overgrazing and foraging of livestock	Implementing grazing plans using rotational grazing to reduce overgrazing and rebuild habitat for herbivores.
Wildlife poaching	Habitat improvements in the Sera Conservancy contributed to the establishment of a Rhino sanctuary by the Kenya Wildlife Service. Declines in elephant caresses found that resulted from illegal killing began with the start of the carbon project from 2012-2015.
Lack of wildlife protections	Game scouts reported sightings of Reticulated Giraffe, Grevy's Zebra, Beisa Oryx, wild dogs, and lions increased from 2012 to 2015 in Melako and Sera Conservancies as these conservancies began implementing planned grazing across full Conservancy in 2014. Game scouts reported sightings of Reticulated Giraffe, Grevy's Zebra, Beisa Oryx, wild dogs, and lions increased from 2012 to 205 in Melako and Sera Conservancies as these conservancies began implementing planned grazing across full Conservancy in 2014. Kenya Wildlife Service will address rhino conservation and Grevy's Zebra Trust will address zebra conservation. GZT specifically focuses on restoring water and forage for zebras.
Violent conflict for limited forage creating greater insecurity and furthering degradation of soil, vegetation, and wildlife poaching	Project activity includes creating and enforcing agreements to allow herders from outside the project area to graze inside the Project Zone when instructed to support the planned rotational grazing.

Minimal data gathered over time to show trends in species biodiversity	Project includes data gathered for wildlife counts of major species as well as species diversity of key indicator groups plants, dung beetles, trees, grasshoppers, and trees.
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Project activities follow the precautionary principle, i.e. the project does not introduce new products or processes whose ultimate effects are disputed or unknown. The project does not employ any new vegetation treatments or restoration procedures other than adjustment of animal husbandry.

5.1.3 Net Positive Biodiversity Impacts (CCB, B2.2, GL1.4)

Project activities that engage rangeland coordinators and game scouts on the landscape have led already to demonstrable biodiversity benefits. Black rhinos (*Dicornis bicerors*) have successfully been introduced into the Sera Conservancy in 2016 and remain dependent on sustained, diverse forages and safety from monitoring by engaged Conservancy members. Illegally killed elephants have sharply declined since the start of the project and three umbrella wildlife species that are endemic to northern Kenya and/or endangered are reticulated giraffe, Grevy's zebra, and beisa oryx, as discussed in section **5.4 Exceptional Biodiversity Benefits** below, have expanded their range and increased in density since the start of the project. These positive responses have occurred in part because Conservancy members are engaged across the wider Conservancy landscape in wildlife monitoring and shifting grazing activities. However, rangeland recovery associated with planned grazing appears to be important as well because the highest densities of wildlife occur in "Core" and Buffer" areas that previously experienced the most protection from grazing and typically were only grazed by livestock in the dry season. As rangeland vegetation recovers in settlement areas, wildlife populations should expand into those areas as well.

The project will not specifically undertake specific management actions to increase any specific component of biodiversity or endangered or vulnerable species. However, other stakeholders and partners with NRT, specifically the Kenya Wildlife Service in addressing rhino conservation and Grevy's Zebra Trust in addressing Grevy's zebra conservation will have their efforts augmented by project activities. Indeed GZT's priority management action for the past 8 years has been to improve livestock grazing practices to restore water and forage for zebras.

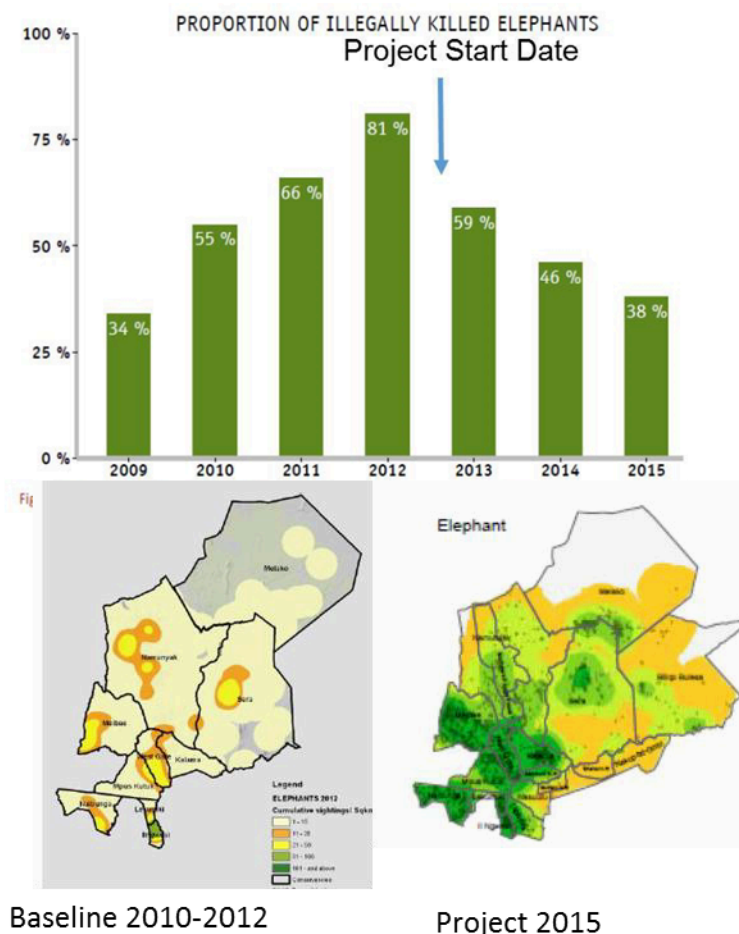


Figure 13. Upper: Proportion of elephant carcasses found that resulted from illegal killing (PIKE) from 2009 – 2015. Note that declines began following the start of the carbon project, again likely due to increased activity of game scouts and herders across the landscape in an effort to implement regional grazing plans in the dry season and to more fully utilize forage within the project area. Other background factors, such as a decline in ivory prices and better international enforcement likely also contribute to this trend. **Lower:** Distribution and abundance of elephant sightings on the NRT Conservancies, 2010-2013, under Baseline conditions compared to Project conditions (2015). Note the expansion of range (any elephants observed) and local density (shift from yellow to green) across the project zone. Note in particular the dramatic increases in numbers in the southern group of Conservancies and along the Melako-Sera Conservancy border.

The project benefits the High Conservation Values associated with the forest reserves and other protected areas adjacent to the project area. The anticipated increase in grazing options within the project area should reduce pressure to use protected areas for forage during dry seasons or droughts. Previous studies in the NRT or northern Kenya area^[1, 43-46] strongly suggest that planned rotational grazing will increase plant diversity^[47-50]. However, the impacts of rangeland restoration and soil carbon storage on other potential conservation values are poorly documented outside major, well-studied protected areas. Project activities that engage rangeland coordinators and game scouts on the landscape have led already to

the demonstrable biodiversity benefits discussed above. During the same period (2005-2015), reflecting baseline conditions, Ogotu et al^[51] report that virtually all species of Kenya wildlife even inside reserves have declined on average by 63%. This contrasts with the considerable increases in elephants (Figure 13) the three endangered species (Figure 14) observed in comparing baseline years 2009-2012 and with reports of wildlife from game scouts (Wildlife CoMMs) in 2015-2016 during the project period.

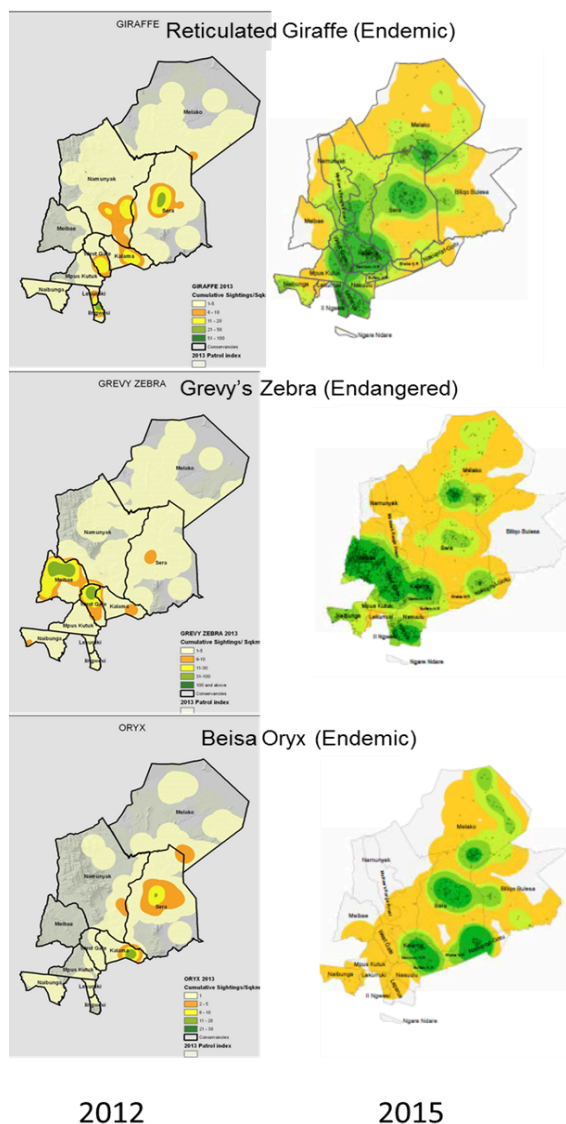
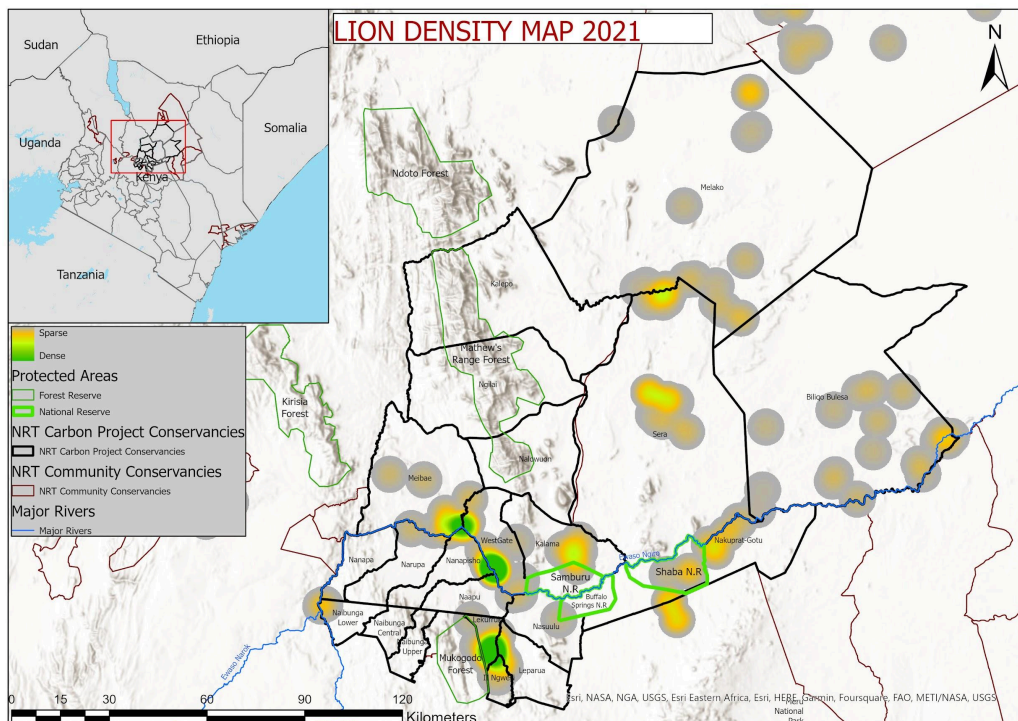
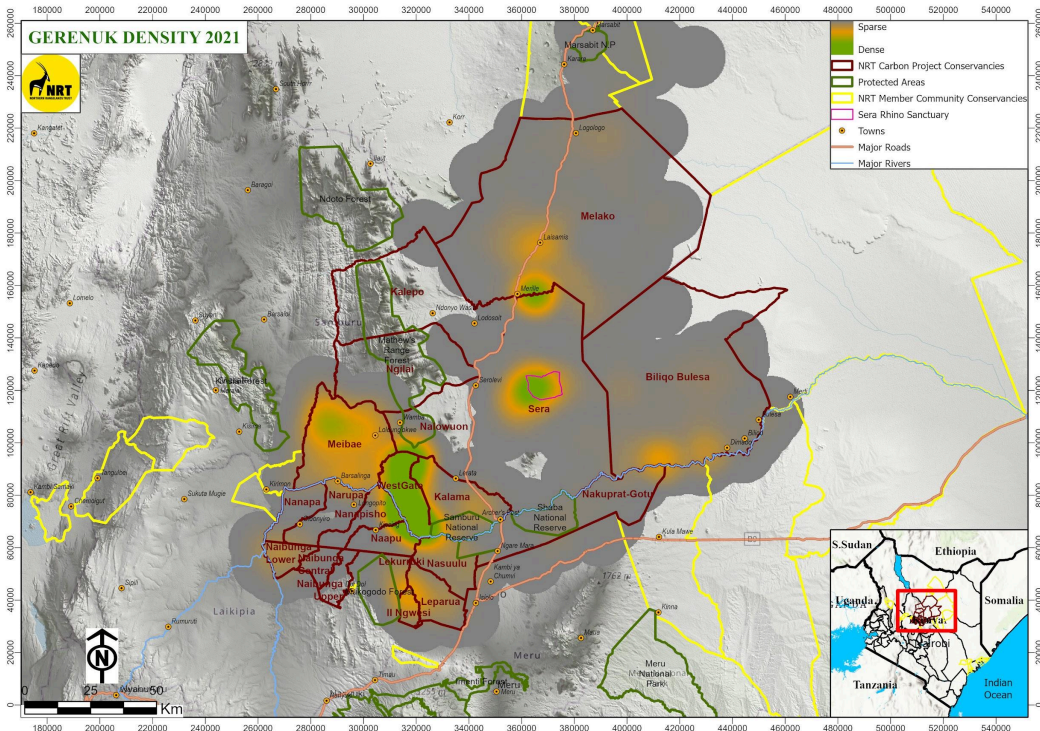
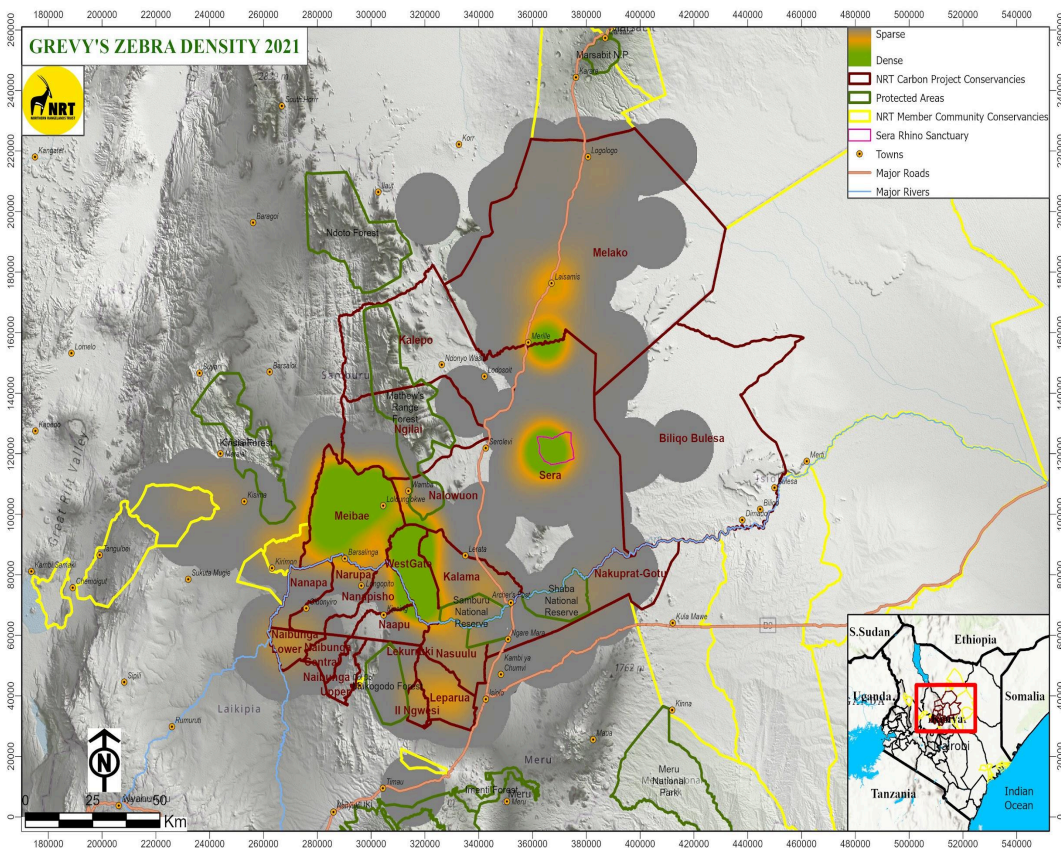
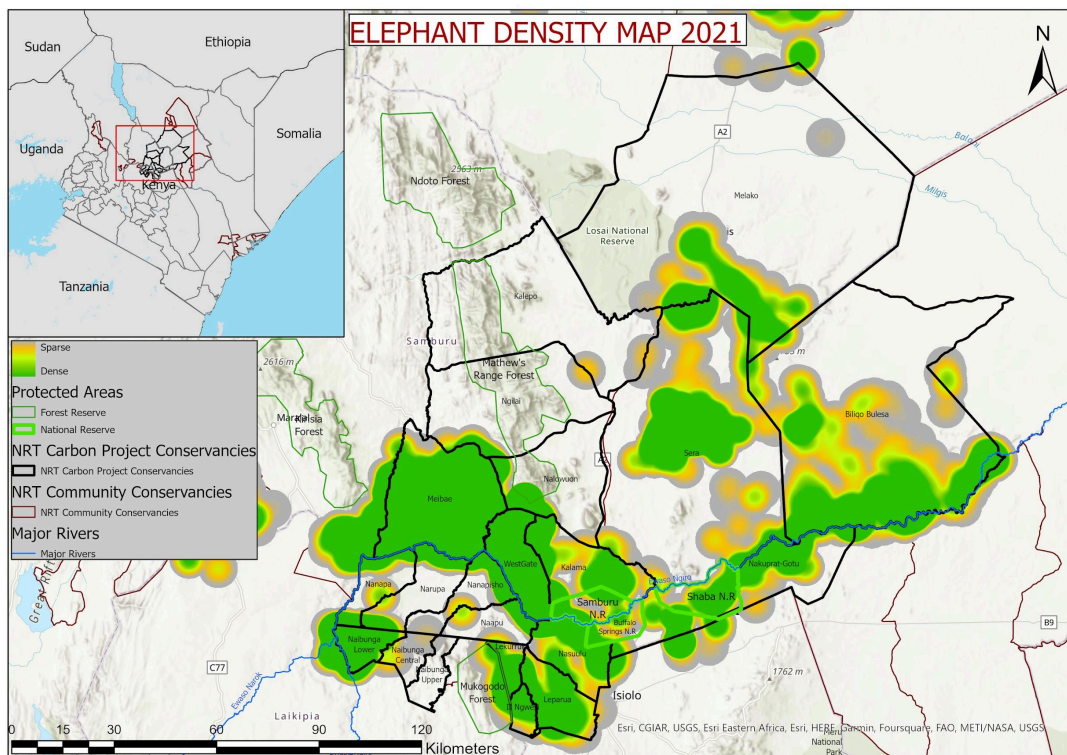


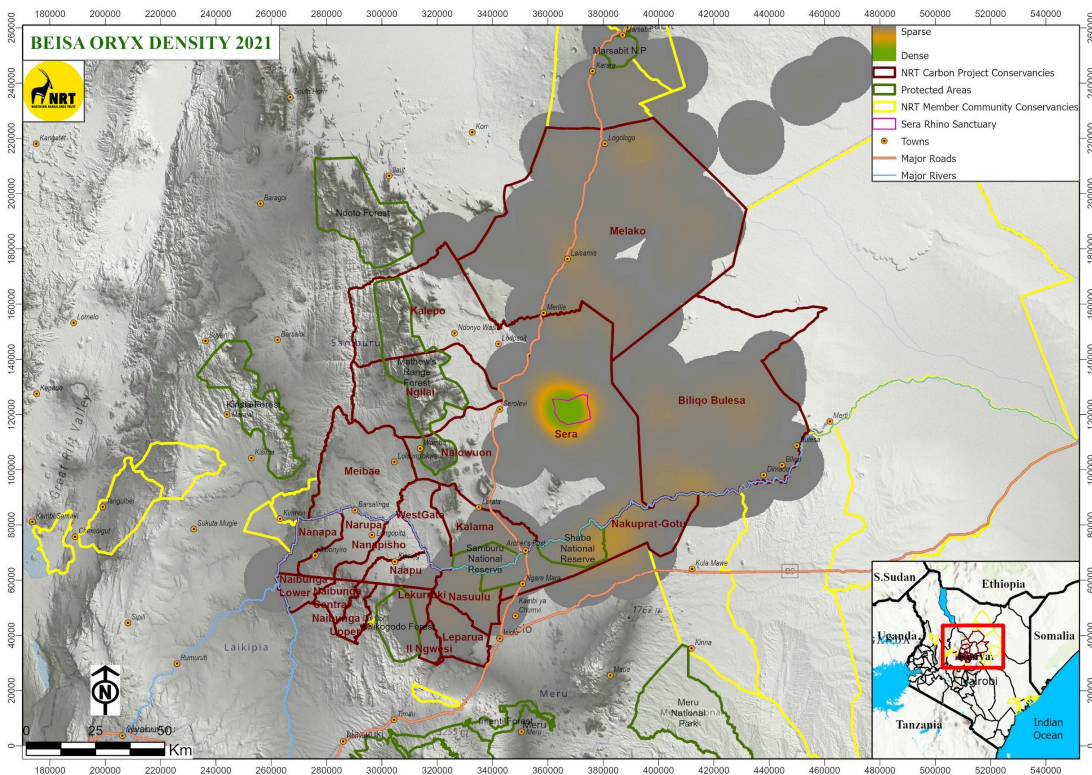
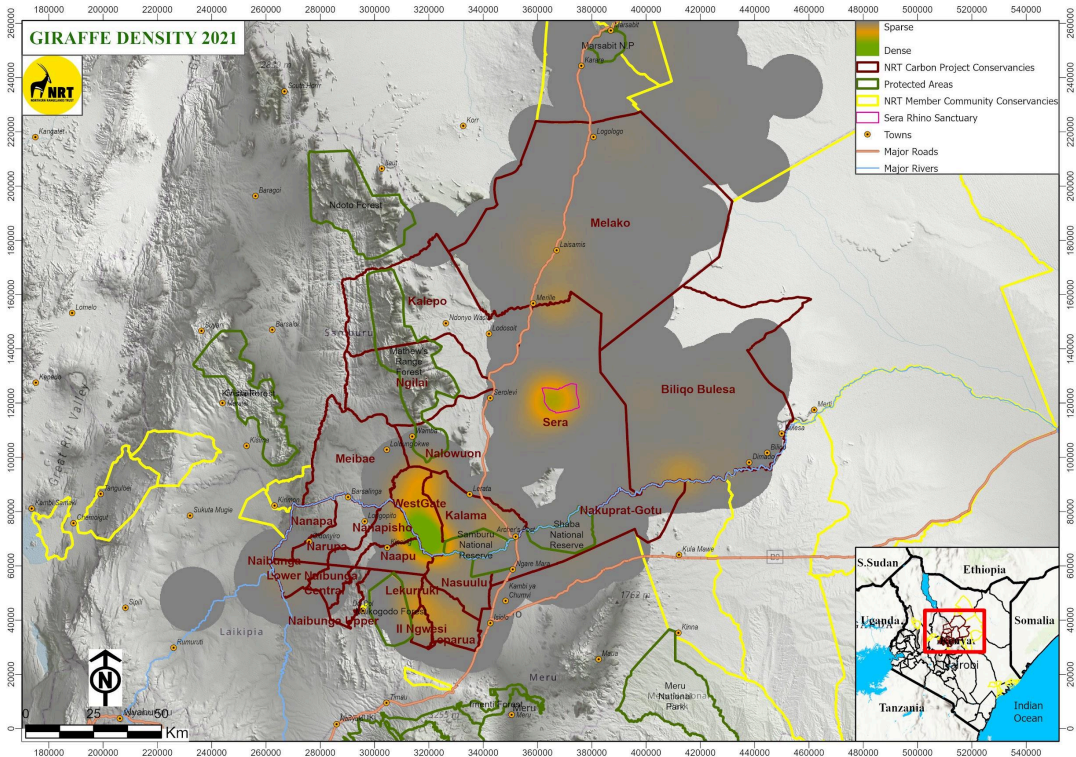
Figure 14. Heat maps of three umbrella, endemic and/or endangered species in the project zone, based on reported sightings by game scouts for 2012 (baseline conditions) and 2015 (project scenario). Green areas indicate the highest density of sightings, yellow and lighter tan areas the lowest density. Note the expansion of higher density areas in Melako and Sera Conservancies as these Conservancies began implementing planned grazing across the full Conservancy in 2014. These results, and similar results for wild dogs and lions (data not shown) show the potential for High Conservation Values in biodiversity as a result of project activities.

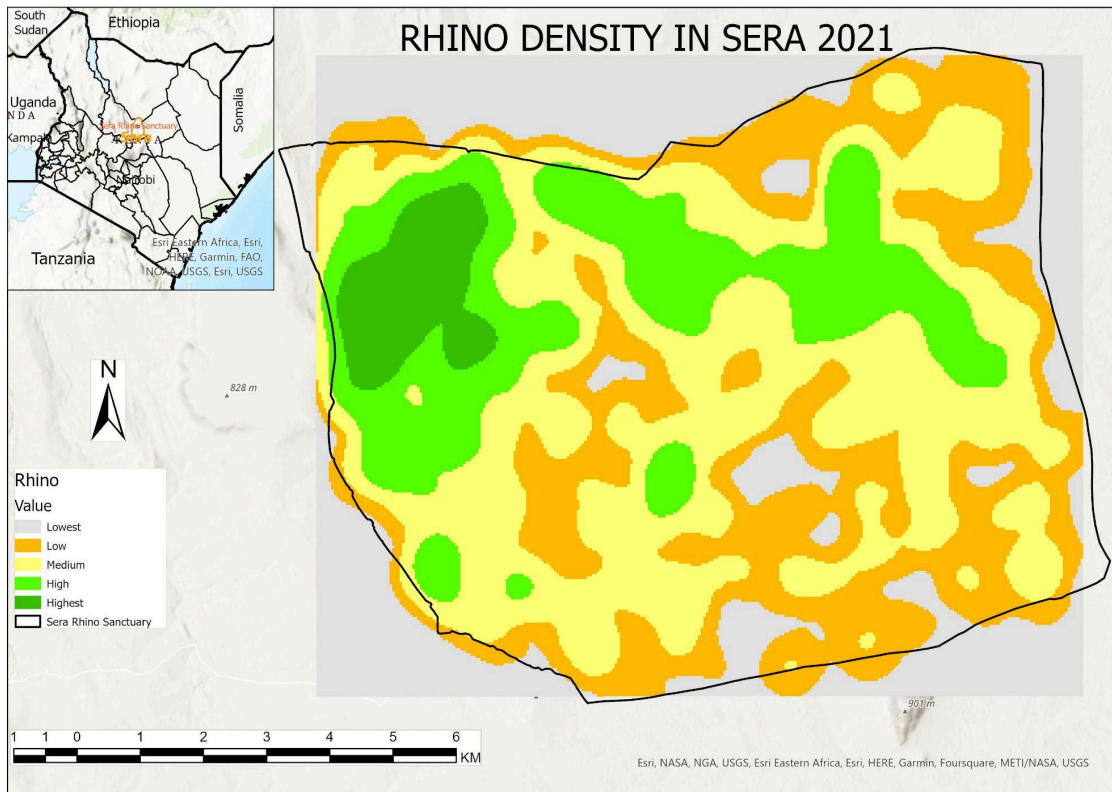
[1] <https://www.lewa.org/wp-content/uploads/2020/09/Kimiti-et-al-2020.-Evaluation-of-Restoration-success.pdf>











Figures 15-20d. Heat maps for the trigger species of Gerenuk, Lion, Wild Dog, Grevys Zebra, Cheetah, Elephant, Reticulated Giraffe (20b), Beisa Oryx (20c), and Black Rhino (20d) from data collected from 2021. As an example, when Grevys Zebra is compared in heat maps that represent data from 2012 and 2015 (shown in Figure 14), there is a noticeable increase in the amount of density represented across the project area.

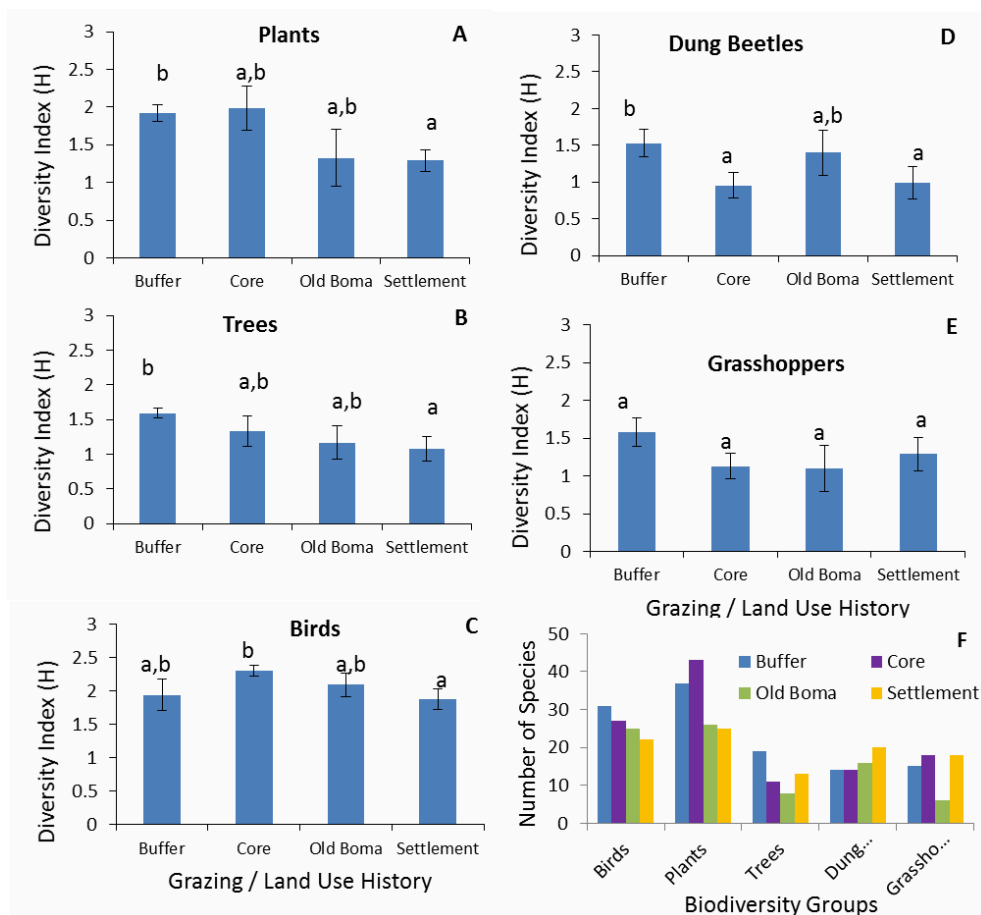


Figure 21A-F. Mean ($\pm$ SE) Shannon-Wiener diversity indices for five different biodiversity indicator groups (A-E) sampled from each of four different land uses/management areas in three Conservancies (Kalama, Westgate, and Sera) in northern Kenya: Buffer areas with only dry season grazing, Core areas with rare dry season grazing only, Old bomas (livestock corrals) abandoned 10-20 years previously, and Settlement areas subject to continuous grazing. F) Cumulative number of species of the different indicator groups across all three Conservancies for each of the different land management/uses.

In addition the project is expected to increase species diversity of multiple taxa, including plants^{[46],[52, 53]}, birds^[48, 54] and arthropods^[47, 48, 50, 55]. Our baseline data for areas with different grazing history suggest that project activities will enhance diversity of multiple previously under-sampled groups.

Baseline measurements of the species diversity of key indicator groups, plants (trees and herbaceous), birds, grasshoppers and dung beetles were made at six sites at each of three different Conservancies: Kalama, Sera, and Westgate (Figure 21). This study demonstrates a proof of concept that relaxed grazing pressure from PRM grazing practices will increase diversity. The species diversity index of all herbaceous plants and trees was significantly higher in Core versus Settlement areas and for dung beetles in protected (Core or Buffer) areas versus Settlement areas. The data clearly show that PRM grazing practices are likely to improve a broad spectrum of biodiversity across the project area. The baseline diversity

study sampled a total of 746 trees, 1481 birds, 1161 dung beetles and 351 grasshoppers. Number of herbaceous plants was not available because cover was measured, rather than individuals. 67 herbaceous plant, 24 tree, 83 bird, 31 grasshopper, and 32 dung beetle species were observed or collected across the 18 sites.

Species richness and Shannon-Wiener diversity indices were calculated for the cumulative observed or captured species for each taxonomic group. The standard Shannon-Wiener Index^[56] was H , where

$$H = - \sum_{i=1}^n p_i \ln(p_i) \quad (44)$$

Where p_i is the relative (proportional) abundance (or in the case of herbaceous plants, canopy cover) of the i th species in the sample.

Mean species richness per site was not significantly associated with land use or past grazing management for any of the taxonomic groups (Figure 21F). Cumulative numbers of species across all three conservancies for each grazing history/land use showed different patterns : birds, herbaceous plants and trees exhibited a greater cumulative number of species across conservancies in Protected areas than in Settlements, while dung beetles exhibited fewer cumulative numbers of species in Buffer and Core areas compared to Settlement areas.

Buffer areas (seasonal grazing only) had higher species richness or diversity indices than Settlement areas in 10 out of 10 tests (sign test, $P = 0.001$, one-tailed test) and Buffer and core areas combined had higher species richness and diversity indices than Settlement areas in 16 of 20 tests ($P = 0.011$, one-tailed tests).

Ten species endemic to northern Kenya or the Horn of Africa (Ethiopia, Djibouti, Eritrea, Somalia, and northern Kenya) were found among birds, dung beetles, and grasshoppers combined. Of these endemic species, 5 occurred in Core areas, 4 in Buffer areas, and 1 in Settlement areas, and thus significantly more frequently than expected by chance ($P = 0.02$) in Protected areas than in Settlement.

Taken together, these baseline data show how project activities enhance biodiversity. Scheduled future biodiversity surveys will be expanded to include Melako Conservancy, thus expanding the range of sampling for indicator taxa of biodiversity.

In 2021, in Westgate, Kalama, and Sera biodiversity camps, a total of 1,362 birds of 73 species from 329 records were observed. Likewise, 992 dung beetles of 51 species were captured. We found almost no grasshoppers, and then only newly hatched nymphs, in 2021. By comparison, the 2014 (baseline) study counted 746 trees, 1,481 birds, 1,161 dung beetles and 351 grasshoppers. Number of herbaceous plants was not available because cover was measured, rather than individuals. The 2014 survey found 67 herbaceous plant, 24 tree, 83 bird, 31 grasshopper, and 32 dung beetle species

These somewhat lower overall animal numbers in 2021 compared to 2014 (baseline) reflect the fact that sampling in 2021 occurred during a time period, April – June, that would normally feature 200-300 mm rainfall. Instead, the project area experienced no rains in the previous short rainy season (November- December) or during the early wet season just prior

to sampling^[1]. The dry conditions that persisted on the project area from June 2020 to May 2021 may have contributed to essentially a failed early wet season hatch of grasshoppers. In addition, locust swarms were widely sprayed throughout the project area during 2019 and 2020^[2]. These insecticide sprays may have dramatically depleted grasshopper populations and egg-laying across virtually all Orthoptera taxa. Thus, a possible change in grasshopper diversity and species richness cannot be determined from 2021 sampling and will await future biodiversity surveys.

The weather pattern, and possibly locust control insecticide spraying, may also explain shifts in bird species composition in our survey. The year 2014 featured 8 more insectivorous flycatcher and sunbird species than did 2021, and large gallinaceous birds like vulturine guineafowl, yellow-necked spurfowl, and black-faced sandgrouse were observed in far fewer transects (4 of 19) in 2021 than in 2014 (13 of 18) and only 151 individuals were observed in 2021 as compared to the 694 observed in 2014. In addition, the frequency of encounters with large colonies of weavers declined from 12 of 19 surveys in 2014 to 4 of 19 in 2021. Other patterns in the bird data suggest that these declines were not related to the project activity. Six species not observed in 2014 and commonly found in tallgrass habitats in East African protected areas were observed in the project area: Plain-backed pipit, Rattling cisticola, Fischer's sparrow lark, Side-spotted barbet, and Steel-blue and Straw-tailed whydahs. This suggests that improvements in vegetative cover observed during the monitoring periods 2013-2016 and 2017-2021 may have attracted tall grass-associated bird species in some areas. Moreover, 8 species endemic to northern Kenya or the Horn of Africa (eastern Ethiopia, Somalia, and northern and eastern Kenya) were found in 2021 but not in 2014.

These observations for the entire project area are consistent with a decline in bird species richness per site from 2014 to 2021 (Fig. 32A). However, bird diversity index H' per site was not significantly different between years (Fig. 33A). This result suggests that the disappearance in our surveys of large, locally abundant gallinaceous birds and large colonies of weavers may have made the bird species we did encounter more similar in abundance, which would yield a higher diversity index for the same number of bird species.

^[1] <https://www.nature.com/articles/d41586-020-02698-3>

^[2] <https://www.theafricareport.com/23745/kenya-opts-for-pesticides-to-contain-locust-infestations/>

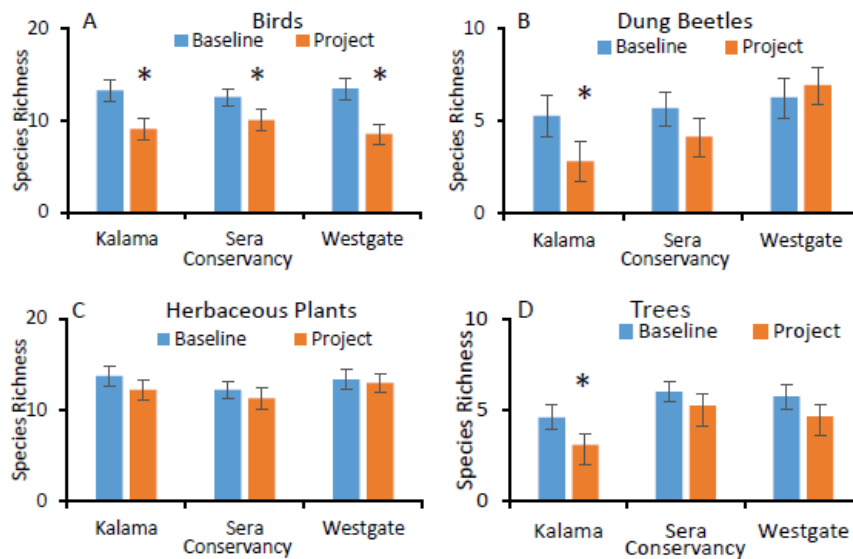


Figure 32. Mean $\pm$ SE species richness of four indicator taxa in 2014 (Baseline) and 2021 (Project) at three different Conservancies in the project area. Asterisks indicate difference significant at $P < 0.05$ in a Generalized Linear Model

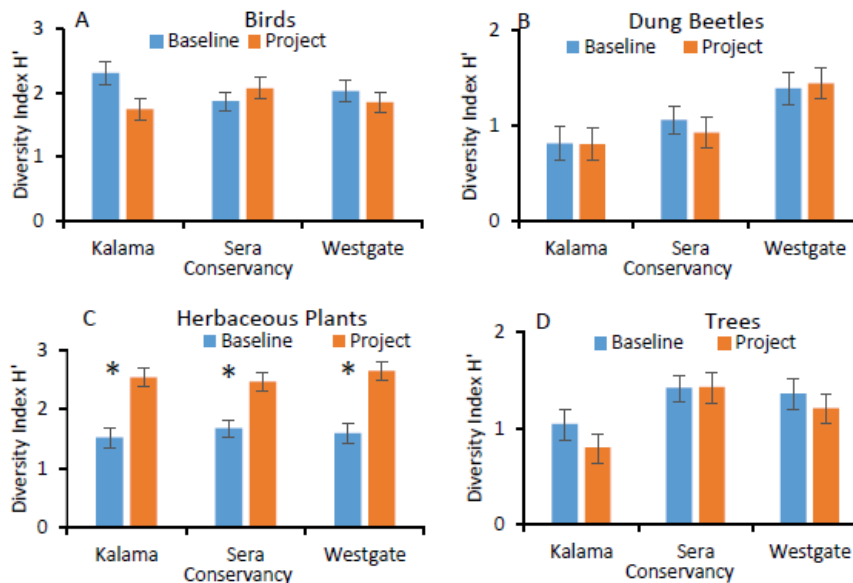


Figure 33. Mean $\pm$ SE Shannon-Wiener diversity Indices for four indicator taxa in 2014 (Baseline) and 2021 (Project). Asterisks indicate differences significant at $P < 0.05$ in a Generalized Linear Model

In contrast with birds, overall species richness of dung beetles was much higher in 2021 (51) than in 2014 (32), a trend that would be expected from the observed improved vegetation cover and expanding wildlife populations associated with project activity. Despite this increase in overall project area species richness, mean dung beetle species richness and diversity index H' per site showed no difference between 2014 and 2021 (Fig. 32B, 33B). These two patterns combined suggest that in comparison to 2014 dung beetle species communities in 2021 differed in composition more strongly between different locations. This is a pattern consistent with higher beta diversity associated with project activity.

Herbaceous plant species richness across all sites increased over time across the entirety of the project area, as we found 81 species in 2021 as compared with 62 species in 2014. This change was associated with an increase in mean perennial grass species richness per site (Figure 32C). In 2014, we detected only 8 perennial grass species across the project area, but found more than 16 species in 2021. Moreover, % cover of perennial species per site increased from an average of 1.2% (± 2.4 SE) in 2014 to 5.2% (± 2.1 SE) in 2021. These changes produced large increases in the diversity index for herbaceous plants at all three Conservancies sampled between 2014 and 2021 (Fig. 33C).

Previously, we compared the diversity of these taxa among areas receiving different livestock use as a proof of concept that relaxed grazing pressure from PRM grazing practices will increase diversity (Fig. 31). In 2014, the species diversity index of all herbaceous plants and trees was significantly higher in Core versus Settlement areas and for dung beetles in protected (Core or Buffer) areas versus Settlement areas. In contrast, in 2021, bird and dung beetle species richness and diversity index showed no difference between protected (previous buffer and core) and settlement areas (Fig. 34). Such a lack of difference may reflect an increase over time in these metrics in settlement areas.

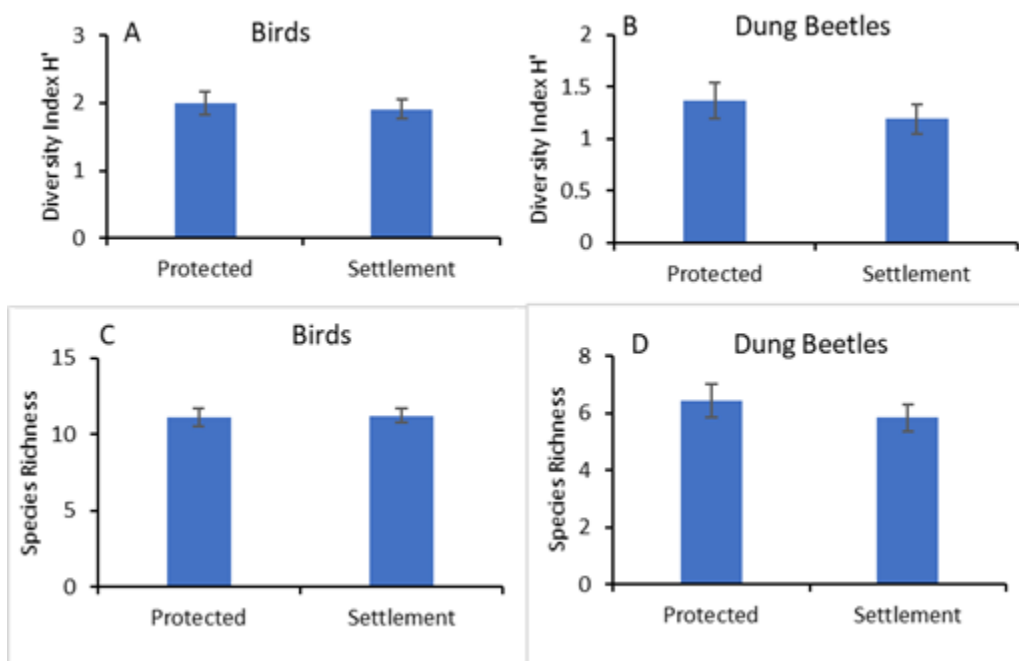


Figure 24A-D. Mean ($\pm$ SE) diversity index (A,B) and species richness (C,D) for birds and dung beetles, two indicator taxa, respectively, in areas with grazing restricted to dry season or short periods in the wet season (Protected) vs areas subject to planned rotational grazing in the wet season (Settlement) under the project activity. Means were not different ($P < 0.05$) in Generalized Linear Models

In contrast, mean herbaceous plant and tree diversity indices and herbaceous plant species richness continued to be higher in “protected” areas with restricted grazing in 2021, as they were in 2014 (Fig. 35).

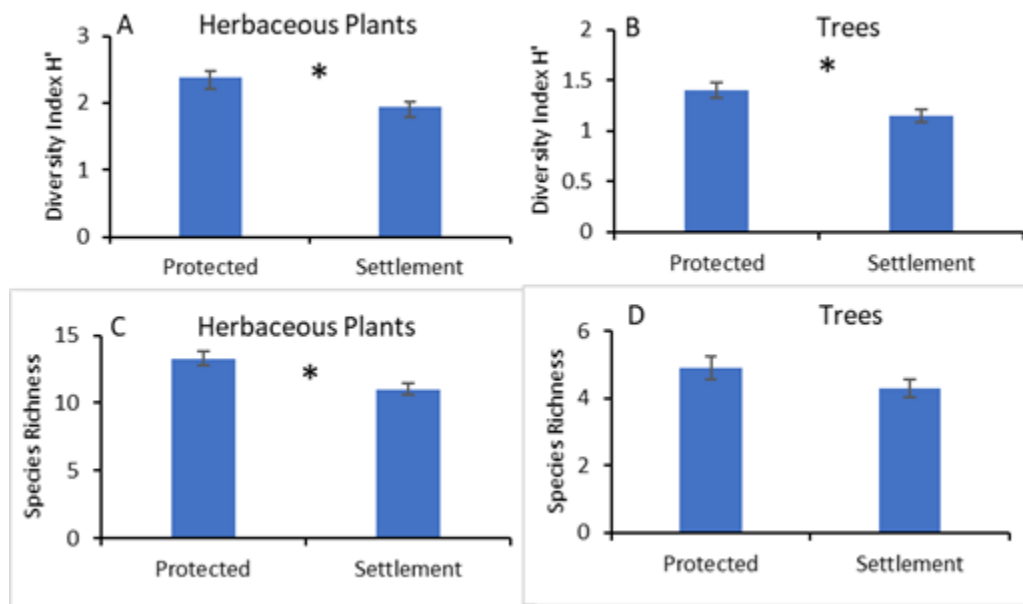


Figure 25. Mean ($\pm$ SE) diversity index (A,B) and species richness (C,D) for birds and dung beetles, two indicator taxa, respectively, in areas with grazing restricted to dry season or short periods in the wet season (Protected) vs areas subject to planned rotational grazing under the project activity grazing in the wet season (Settlement) under the project activity. Asterisks indicate significantly different ($P < 0.05$) differences in Generalized Linear Models.

These persistent differences due to grazing management that occurred prior to the onset of the project activity suggest that vegetation is recovering in both types of areas from poor grazing practices that occurred decades before grazing restrictions were imposed. Fig. 33C shows that herbaceous plant species diversity indices are increasing in the project area and the Generalized Linear Model suggests that diversity indices increased similarly under the project activity in protected and settlement areas (interaction between management and year is not significant, $X^2_{2,29} = 0.672$, $P = 0.715$).

Tree numbers and species richness would be expected to change slowly with time, and with the exception of tree species richness in Kalama Conservancy, we observed no difference in tree species richness between 2014 and 2021 or between protected and settlement grazing management histories (Fig. 32D, 33D, 34D, 35D). The decline in tree species richness in

Kalama Conservancy may be due to attempted control of the invasive tree *Acacia reficiens* at several of the sampling sites^[1].

Overall, our results suggest that, given the prolonged low rainfall conditions and the prevalence of insecticide spraying to control locusts prior to sampling, the 2021 biodiversity survey may not provide a worthy evaluation of change in biodiversity of animal indicator taxa. Nevertheless, other data suggest the project activity is increasing species richness of dung beetles and herbaceous plants over the entire project area. Herbaceous plant diversity indices were substantially higher in 2021 than in 2014 and this change was driven by an increase in the number of perennial grass species detected. Results for total species numbers across the three Conservancies sampled in both 2014 and 2021 suggest that the project activity is associated with higher beta diversity (greater variation in species composition across sites) of plants and dung beetles. Even for birds, we found increased numbers of species of endemics and species associated with tall grass habitats that were absent from the 2014 survey. Thus we conclude that the project activity has improved several important aspects of biodiversity of these indicator taxa relative to baseline conditions.

Species and habitat	Grevy's zebra, elephant, beisa oryx, black rhino, gerenuk, and reticulated giraffe The project aims to reduce adverse impacts on species habitats. The project utilizes the Wildlife-CoMMS system. The Wildlife-Conservancy Management Monitoring System (Wildlife-CoMMS) was developed by the Northern Rangelands Trust (NRT) to support community conservancies in Northern Kenya. Functioning as a ranger-based monitoring system, Wildlife-CoMMS plays a crucial role in tracking wildlife and detecting illegal activities. Wildlife-CoMMS has proven to be a highly effective tool in safeguarding wildlife and combating illegal activities within community conservancies across Northern Kenya. The system aims to provide simple and practical information on wildlife trends, distribution, and incidents of illegal activities. Importantly, it empowers conservancies to manage the system independently, from data collection to analysis and reporting, reducing dependence on external scientific assistance in the long run. Monitored positive change in increased populations of the trigger species referenced above, since the project start, has occurred through the implementation of Wildlife-CoMMS paired with increased protection by project-employed game scouts, who also help manage leakage and livestock incursions from outside the project area.
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5.1.4 High Conservation Values Protected (CCB, B2.4)

Use of Biodiversity HCV sites by livestock has been decreased by project activities (Figure 11). Livestock monitoring data and analysis of satellite imagery for relative change in NDVI for 2011-2012 (Baseline) and the monitoring period 2013 – 2021 (Project) indicate a clear reversal of trend in reducing encroachment into the Mathews Forest (Figure 11, 22). Even in a drought year, when encroachment is usually at its worst, livestock produced negative impacts on only 20% of the Mathews Forest in 2014 and approximately 15% in 2017 (Figure 12). Impacts during the project (e.g., 2014) have mostly occurred on the edges where degradation has already occurred, compared to over 30% of the forest negatively impacted, and mostly in the high biodiversity rainforests at high elevation, that occurred in 2008, a comparatively wet year (Figure 12). As rotational practices continue and grass banks are established across a wide range of locations in the project area, encroachment should decrease further, thus addressing the number one threat to the Mathews Forest.

5.1.5 Species Used (VCS, 3.19; CCB, B2.5, 2.6)

No non-native species have been introduced or used as part of project activities. Small-scale grass restoration efforts have used a native grass *Cenchrus ciliaris* in these restoration efforts.

Species introduced	Classification	Justification for use	Adverse effects and mitigation

5.1.6 Invasive Species (VCS, 3.19; CCB, B2.5)

No invasive species have been introduced into the project zone, nor have established invasive species spread through the project zone, due to project activities.

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species

5.1.7 GMO Exclusion (CCB, B2.7)

No GMOs have been used to generate GHG emissions reductions or removals, as re-seeding or plantings are not a project activity. Project activities merely adjust the way in

which livestock are herded and moved across the landscape.

5.1.8 Inputs Justification (VCS, 3.19; CCB, B2.8)

No fertilizers are allowed according to the applicability conditions of methodology VM0032. Project activities do not include addition of pesticides or other potentially harmful chemicals into the environment.

5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impacts (CCB, B3.1) and Mitigation Actions (CCB, B3.2)

Through 2021, there have been no negative offsite biodiversity impacts related to project activities. Our efforts to reduce herder excursions off the project area have more than likely reduced impacts on off-project forest areas (see section 5.1).

5.2.2 Net Offsite Biodiversity Benefits (VCS, 3.19; CCB, B3.3)

No explicit evaluation of biodiversity off the project area is available. However, given the rapid expansion of areas used by the six trigger species (Figures. 25-30) and possible dispersal of animals off the project area, it is likely that project activities increased biodiversity off the project area.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

Biodiversity of indicator taxa birds, dung beetles, grasshoppers, herbaceous plants and trees was measured in April and May 2021 at 6-7 sites at each of five biodiversity camps in the project area (Fig. 36). Measurements of the species diversity of key indicator groups, plants (trees and herbaceous), birds, grasshoppers and dung beetles were made at 6-7 sites at each of five different Conservancies: Kalama, Sera, Il Ngwesi, Melako, and Westgate, for a total of 31 sites sampled. We sampled at the same sites (approximately) as were sampled in Sera, Kalama, and Westgate in 2014, which allowed for a time comparison over the course of the monitoring period for those three Conservancies.

Monitoring biodiversity occurs primarily through 1) wildlife counts by game scouts during patrols on their home Conservancies as part of the Conservancy's community monitored and management activities (Wildlife CoMMs), and 2) through periodic sampling of indicator taxa (insects, plants, birds) in the vicinity of five biodiversity camps distributed across the rainfall gradient in the project area.

The results of these monitoring efforts for the monitoring period were all presented in section 5.1.3 (Figures 25-35).

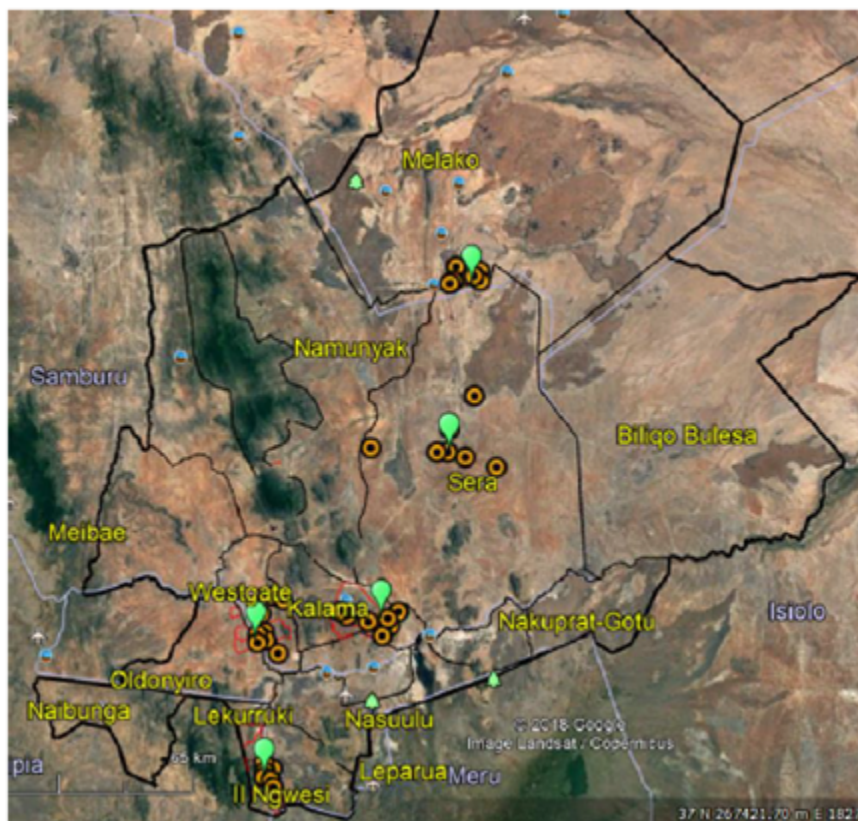


Figure 26. Map of locations of biodiversity sampling across the project area in 2021. Each point is clustered around a biodiversity “camp” from which measurements are made at each of 6-7 sites that include both sandy and black soils and areas with restricted (protected) and unrestricted grazing (settlement). Five biodiversity camps were sampled that encompassed the project-wide mean annual rainfall gradient from 300 mm in the north (Melako) to 800 mm (Il Ngwesi) in the south.

Species richness (number of species per site) and Shannon-Wiener diversity indices were calculated for the cumulative observed or captured species for each taxonomic group. The standard Shannon-Wiener Index^[56] was H , where

$$H = \sum_{i=1}^n p_i (-\ln(p_i)) \quad (44)$$

where p_i is the relative (proportional) abundance (or in the case of herbaceous plants, canopy cover) of the i th species in the sample.

5.3.2 Biodiversity Monitoring Plan Dissemination (CCB, B4.3)

In September 2016, we held a Conservancy-wide Project Meeting attended by Conservancy chairs, community grazing chairpersons, grazing coordinators and interested public. This meeting was announced on the NRT website and notices were posted in major community kiosks and markets. At the meeting, a report of the previous 12 months' monitoring results for both implementation of the grazing plans and of wildlife and indicator taxa monitoring was presented. Results of the previous Social CoMMs surveys were also presented. Following this presentation, plans for managing grazing during the ensuing 12 months were discussed, with feedback from Conservancy managers and the community attendees.

Since 2016, and as mentioned in Section 3.14 'Dissemination of Monitoring Plan and Results,' NRT continues to share communications notifying conservancies of upcoming biodiversity surveys. The results of these surveys, including 2021, are shared with the conservancies who then communicate the results to communities. All conservancies have been briefed on the project and its ongoing monitoring requirements in terms of soil sampling, biodiversity surveys, remote sensing, livestock movements and numbers, etc. At the end of every year, each conservancy receives a summary of their Wildlife CoMMS data that highlights their progress.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

The project activities will provide exceptional biodiversity benefits. Project activities protect and promote populations of trigger species, protect sky island habitat for forest endemic species and reduce livestock pressure on permanent wildlife water sources protected by national game reserves.

5.4.1 Trigger Species Population Trends (CCB, GL3.2, GL3.3)

In the *project scenario*, a specific rhino restoration project begun in 2015 in Sera Conservancy has now yielded 20 rhinos. Active conservation methods and range improvements resulting from grazing management education by stakeholder Grevy's Zebra Trust²⁵ are associated with expansion of area used and number of sightings of Grevy's zebras (Fig. 24 in section 5.1.3) The Conservancies represent the only growing population of Grevy's zebra in Kenya (see section 5.1.2 and 5.1.3). Trends for gerenuk and elephants are positive in the project zone (Figs. 25, 30) in contrast to negative trends even in protected Reserves found in the Ogutu et al. (2018) study. Active conservation measures in the project zone include persistent patrols of community game scouts, which have reduced elephant mortality in the project zone by more than 60% over the past five years. Rangeland recovery spurred by planned rotational and migratory grazing should further aid conservation efforts through habitat restoration, as shown by the dramatic increase in wildlife species in livestock "refuges" (core and buffer areas) since the project start.

²⁵ Grevy's Zebra Trust is a local stakeholder in the project, <http://www.grevyszebratrust.org>

Trigger Species	Black rhino (<i>Diceros bicornis</i>)
Population Trend at Start of Project	No animals on project area
Without-project Scenario	Continued lack of presence on the project area and further declines in nearby National Reserves and Lewa Conservancy
With-project Scenario	A specific rhino restoration project begun in 2015 in Sera Conservancy established the Sera has now yielded 20 rhinos and reproduction had been documented for this project as heat maps until 2017. Community-based game scouts have successfully protected the sanctuary for more than two years. See 2013-2016 Monitoring Report for further data on the Black rhino.

Trigger Species	Elephant (<i>Loxodonta africana</i>)
Population Trend at Start of Project	Small herds near Samburu National Reserve and Mathews Forest, declining.
Without-project Scenario	Habitat degradation between preferred elephant habitats further isolates maternal groups, reducing migration and gene flow and increasing risk of mortality during drought and poaching in more restricted migration routes.
With-project Scenario	Improvement in wet season grass forage biomass and project-sponsored increases in game scouts creates safer habitats for elephants. Percent of elephant carcasses poached has declined from 81% to 30% since project start. Elephant sightings now occur throughout the project area with > 450 mm annual rainfall.

Trigger Species	Oryx (<i>Oryx beisa</i>)
Population Trend at Start of Project	Scattered small herds in remote areas of Sera Conservancy and near Samburu National Reserve, declining.
Without-project Scenario	Continued lack of presence on the project area and further declines in nearby National Reserves and Lewa Conservancy.
With-project Scenario	Protection from game scouts and improved forage increased sightings 20-fold. Found predominantly in project area < 400 mm annual rainfall in Sera, Melako, eastern Namunyak, and Biliqo Bulesa.

Trigger Species	Reticulated giraffe (<i>Giraffa camelopardus reticulata</i>)
Population Trend at Start of Project	Scattered small herds in remote areas of Sera Conservancy and Namunyak Conservancy, declining.
Without-project Scenario	Continued range degradation and poaching without game scouts.
With-project Scenario	Protection from game scouts and improved forage increased sightings 50-fold. Found in all 13 Conservancies but predominantly in project area > 450 mm annual rainfall in Namunyak, Kalama, Meibae, Oldonyiro.

Trigger Species	Gerenuk (<i>Litocranius walleri</i>)
Population Trend at Start of Project	Scattered small herds in remote areas of Sera Conservancy and eastern Namunyak Conservancy, declining.
Without-project Scenario	Continued range degradation and poaching without game scouts.
With-project Scenario	Protection from game scouts and improved forage increased sightings 20-fold. Found throughout the project area < 450 mm annual rainfall in Nakuprat-Gotu, Sera, Melako and Biliqo Bulesa Conservancies.

Trigger Species	Carnivores, lion (<i>Panthera leo</i>), cheetah (<i>Acinonyx jubatus</i>) and wild dog (<i>Lycaon pictus</i>)
Population Trend at Start of Project	Occasional sightings in Sera Conservancy and Kalama Conservancy near Samburu National Reserve, declining.
Without-project Scenario	Continued prey declines and poaching without game scouts. Continued persecution by pastoralists due to loss of wild prey, forcing predation on livestock.
With-project Scenario	Protection from game scouts and improved prey abundance increased sightings 10-fold. Found in remote areas of Melako, Sera, and Namunyak Conservancies and near Ewaso Nyiro River (Oldonyiro, Westgate, Nasuulu, Kalama).

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APPENDIX 2: ADDITIONAL INFORMATION

Summary

For the Northern Kenya Rangeland Carbon Project, a remote sensing method is employed to verify the degree of successful implementation of the project activity and subsequently adjust the modeled output of the SNAP soil organic carbon model. The remote sensing method was calibrated during a pre-study following baseline soil sampling and during baseline biodiversity sampling in April - June 2014. The analysis evaluates the annual changes in green biomass at a site relative to that expected if there was no grazing. This annual relative change is calibrated relative to known temporal patterns of grazing impacts, and a threshold of annual relative change is determined that corresponds to more than one major grazing event per year, and thus unsuccessful implementation of project activities. The area that exhibits annual relative change in excess of the criterion is subtracted from the project area in determining total modeled estimates of soil carbon sequestration, this making claimed removals more conservative.

Initially presented for the 2017–2020 monitoring period, there were two deviations to the remote sensing method as compared to the 2013–2016 Monitoring Report. First, an adjusted set of ungrazed reference polygons is used for calculating annual relative change in biomass. This is justified due to conservancies with lower mean annual rainfall (relative to wetter reference polygons) exhibiting high relative changes in NDVI and thus being judged in error as unsuccessful. Secondly, the threshold value of annual relative change in NDVI is adjusted from 0.05 to 0.1. This is justified because a threshold of 0.1 more accurately reflects the occurrence of single grazing events in both wet and dry-season grazing areas.

Assessment of successful pastoralist grazing plans - full methodological details

Remote sensing of grazing and implementation of grazing plans

In NRT conservancies, data to verify livestock activity were obtained from 72 MODIS satellite images and 468 monthly grazing reports for each conservancy during each month and year (2014-2016). These data include satellite images showing bi-monthly 16-day composite NDVI calculations and used to calculate mean NDVI scores for every 16-day period each year within designated grazing blocks within each Conservancy over the monitoring period.

NDVI was calibrated to actual biomass measures during the pre-study in July and August 2014 following May-June rains by clipping annually produced biomass (not gray litter) from thirty 0.125 m² quadrats within each of thirty separate MODIS pixels during a specific 16-day window that corresponded to a single 16-day composite NDVI image. Measured biomass was strongly correlated with NDVI for the designated pixel both before and after cattle and sheep herds moved through the pixel (Figure A1). Importantly, NDVI measurements estimated biomass across a relatively low range

of biomass encountered across the degraded, high-percentage bare ground landscape in the project area.

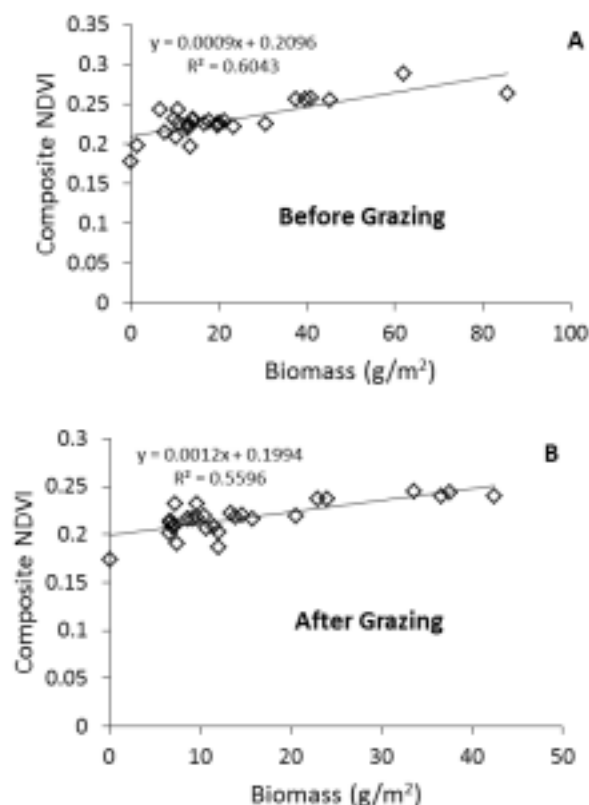


Figure A1. Relationships between NDVI measurements from individual pixels (250 x 250 m) of 15-day composite MODIS NDVI and measured mean forage biomass (shrub leaves and herbaceous plant biomass) before and after grazing. Forage biomass is from thirty 0.125 m² quadrats placed randomly inside each pixel in August 2014 in Kalama and Westgate Conservancies. Relationships are shown for A. before grazing by a herd of 30 or more cattle and/or additional sheep and goats, and B. immediately after grazing. These relationships demonstrate the strong connection between satellite based measures and actual forage biomass and support the use of satellite imagery in monitoring range condition and grazing impact. The relationships imply that a reduction in forage biomass from grazing results in a detectable reduction in NDVI as predicted by the regression lines in each graph, which are very similar. The y intercepts in each relationship reflect the NDVI signal of trees and shrubs, since forage biomass = 0.

To measure livestock grazing impacts, a monthly change in NDVI ($\Delta\text{NDVI}_{\text{Im}}$) was calculated by subtracting the later NDVI image from the earlier NDVI image, with more positive values indicating stronger reduction in forage biomass (Fig. A2; see below); this serves as a measure of livestock impact on forage during the 15 days between composite images. Further, a relative monthly change in NDVI ($\Delta\text{NDVI}_{\text{Im,rel}}$) (Eq. 1) was calculated by subtracting the $\Delta\text{NDVI}_{\text{Im}}$ for the same period from the mean $\Delta\text{NDVI}_{\text{Im}}$ ($\text{mean}\Delta\text{NDVI}_{\text{Im,REF}}$) across one or more ungrazed reference areas (inaccessible plateaus, fenced rhino sanctuaries, ecotourism areas from which livestock were excluded) to correct for expected changes due to recent rainfall or drying.

$$\Delta\text{NDVI}_{\text{Im,rel}} = \Delta\text{NDVI}_{\text{Im}} - \text{mean}\Delta\text{NDVI}_{\text{Im,REF}} \quad (1)$$

$\Delta\text{NDVI}_{\text{Im,rel}}$ for a MODIS pixel, was consistently correlated with change in forage biomass for that pixel (Figure A2), confirming the ability of the remote sensing method to estimate forage availability on the ground.

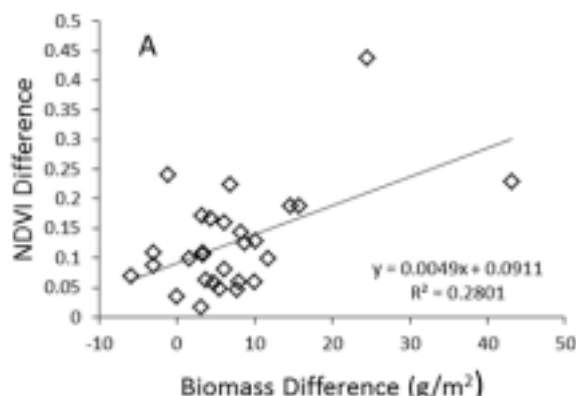


Figure A2. Relationship between measured forage biomass difference before and after grazing by 50 or more cattle and relative change in NDVI, $\Delta NDVI_{m,rel}$ (NDVI difference), between 28 July and 12 August, 2014 for thirty MODIS pixels within Kalama and Westgate Conservancies.

Monthly relative $\Delta NDVI_{m,rel}$ was corroborated with observed distributions of livestock by Conservancy Grazing Coordinators (CGCs) during the May-July 2014 pre-study. Random points were selected from within monthly $\Delta NDVI_{m,rel}$ maps of each conservancy to obtain estimates of $\Delta NDVI_{m,rel}$ for individual pixels, which were then compared to distribution maps of livestock occurrence submitted by grazing coordinators. Livestock occurrence values of 0 for livestock absent or 1 for livestock present were assigned to each pixel and thus to each value of $\Delta NDVI_{m,rel}$. This analysis produced a logistic model in which $\Delta NDVI_{m,rel}$ predicted absence/presence of livestock (Figure A3C).

Given the many potential sources of error in reporting by CGCs, such as mismatch in the period of days assessed by CGCs relative to satellite analysis, imprecision in the drawing of polygons relative to MODIS pixel sizes (Figure A4B), lack of knowledge of livestock distributions in relatively inaccessible areas, e.g. northern Namunyak Conservancy (Figure 4A), and the possibility of encroachment by herders from communities outside the project area, the strength of the logistic regression indicates a strong signal of livestock grazing impacts detected by the $\Delta NDVI_{m,rel}$ measurement.

In the initial ground truthing study in July-August 2014 in Kalama and Westgate Conservancies, $\Delta NDVI_{m,rel}$ was reasonably well correlated with biomass removals by a single livestock grazing event (Figure A2). Consequently, it was concluded that $\Delta NDVI_{m,rel}$ sufficiently represented a measure of rainfall-corrected monthly grazing impacts by livestock to determine success or lack thereof in implementing conservancy grazing plans.

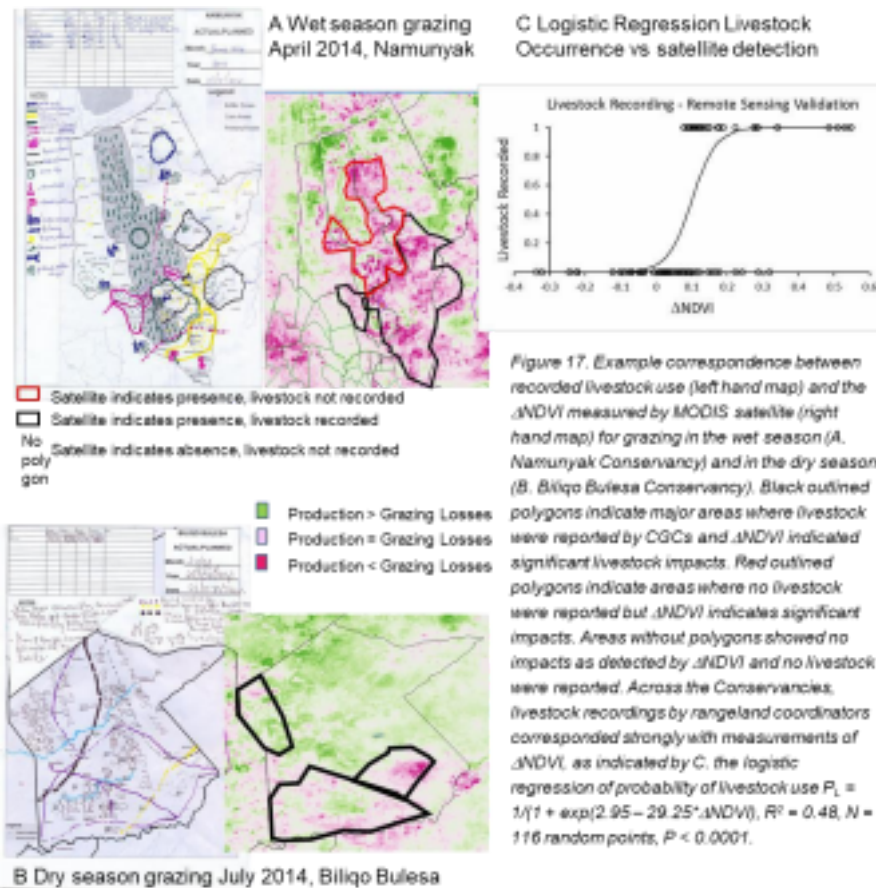


Figure A3. Example correspondence between recorded livestock use (left hand map) and the $\Delta NDVI$ measured by MODIS satellite (right hand map) for grazing in the wet season (A. Namunyak Conservancy) and in the dry season (B. Biliqo Bulesa Conservancy). Black outlined polygons indicate major areas where livestock were reported by CGCs and monthly relative $\Delta NDVI$ indicated significant livestock impacts. Red outlined polygons indicate areas where no livestock were reported but monthly relative $\Delta NDVI$ indicates significant impacts. Areas without polygons showed no impacts as detected by

relative monthly $\Delta NDVI_{m,rel}$ and no livestock were reported. Across the conservancies, livestock recordings by rangeland coordinators corresponded strongly with measurements of $\Delta NDVI_{m,rel}$ as indicated by C., the logistic regression of probability of livestock use $PL = 1/(1 + \exp(2.95 - 29.25 * \Delta NDVI_{m,rel}))$, $R^2 = 0.48$, $N = 116$ random points, $P < 0.0001$.

Method for assessing successful implementation of project activities.

For Meibae, Westgate, Kalama and southern Namunyak Conservancies, monthly grazing impact (not relative) $\Delta NDVI_m$ was calculated for each 17 grazing blocks **for each month of 2014** (see examples in Figure A5) and summed to yield $SUM\Delta NDVI_m$. A similar annual reference sum of NDVI differences, $meanSUM\Delta NDVI_{m,REF}$, was calculated as the sum of mean monthly $\Delta NDVI_m$ across the reference polygons. An annual estimate of grazing impacts, $SUM\Delta NDVI_{rel}$, for each MODIS pixel was calculated as:

$$SUM\Delta NDVI_{rel} = SUM\Delta NDVI_m - meanSUM\Delta NDVI_{m,REF} \quad (2)$$

$SUM\Delta NDVI_{rel}$, the **annual relative change in NDVI** for a MODIS pixel, was strongly correlated with whether a MODIS pixel experienced successful implementation of project activities, i.e., a single major grazing event each year, as reflected by a single month (two consecutive 16-day composite images) in which $\Delta NDVI_{m,rel} > 0.1$ (as suggested by the logistic regression in Fig. A3C) This correspondence was measured by logistic regression of $SUM\Delta NDVI_{rel}$ versus whether one or no grazing events occurred (score 0) or (Figure A5B). The regression suggested a threshold value of

$SUM\Delta NDVI_{rel} < 0.08$ was associated with a probability of successful grazing implementation > 0.5 , so, conservatively, we used $SUM\Delta NDVI_{rel} < 0.05$ as the threshold for successful implementation of rotational grazing. Note that this criterion applies to situations where rotations went as planned, as in the grazing block examples (Figure A4) or if livestock were grouped and moved around a larger regional landscape, as occurred during each dry season.

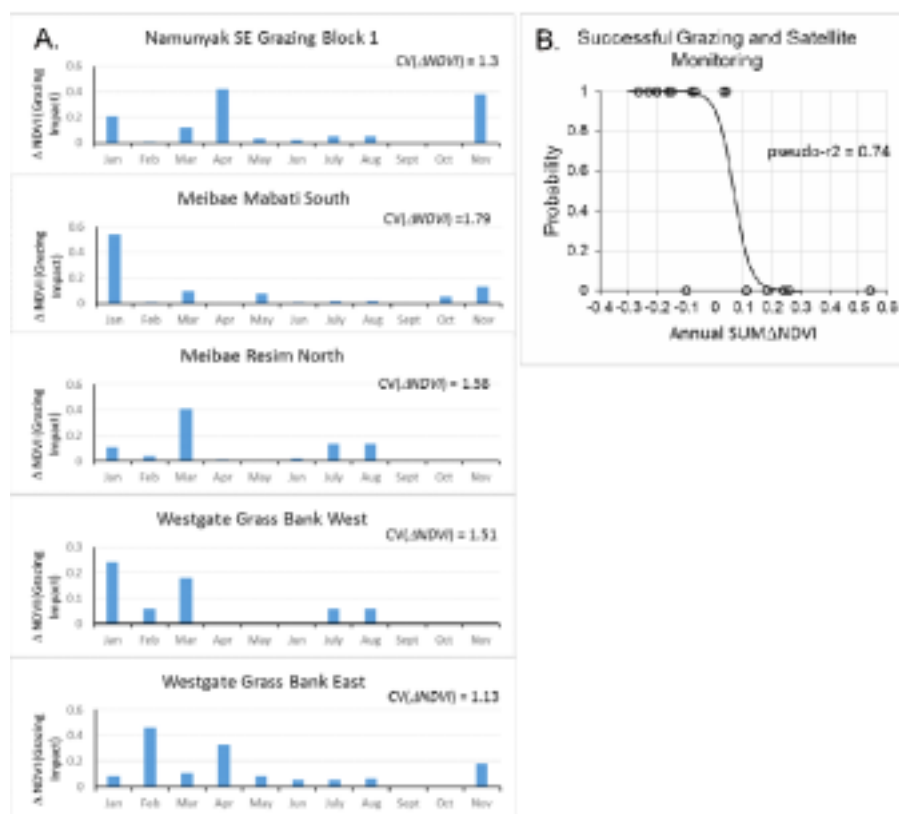


Figure A4. A. Example temporal patterns of monthly grazing impacts, measured as relative monthly change in NDVI ($\Delta NDVI_{m,rel}$), on selected grazing blocks with high quality grazing reports in southern Samburu County, Kenya from Meibae, Westgate, and Namunyak Conservancies for 2014. These examples clearly show the difference in temporal pattern between unsuccessful and successful rotational grazing practices. Unsuccessful patterns include Namunyak SE Grazing Block 1 and Westgate Grass Bank East, which were

grazed ($\Delta NDVI > 0.1$) three times during the year, and especially twice during the wet season. Successful patterns included Meibae Mabati South and Meibae Resim North, which were grazed only once. B. Logistic regression to determine probability of successful grazing implementation over the whole year, p , as a function of annual $SUM\Delta NDVI_{rel}$: $p/(1-p) = 2.3 - 35.2(SUM\Delta NDVI_{rel})$, approximately 74% of the variation explained.

Estimating area of successful implementation

To estimate the area of successful implementation of grazing plans **in the first monitoring period 2013-2016**, we used the criterion of $SUM\Delta NDVI_{rel} < 0.05$ to identify MODIS pixels (250 x 250 m) ((from NASA LP DAAC <https://lpdaac.usgs.gov/>) note that the Google Earth Engine database did not exist at that time) where rotational grazing practices were successful in each year using the Raster->Processes->Geocalgorithms->Zonal Statistics feature in QGIS 3.4 (2013-2016) and are now calculated using Google Earth Engine (2017-present). The proportion of area that was unsuccessful

is the proportion of pixels of $SUM\Delta NDVI_{rel} > 0.05$ to total number of pixels for each Conservancy; this proportion discounts the total area used in the SNAP soil carbon model. Areas of unsuccessful implementation of rotational grazing practices for the 2013-2016 monitoring period are shown in Figure A5.

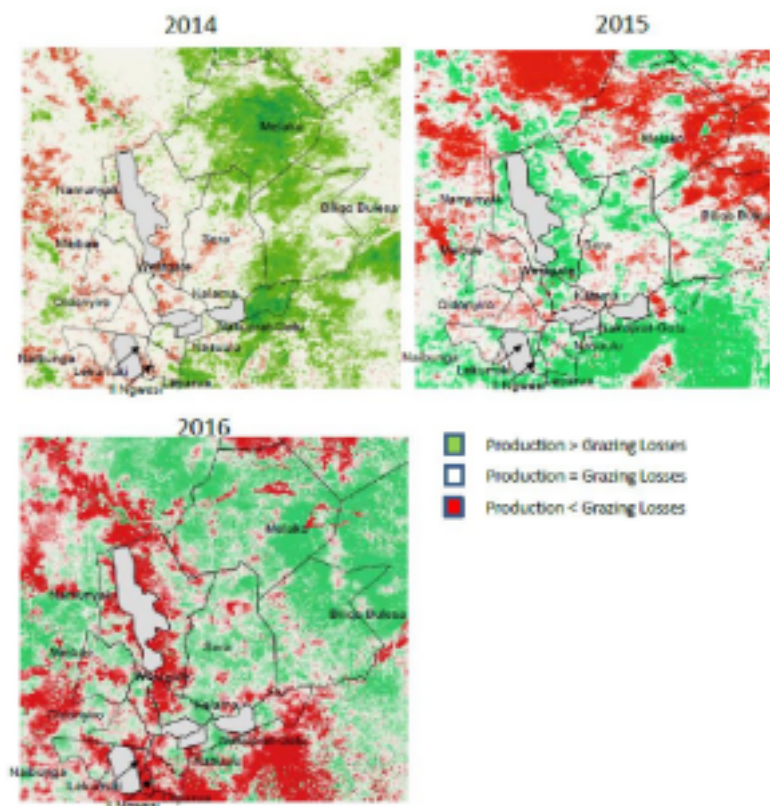


Figure A5. Evidence of successful project implementation (green and white areas) for 2014-2016 in the project zone

Justification for Adjusted sets of reference polygons for calculating successful implementation of grazing plans during 2017-2020 and future monitoring periods.

The method of assessing annual grazing plan implementation depends on comparing annual relative changes in NDVI, $SUM\Delta NDVI_{rel}$, determined by summing monthly $\Delta NDVI_m$ across the twenty-four 16-day intervals for each year, $SUM\Delta NDVI_m$ (Eq. 2 above) at each MODIS pixel, as compared with mean annual sums of $\Delta NDVI_m$ across the same time intervals ($meanSUM\Delta NDVI_{REF}$) from reference polygons that are not grazed or only slightly grazed by livestock (see Eq. 2 above). These reference locations are embodied in multiple polygons that span the project area, including “Core” areas of Il Ngwesi and Kalama Conservancies, the tops of plateaus in Kalama Conservancy that are relatively inaccessible to livestock, and a large ~1,000 ha fenced rhino sanctuary in Sera Conservancy. In most years, the mean annual relative change in NDVI ($meanSUM\Delta NDVI_{REF}$) in these reference areas is relatively similar despite differences in mean annual rainfall. Consequently, the Google Earth Engine (GEE) calculations of annual relative change in NDVI ($SUM\Delta NDVI_{rel}$) for individual pixels represent a good estimate of livestock impacts in drought years (2017, 2018). However, in wet years following droughts, such as occurred in 2019 relative to 2018, the $meanSUM\Delta NDVI_{REF}$ in Il

Ngwesi Core area and Kalama Plateaus are much bigger than the values for Kalama Core and Sera Rhino Sanctuary reference polygons. Averaging in these higher $meanSUM\Delta NDVI_{REF}$ from those reference polygons virtually guarantees that conservancies with lower mean annual rainfall will exhibit frequent pixels with high $SUM\Delta NDVI_{rel}$ and thus be judged in error as unsuccessful. Therefore, reference polygons appropriate to the mean annual rainfall of the different Conservancies should be used for each year (2017-2020 & future monitoring period years). Reference polygon sets are broken into higher rainfall (wetter) and lower rainfall (drier) conservancies:

Reference set 1 is to be used with the higher rainfall conservancies – Il Ngwesi, Naibunga, Meibae, Nanapishu (Oldonyiro), Leparua, Nasuulu, Namunyak

Reference set 2 is to be used with the lower rainfall conservancies – Kalama, Westgate, Sera, Biliqo Bulesa, Melako, Nakuprat-Gotu

Thus, we calculated **relative** change in NDVI, $SUM\Delta NDVI_{rel}$ (Eq. 1) for the drier Conservancies Melako, Sera, Namunyak, Biliqo Bulesa, Nakuprat-Gotu, Nasuulu, Kalama, Westgate, and Leparua Conservancies using only the reference polygons from Kalama Core and the Sera Rhino Conservancy. Calculations for wetter Conservancies Il Ngwesi, Naibunga, Meibae, and Oldonyiro include all the reference polygons, as is coded in the current GEE model.

Justification for a Shift in the Threshold Value of Annual Relative $\Delta NDVI$

The threshold for unsuccessful implementation was obtained from two logistic regressions, which indicate the estimated $\Delta NDVI$ values that correspond to whether substantial livestock were present in a grazing block (Fig. A3), and whether substantial livestock were present for more than one two-week period during the year (unsuccessful implementation).

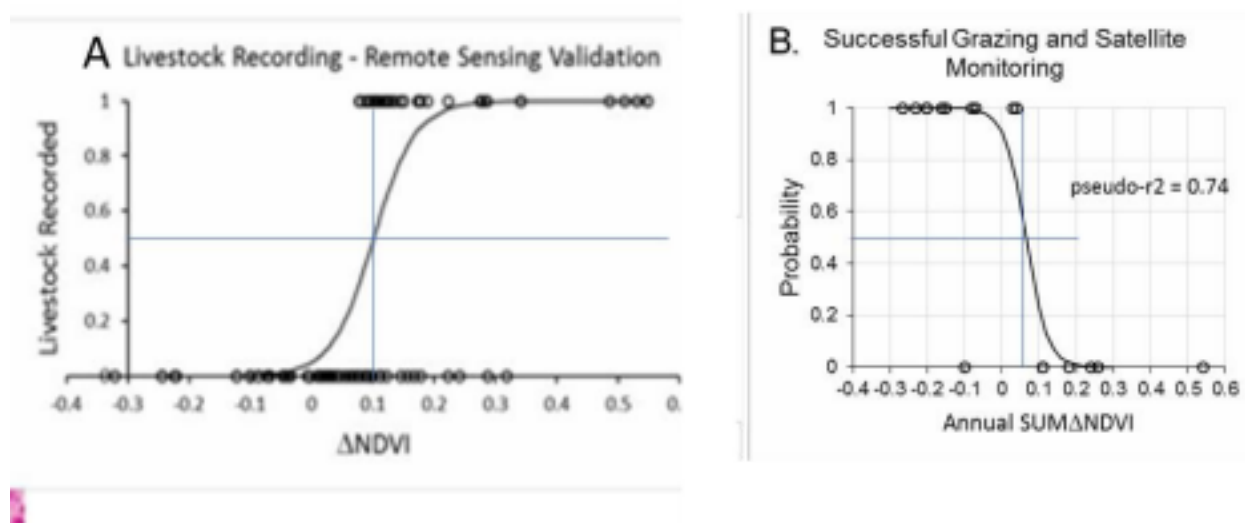


Fig. A1. Logistic regressions to assess successful implementation of grazing plans in the Northern Kenya Grasslands Carbon Project. Fig. A1.B. The logistic regression of livestock presence (livestock recorded = 1) and

absence (livestock recorded = 0) versus calculated **monthly relative $\Delta NDVI$** ($\Delta NDVI_{m,rel}$, Eq. 1), the value of monthly relative $\Delta NDVI$ at which there was greater than a 50% probability (livestock recorded > 0.5) of recording substantial livestock present is 0.1 (intersection of two perpendicular lines). This logistic regression was previously shown in Fig. A3C. Also, livestock were only rarely encountered in map locations where $\Delta NDVI_{m,rel} < 0.1$. Fig. A1.B. At 17 grazing blocks where presence/absence of livestock was monitored over an entire year, a logistic regression of the occurrence of only a single grazing event (probability = 1) or more than one grazing event (probability = 0) versus the relative annual sum of $\Delta NDVI$ for the 2014 calendar year (**annual relative $\Delta NDVI$, $SUM\Delta NDVI_{rel}$ from Eq. 2**). This logistic regression was previously shown in Fig. A4B.

The threshold of 50% probability of unsuccessful implementation occurs at approximately annual relative $\Delta NDVI = 0.08$, but, due to the small sample size of grazing blocks, there is a large confidence interval around that estimate. Since **monthly relative $\Delta NDVI$ ($\Delta NDVI_{m,rel}$)** must be > 0.1 for livestock to be detected at a location then it is reasonable that the annual sum, or **annual relative $\Delta NDVI$ ($SUM\Delta NDVI_{rel}$)** should be > 0.1. This is especially true for dry season grazing areas because they do not regrow immediately after grazing. Since it requires a relative change in NDVI of 0.1 to detect a grazing event (Fig. A1.A.), it is difficult for dry-season grazing areas to show a relative delta NDVI < 0.1 even if they are grazed only once. Thus using a threshold of 0.1 more accurately reflects the occurrence of single grazing events in **both** wet and dry season grazing areas.

This justifies a threshold of **annual relative $\Delta NDVI$ ($SUM\Delta NDVI_{rel} = 0.1$)** for assessing whether grazing plans were successfully implemented at a location. Thus, where pixels have $SUM\Delta NDVI_{rel} \leq 0.1$, these areas are deemed to be successfully grazed in each year 2017-2020. Pixels with $SUM\Delta NDVI_{rel} > 0.1$, are deemed to be unsuccessfully grazed in each year 2017-2020 and future monitoring period years.